

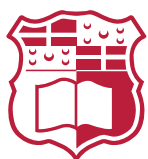
Uncertainty-Conditioned Reinforcement Learning with Evidential Policy Networks for Autonomous Systems

Antonio Galdes

Supervisor: Dr Josef Bajada

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L-Università ta' Malta

Faculty of Information &
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Abstract

Reinforcement Learning (RL) parking policies are usually developed and tested on the assumption that localisation is reliable. In operational parking areas, however, the accuracy of Real-Time Kinematic (RTK) Global Navigation Satellite System (GNSS) positioning can fall silently from centimetres to metres during a manoeuvre. The pose estimate remains available and plausible throughout, so a vehicle with no channel to carry that change commits to the manoeuvre regardless.

Two uncertainty mechanisms are therefore examined on a Proximal Policy Optimisation (PPO) parking agent, one supplying uncertainty and the other letting the actor use it. On the input side, an Extended Kalman Filter (EKF) fusing RTK GNSS with an Inertial Measurement Unit (IMU) passes its posterior covariance into the observation, quantifying how far the pose estimate can be relied on. On the output side, that signal is consumed by an evidential Normal-Inverse-Gamma (NIG) actor head, which produces per-action epistemic and aleatoric estimates in a single forward pass. Training runs in the CARLA simulator under an always-on GNSS fix-state Markov chain, with a reward that contains no uncertainty term. The covariance is thus observed but never rewarded, so any behaviour that depends on it must emerge on its own.

A 2×2 factorial ablation switches each mechanism on and off across three training seeds. Incorporating the covariance into the input of the evidential actor increased parking success by 26.8 to 36.2 percentage points (pp) on the trained lot, more than doubling the rate of the identical actor denied that input, which parked in under 30% of episodes. Each 95% bootstrap Confidence Interval (CI) excludes zero, and the arm receiving the covariance is favoured under all three seeds. Neither mechanism alone explains the gain, as the same input supplied to a standard Gaussian actor yields no measurable benefit, so the signal is made usable by the evidential head, most plausibly through its state-conditioned action variance, which the factorial does not isolate. The combination operates through emergent caution, with braking intensifying as the filter reports greater positional uncertainty, a Spearman rank correlation of +0.41 against +0.22 for the covariance-blind arm.

The uncertainty the head reports is of less use than the uncertainty supplied to it. In a single-head NIG actor the epistemic estimate is the aleatoric one divided by the evidence parameter, so the two channels separate only when that divisor varies across states. Model-free RL contains no term that supervises that divisor, as a sampled action is not its own label, leaving the two channels inseparable in practice.

The contribution is nonetheless the first placement of a NIG evidential head on the actor of a continuous-control policy-gradient agent, alongside the first use of raw EKF covariance as observation input for parking control.

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List of Abbreviations

ADR Automatic Domain Randomisation.

AUC Area Under the Curve.

CAN Controller Area Network.

CEP Circular Error Probable.

CI Confidence Interval.

COG Course Over Ground.

DDS Data Distribution Service.

DER Deep Evidential Regression.

DQN Deep Q-Network.

DRL Deep Reinforcement Learning.

DRMS Distance Root Mean Square.

EDL Evidential Deep Learning.

EKF Extended Kalman Filter.

GAE Generalised Advantage Estimation.

GNSS Global Navigation Satellite System.

GPU Graphics Processing Unit.

HPO Hyperparameter Optimisation.

HUA High Uncertainty Area.

IMU Inertial Measurement Unit.

JSON JavaScript Object Notation.

KL Kullback–Leibler.

LiDAR Light Detection and Ranging.

MDP Markov Decision Process.

MPC Model Predictive Control.

NGC NVIDIA GPU Cloud.

NIG Normal-Inverse-Gamma.

OOD Out-of-Distribution.

pp percentage points.

PPO Proximal Policy Optimisation.

ReLU Rectified Linear Unit.

RL Reinforcement Learning.

ROC Receiver Operating Characteristic.

ROS Robot Operating System.

RTK Real-Time Kinematic.

SAC Soft Actor-Critic.

SB3 Stable-Baselines3.

SLAM Simultaneous Localisation and Mapping.

TD Temporal-Difference.

TSUM Task-Specific Uncertainty Map.

YAML YAML Ain't Markup Language.

ZUPT Zero-Velocity Update.

Glossary of Symbols

a Action; a_t the action at decision t and a_t^{cmd} the command delivered after rate limiting.

$\hat{A}$ Generalised advantage estimate.

α Shape parameter of the Inverse-Gamma component of the evidential head.

β Scale parameter of the Inverse-Gamma component of the evidential head.

c_1 Weighting coefficient of the value-function loss within the PPO objective.

c_2 Weighting coefficient of the entropy bonus within the PPO objective, decayed to a floor within each stage.

$\Delta x, \Delta y, \Delta \psi$ Forward offset, lateral offset and heading discrepancy of the target bay in the ego frame, observation dimensions 5 to 7.

γ Discount factor of the Markov decision process, weighting future reward against immediate reward.

$\mathcal{H}$ Policy entropy, rewarded by the PPO objective to maintain stochasticity.

L^{CLIP} Clipped surrogate objective of PPO.

L^V Value-function loss of the PPO objective.

λ Weight of generalised advantage estimation, trading the bias of the value baseline against the variance of further reward terms.

λ_{reg} Coefficient weighting the evidential regulariser against the likelihood in the total loss, applied in this work to the prior-anchoring term and raised linearly from zero to its target at the start of each stage.

M Number of forward passes per prediction, being the Monte Carlo dropout samples or the members of an ensemble.

μ_0 Location parameter of the Normal-Inverse-Gamma evidential distribution, giving the predicted mean (the action mean on the evidential actor).

ν Virtual-evidence (pseudo-observation) count of the Normal-Inverse-Gamma distribution, governing confidence in the mean.

P State estimate covariance matrix of the extended Kalman filter.

- Φ Shaping potential over states. In this work the bay-frame corridor potential: the negative weighted sum of the along-track magnitude, cross-track magnitude and heading error, zero at the perfectly parked pose and negative elsewhere.
- π_θ Stochastic control policy with parameters θ .
- q Per-episode occupancy rate of the non-target bays, drawn uniformly between the stage bounds $q_{\min}$ and $q_{\max}$.
- Q Process noise covariance matrix.
- R Measurement noise covariance matrix.
- r_t Total per-decision reward: the additive sum of the progress, endgame, clearance, out-of-bounds and terminal terms, with no uncertainty term.
- s State of the Markov decision process; s_t the state at step t and s' the successor state.
- σ_{alea}^2 Aleatoric (data) variance estimate produced by the evidential head.
- σ_{epi}^2 Epistemic (model) variance estimate produced by the evidential head.
- σ_{total}^2 Total predictive variance, the sum of the aleatoric and epistemic components.
- σ_x Posterior standard deviation of the position estimate along the x axis, exposed to the policy as an observation dimension.
- σ_y Posterior standard deviation of the position estimate along the y axis, exposed to the policy as an observation dimension.
- σ_ψ Posterior standard deviation of the heading estimate, exposed to the policy as an observation dimension.
- V_ω Learned state-value function (the critic) with parameters ω .
- y^* Regression target (ground-truth label) of supervised deep evidential regression.

1 Introduction

Autonomous parking controllers are conventionally designed under the assumption of dependable localisation. The estimator delivering that position also reports its reliability, an output that existing designs leave unused and that this dissertation builds on. External influences degrade the localisation partway through a manoeuvre, yet the reported position arrives on schedule and in the expected form, carrying no mark of the error it has acquired. A policy developed on that position alone therefore cannot tell an accurate estimate from one wrong by several metres.

That missing distinction is supplied by the reinforcement learning architecture devised and assessed in this dissertation, which incorporates uncertainty at both ends of the policy. At the input, it receives the posterior covariance of an Extended Kalman Filter (EKF). Any sensor suite reports its uncertainty in that one form, and the noise models of the sensors fused set how large it grows. At the output, an evidential head generates a confidence metric for every selected action. The two mechanisms are subsequently crossed in a factorial ablation under degraded Real-Time Kinematic (RTK) Global Navigation Satellite System (GNSS) positioning, allowing each contribution to be attributed separately.

1.1 Problem Definition

Autonomous parking is treated as a continuous control problem in which a vehicle must come to rest within the markings of a conventional perpendicular bay 2.5 m wide. The scenario examined is forward perpendicular entry into a mapped site, so the geometry of the lot is known in advance and is never discovered during the manoeuvre. The critical factor is instead the accuracy of the vehicle's position estimate at every instant of the approach.

The estimate is generated by RTK-corrected GNSS fused with an Inertial Measurement Unit (IMU) through an EKF. Its accuracy is not constant but follows the quality of the satellite solution available at each moment. The target receiver is the u-blox ZED-F9P, with the fixed and standalone accuracy figures taken from its datasheet [1]. The position is accurate to within 1 cm while the satellite solution holds, then only to within a decimetre and afterwards a metre as that solution is lost. Those tiers form a hierarchy of fix states, ordered by the quality of the correction the receiver retains.

A policy trained on that estimate alone receives no evidence of a decline along the hierarchy, so the bay entry relies on a position that may be several metres inaccurate. A failure of that kind is judged by more than its consequences, since the

hazard analysis that ISO 26262 [2] mandates for road vehicles also scores its controllability, meaning whether a driver or operator can intervene in time to prevent harm. An unregistered decline is uncontrollable in that sense, as nothing prompts the intervention a favourable score assumes.

The information needed to mitigate this is nonetheless available, as the filter already reports the covariance of its own estimate, a direct measure of the position's reliability. Supplying that covariance as an observation therefore closes the gap at the input. A second gap remains at the output, where a standard actor emits a control with no indication of its reliability, whereas an evidential head attaches such a measure to each action. Both remedies are available in principle, yet neither has been tested on a parking policy while the satellite solution degrades mid-manoeuve (Section 2.3), leaving the following question open:

Does conditioning a continuous-control parking policy on its localisation uncertainty and incorporating an evidential output head measurably improve parking success and precision as localisation quality degrades, relative to the same architecture with each mechanism removed?

The question is framed comparatively, so each of the two separable mechanisms is measured against a control that differs from it by exactly one characteristic. The answer is accordingly reported in Chapter 4 as a set of controlled contrasts and not as a single performance metric.

1.2 Motivation

Conditioning a driving or navigation policy on its localisation uncertainty, the first mechanism named in the research question, has precedent, although the systems discussed in Section 2.2 each derive that signal from an indirect source. The covariance reported by the localiser at every step therefore remains unutilised. This study draws on it directly, taking the standard deviations of the two position axes and the heading as three features available on any vehicle equipped with the filter.

The uncertainty a policy reports over its own actions is separate from the one the filter supplies, and its two sources call for different responses. Noise inherent to the sensors is best answered by slowing the vehicle, since no amount of further data removes it, whereas a situation seldom encountered in training warrants handing control to an operator instead. A handover needs someone to receive it, and the proposed deployment environment is an instrumented outdoor parking facility with an operator on hand. The dependability of the reported estimate then governs whether it can trigger that handover, a question assessed in Section 4.8.2.

Reporting both components is nonetheless constrained by what a parking controller can afford to compute, as decisions are issued at a fixed rate on embedded hardware shared with perception, planning and the rest of the control stack. The sampling estimators discussed in Section 2.1.2 multiply the inference cost of every decision by the number of forward passes drawn, whereas Evidential Deep Learning (EDL) requires only a single pass [3], and the head constructed upon it is detailed in Section 3.5.

1.3 Aims and Objectives

The filter covariance at the input and the single-pass evidential head at the output together shape the aim of this work, which is to develop and evaluate an uncertainty-conditioned reinforcement learning architecture that enables an autonomous parking agent to degrade gracefully as localisation quality declines. That aim is decomposed into four objectives, each depending on its predecessor and progressing from the components developed to the experiments that isolate their contributions.

- O1** *Honest localisation uncertainty.* An EKF localisation framework is implemented in the simulator, integrating RTK-GNSS and IMU measurements under datasheet-derived noise models, such that the covariance it reports is an honest indicator of its true pose error. The objective is assessed in Section 4.2 by a calibration analysis comparing the reported position standard deviation with the actual position error.
- O2** *An evidential actor.* An evidential variant of Proximal Policy Optimisation (PPO) is developed in which the actor produces Normal-Inverse-Gamma (NIG) parameters, providing the single-pass uncertainty estimates motivated in Section 1.2. Training such a variant to convergence first requires mitigating the variance collapse inherent to this design, addressed by a purpose-built head construction (Section 3.5). Convergence alone is insufficient, as the outputs of the trained head must also be informative, and the objective is therefore assessed on three counts. The training record set out in Section 3.8.3 is first checked for stability across all seeds and curriculum stages, alongside the curriculum that produced it in Section 3.7. The behaviour of the uncertainty reported at runtime is then examined in Section 4.3, and the epistemic and aleatoric estimates are finally tested for separability in Section 4.3.2.
- O3** *Isolated contributions.* The contribution of each uncertainty layer is isolated through a two-by-two factorial ablation that crosses covariance observation with

the evidential head, producing four de-confounded arms, each trained under three seeds. Every arm is then evaluated according to parking success, final position error and an outcome taxonomy. The objective is therefore assessed by pooling those arms into covariance contrasts reported with bootstrap confidence intervals in Section 4.4, and the same contrasts are then broken down by seed in Section 4.7 to test whether the intervals survive retraining.

O4 *The mechanism.* Any performance gain is shown to operate through uncertainty-conditioned behaviour, in that the vehicle measurably slows, brakes harder and turns less sharply as EKF uncertainty rises. Such behaviour is not instructed by the reward, which contains no term referring to uncertainty, and any conditioning of this kind is therefore emergent rather than imposed. The claim is assessed in Section 4.6.1 by grouping brake command, speed, yaw rate and action jerk into bands of EKF standard deviation and comparing their correlation and mean across arms.

1.4 Proposed Solution

The system that realises the four objectives above is a dual-layer uncertainty architecture, depicted in Fig. 1.1 and developed in full in Chapter 3. Its two layers align with the two mechanisms motivated in Section 1.2, as the input layer incorporates localisation uncertainty into the observation, while the output layer represents uncertainty over the selected action.

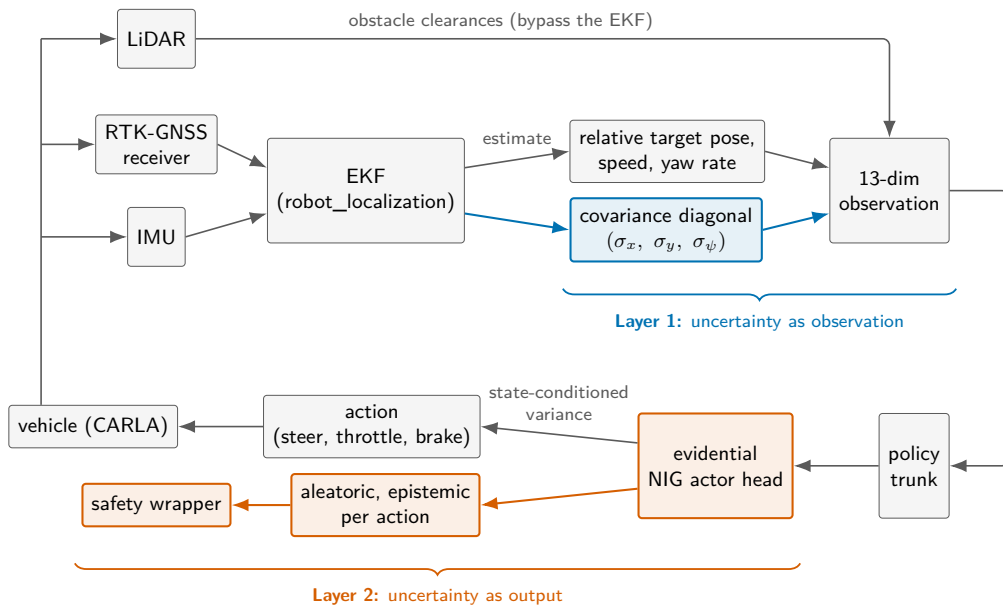


Figure 1.1 The two-layer uncertainty architecture.

On the input side, three of the thirteen dimensions of the observation vector are pose standard deviations derived from the EKF covariance diagonal. The two positional axes give σ_x and σ_y , and the heading gives σ_ψ . The remaining ten convey the vehicle state, the relative target pose and five Light Detection and Ranging (LiDAR) obstacle clearances. These three deviations arise from a standard Robot Operating System (ROS) EKF node operating live within a training loop, fusing the noisy GNSS and IMU streams rather than sampling a prescribed noise model. The covariance observed by the policy is consequently generated by the same estimation machinery that a deployed vehicle would carry. The complete observation specification and the localisation pipeline behind it are detailed in Sections 3.3 and 3.4.

On the output side, the standard Gaussian actor of PPO is set against an evidential head [3], a construction described in Section 2.1.3. It outputs the four parameters of a NIG distribution for each of the three action dimensions: steering, throttle and braking. Under the evidential arm, the variance used to sample actions is aleatoric and varies with the state, unlike the state-independent variance of the Gaussian actor. The same forward pass simultaneously produces an epistemic estimate, and a runtime safety wrapper uses both to slow the vehicle or hand control to the on-site operator. Adapting the evidential formulation to model-free reinforcement learning, in which no ground-truth action target exists, required a redesigned regulariser and a new initialisation protocol besides the variance-collapse mitigation already noted, all documented in Section 3.5.

Both layers are trained together in CARLA 0.9.16 [4] within a bespoke flat parking environment, under a six-stage single-phase curriculum (Section 3.7) that widens one difficulty axis at a time. Every observation channel is present from the initial stage, with the GNSS fix-state chain driving the covariance features from the outset, so successive stages increase the difficulty of the manoeuvre rather than the information available to the policy. Both uncertainty layers are nonetheless observed rather than rewarded during training, as the reward scores progress and outcome but contains no uncertainty term (Section 3.6), and the safety wrapper acting on their output is active in deployment alone, disabled throughout training and evaluation (Section 3.9).

1.5 Summary of Results and Contributions

Incorporating the EKF covariance into the evidential actor enhanced parking success by 26.8 to 36.2 percentage points (pp) on the trained lot, and the covariance was favoured under every seed, with a margin of +16.0 pp in the closest pairing. The same input, meanwhile, yielded no measurable benefit on a standard Gaussian actor, a null

result reported with the headline contrasts and the interaction analysis in Sections 4.4 and 4.4.2. The uncertainty that the evidential head reports for its own actions proves less tractable than the uncertainty supplied to it (Section 4.3.1). The sum of its aleatoric and epistemic estimates warns of failure in an empty lot, with an Area Under the Curve (AUC) of 0.88, and loses that warning near parked vehicles, where the figure falls to 0.44. These obstacles raise the reported uncertainty on successful episodes to the level already recorded on failing ones, so the estimate tracks obstacles and not failure. The two channels also fail to separate under every condition, because model-free reinforcement learning provides no supervisory signal for the evidence parameter that divides them. The remedy is consequently architectural, requiring a head grounded in the training distribution, which is proposed as future work in Section 5.2.

Together these findings bear on each of the four objectives, and Section 5.1.1 weighs each against the evidence obtained for it. The same results in turn support five contributions, stated in full in Section 5.1.2.

1.6 Document Structure

The foundational theory for each layer of the architecture is established in Chapter 2, covering reinforcement learning, uncertainty quantification and EKF localisation, against which three strands of existing work are evaluated. None of them couples a localisation-uncertainty observation with an actor capable of responding to it, and that deficiency is the research gap (Section 2.3) addressed by the proposed architecture.

The design decisions mandated by the gap are set out with their justifications in Chapter 3, and the assembled system is assessed against objectives O1 to O4 in Chapter 4. Each objective is then weighed against the resulting evidence in Chapter 5, where five contributions are derived from it and the questions the results leave unanswered become the closing proposals for further work.

2 Background and Literature Review

Six pillars support every component of the system, with the basis for each set out in Section 2.1, and the same pillars provide the axes for comparing prior work in Section 2.2. Omissions on the input and output sides of a learned controller emerge from that comparison, and both are consolidated into a single gap statement in Section 2.3.

2.1 Background

2.1.1 Reinforcement Learning and Proximal Policy Optimisation

Reinforcement Learning (RL) treats sequential decision-making as a Markov Decision Process (MDP), characterised by a state space $\mathcal{S}$, an action space $\mathcal{A}$, dynamics $p(s', r \mid s, a)$ that determine the transition probabilities and expected rewards, and a discount factor γ constrained by $0 \leq \gamma \leq 1$ [5]. A policy $\pi_\theta(a \mid s)$ maps states to actions through neural-network weights θ , which are optimised to maximise the expected discounted return $\mathbb{E}[\sum_t \gamma^t r_t]$. Policy-gradient techniques achieve this by modifying θ according to an estimated return gradient, computed with respect to the action representation the network parameterises. In the continuous-control setting adopted in this work, that representation is a distribution over actions rather than a probability for each action. The primary vulnerability of such techniques is that the gradient is derived from data gathered by the policy itself: the estimate is valid only while θ stays near its value at collection time, a constraint that a substantial update contravenes.

This instability is mitigated in PPO [6] using a clipped surrogate objective. The probability ratio $\rho_t(\theta) = \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$ measures how far an update has moved the policy, and the objective bounds the benefit attributable to any deviation of the ratio from unity:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right] \quad (2.1)$$

where $\hat{A}_t$ estimates the advantage of action a_t , and ϵ bounds the probability ratio $\rho_t(\theta)$ to the range $[1 - \epsilon, 1 + \epsilon]$. Once the ratio leaves that range, the lesser of the clipped and unclipped terms is taken, and no further gain is credited while the associated penalty applies in full. A batch of data can thus be reused across several gradient epochs without the policy deviating significantly from the distribution that generated it.

The advantage $\hat{A}_t$ entering Eq. (2.1) is itself produced by Generalised Advantage Estimation (GAE) [7], which blends multi-step Temporal-Difference (TD) errors into a

single exponentially weighted sum:

$$\hat{A}_t^{\text{GAE}(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l} \quad \delta_t = r_t + \gamma V_{\omega}(s_{t+1}) - V_{\omega}(s_t) \quad (2.2)$$

where V_{ω} is a learned value function estimating the expected return from a given state, and the weight $\lambda \in [0, 1]$ governs the decay of later terms in the summation, trading the bias of reliance on V_{ω} against the variance of further reward terms. This weighting matters in long-horizon tasks for which a terminal outcome must propagate back through numerous decisions. The value function is learned alongside the policy, its weights ω adjusted to minimise a regression loss against a return target:

$$L^V(\omega) = \mathbb{E}_t \left[(V_{\omega}(s_t) - \hat{G}_t)^2 \right] \quad (2.3)$$

where $\hat{G}_t = \hat{A}_t + V_{\omega}(s_t)$ is the return implied by the advantage estimate, so the same λ governs both the advantage and the regression target. The three components are subsequently combined into the full PPO objective:

$$L^{\text{PPO}}(\theta, \omega) = \mathbb{E}_t \left[L^{\text{CLIP}}(\theta) - c_1 L^V(\omega) + c_2 \mathcal{H}[\pi_{\theta}](s_t) \right] \quad (2.4)$$

Policy entropy is denoted by $\mathcal{H}$, while c_1 and c_2 are the weighting coefficients, and the policy and value parameters θ and ω are updated in one gradient step rather than in alternating phases.

The entropy term, introduced by Williams and Peng [8] and incorporated into deep actor-critic training by Mnih et al. [9], rewards a policy for maintaining stochasticity and prevents early convergence to a suboptimal deterministic one. Entropy thus sustains exploration, whereas the size of each update is bounded separately, by the clipping in $L^{\text{CLIP}}(\theta)$ and by an early stop on the Kullback–Leibler (KL) divergence. The clip constrains the probability ratio of each action independently, so numerous individually permitted shifts can still accumulate into a large change in the policy as a whole. The early stop guards against this accumulation from the training code instead of the objective. Under the Stable-Baselines3 (SB3) implementation [10, 11], it discards any remaining minibatches of an update once the approximate KL divergence between the data-collecting and updated policies exceeds 1.5 times a configured target.

The clip threshold ϵ remains the one unspecified quantity governing the update. A large-scale empirical study by Andrychowicz et al. [12] found a moderate ϵ to perform reasonably well across five continuous-control environments, supporting a single threshold fixed across every curriculum stage, the choice later set out in Chapter 3.

Two further empirical findings bear directly on the proposed design, both concerning the configuration around the objective rather than the objective itself. The first comes from Engstrom et al. [13], who found that code-level optimisations missing from the published algorithm, such as learning-rate annealing, produce most of the reward gain of PPO over its trust-region predecessor. The second comes from Andrychowicz et al. [12], who trained over 250 000 agents in five environments to measure the effect of more than 50 on-policy training choices. Normalising the observations proved crucial in almost every environment, and the learning rate ranked among the most influential parameters evaluated individually, well ahead of its decay schedule. Such parameters must therefore remain fixed across policy variants, which motivates the shared normalisation and learning-rate settings adopted later in Section 3.3.2. The same finding on the learning rate applies at a curriculum boundary, where the value at which the rate restarts, rather than the schedule state inherited from the previous stage, should define the configuration carried across a stage, as in the curriculum set out in Section 3.7.

The objective itself, irrespective of its configuration, admits a provably safe form of reward shaping. Ng et al. [14] showed that an extra reward term leaves the optimal policy of the underlying task unchanged whenever it takes the form $\gamma\Phi(s') - \Phi(s)$ at a transition from state s to state s' , for some potential Φ defined over states. The same analysis establishes that a bonus outside the potential-based framework can instead introduce cycles of positive net return, letting a policy accumulate reward indefinitely without terminating. The progress reward is designed in Section 3.6 to preclude that outcome.

Beyond training itself lies a constraint on what the actor can express, since the action spread is set by the architecture of the standard actor rather than the state. In the default Gaussian head of SB3 [10, 11], the action mean is a function of the state, whereas the log standard deviation is a single learned parameter vector independent of the input. A single global variance therefore governs exploration across all states, preventing the head from narrowing its action distribution in familiar states while widening it in unfamiliar ones. The limitation is a property of this particular head, not of policy-gradient methods as a class. Soft Actor-Critic (SAC) [15] conditions both moments of its action distribution on the input, so its spread varies from one situation to the next. A spread responsive to the vehicle’s circumstances therefore requires replacing the head, the purpose of the evidential actor developed in Section 3.5. The standard head is nevertheless retained as the comparison arm of a factorial ablation set out in Section 3.8, since holding it fixed while the input varies is what isolates the contribution of the replacement. A localisation covariance supplied to that unchanged head can shift the action mean but cannot widen or narrow the action distribution, and the consequences of the pairing are measured in Section 4.4.2.

2.1.2 Aleatoric and Epistemic Uncertainty in Deep Learning

Predictive uncertainty in deep learning is categorised into aleatoric and epistemic components. The distinction was initially established in risk and reliability analysis [16] and subsequently integrated into deep vision models by Kendall and Gal [17], who first estimated both within a single network. Aleatoric uncertainty refers to the intrinsic noise present in observations, which originates from sensors and stochastic disturbances in a driving context and cannot be reduced by gathering additional data. Epistemic uncertainty instead refers to the model's lack of knowledge regarding the function that fits the data, diminishing as more data is collected and increasing for inputs that deviate from the training distribution. Both components are set out in Table 2.1 alongside their sources and the response each warrants, since irreducible noise can only be accommodated in the manoeuvre itself, whereas a gap in the model's knowledge can be closed with further experience.

Table 2.1 The aleatoric-epistemic taxonomy.

Type	Source	Reducible	Operational response
Aleatoric	Noise inherent in the observations and dynamics (sensor noise, stochastic disturbance)	No	Act more conservatively; slow down; widen margins
Epistemic	Model ignorance; inputs outside the training distribution	Yes, with data	Defer or hand off the decision; collect the missing experience

For a regression output y^* , the two components add to the total predictive variance:

$$\sigma_{\text{total}}^2 = \underbrace{\sigma_{\text{alea}}^2}_{\text{data noise}} + \underbrace{\sigma_{\text{epi}}^2}_{\text{model ignorance}} \quad (2.5)$$

In conventional estimators, the epistemic component is obtained through model sampling. Under Monte Carlo dropout [18], the random masks stay active at test time and M stochastic forward passes are interpreted as samples from an approximate posterior over the weights. This sampling accounts for the epistemic component alone, so the aleatoric noise term remains a single calibrated constant rather than a per-input estimate. Deep ensembles [19] instead recover both components by training M independently initialised networks, extracting the epistemic component from disagreement among their predicted means and the aleatoric component from the variance predicted by each member. Disagreement obtained this way carries no uncertainty beyond what the observed data supplies, so Osband et al. [20] add a fixed random prior network to each member and train it against the residual. The networks

are driven into agreement wherever data constrains them, while retaining the diversity of their priors elsewhere. Both methods share a structural cost, as uncertainty characterised across samples entails M forward passes, whereas a deterministic network requires only one. For a controller operating at a fixed decision rate on embedded hardware, the sample count multiplies the per-decision inference budget, the primary rationale for the single-pass approach adopted in Section 3.5.

2.1.3 Evidential Deep Learning

EDL acquires the same two components through a single forward pass by predicting the parameters of a higher-order distribution over the likelihood. No model sampling is necessary, so an uncertainty estimate costs the same as a deterministic network. Sensoy et al. [21] proposed a classification framework in which the evidence produced by a network sets the concentration parameters of a Dirichlet distribution over class probabilities. The total evidence nonetheless collapses to a single scalar uncertainty mass, so the Dirichlet formulation cannot separate the two components defined in Eq. (2.5). Amini et al. [3] closed that gap by extending the mechanism to continuous targets as Deep Evidential Regression (DER), imposing a NIG prior on the parameters of a Gaussian likelihood to reinstate the separation:

$$y^* \sim \mathcal{N}(\mu, \sigma^2) \quad \mu \sim \mathcal{N}\left(\mu_0, \frac{\sigma^2}{\nu}\right) \quad \sigma^2 \sim \Gamma^{-1}(\alpha, \beta) \quad (2.6)$$

where μ and σ^2 represent the unknown mean and variance of the likelihood over the target y^* , and Γ^{-1} denotes the inverse-gamma distribution.

A network trained with this prior generates the four NIG parameters $(\mu_0, \nu, \alpha, \beta)$ for each output dimension, which are interpreted as virtual observations. The predicted mean is μ_0 , and ν counts the observations supporting it. The noise variance is described by the remaining pair, α counting the observations supporting it and β setting its scale. Since counts and a scale cannot take negative values, μ_0 alone is produced by a linear unit, while the remaining three pass through softplus activations that constrain them to their valid ranges ($\nu > 0, \alpha > 1, \beta > 0$). A softplus returns a positive value that can fall below one, so the activation on α carries an additive offset to meet its lower bound.

The two counts combine into the total evidence $2\nu + \alpha$, from which the prediction and both uncertainty components are derived in closed form as the NIG moments:

$$\mathbb{E}[\mu] = \mu_0 \quad \underbrace{\mathbb{E}[\sigma^2] = \frac{\beta}{\alpha - 1}}_{\text{aleatoric}} \quad \underbrace{\text{Var}[\mu] = \frac{\beta}{\nu(\alpha - 1)}}_{\text{epistemic}} \quad (2.7)$$

Aleatoric uncertainty is therefore the expected data noise, while epistemic uncertainty is the variance of the predicted mean. The two differ by a factor of $1/\nu$, so the

epistemic component diminishes as ν increases, whereas the aleatoric estimate remains unaffected. All four parameters are emitted from a shared feature trunk, as illustrated in Fig. 2.1.

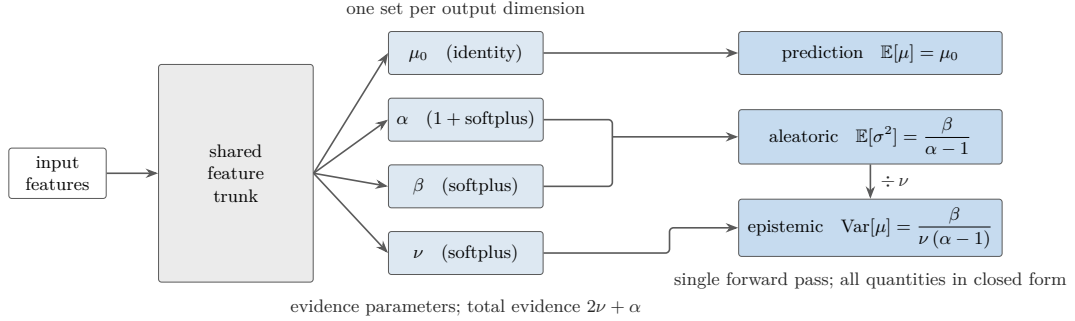


Figure 2.1 The NIG evidential regression head, after Amini et al. [3].

Fitting that head requires a target to regress against, which the supervised setting of Amini et al. [3] supplies as a label. Training then optimises the marginal likelihood of the label under the NIG prior, available in closed form as a Student- t density. Fitting the likelihood alone leaves the estimate over-confident, so a supplementary evidence regulariser penalises confidence in incorrect predictions:

$$L^R(\theta) = |y^* - \mu_0| (2\nu + \alpha) \quad (2.8)$$

The total evidence consequently scales the prediction error, and this scaled penalty enters the total loss weighted by a coefficient λ_{reg} [3]:

$$L^{\text{DER}}(\theta) = L^{\text{NLL}}(\theta) + \lambda_{\text{reg}} L^R(\theta) \quad (2.9)$$

where $L^{\text{NLL}}(\theta)$ denotes the negative log of the Student- t marginal likelihood, and λ_{reg} sets how strongly the regulariser is weighted against it. The likelihood and the regulariser nonetheless both depend on the ground-truth target y^* , since the likelihood scores the fit of the predicted distribution to the label and the regulariser withdraws evidence in proportion to the prediction error. Such a target is therefore a structural premise of the framework, and its absence proves decisive in the reinforcement-learning adaptation examined in Section 3.5.2.

Three acknowledged limitations restrict how the NIG output may legitimately be used. Meinert et al. [22] establish the first of these, identifying three shortcomings that render DER a heuristic rather than an exact method for quantifying uncertainty. The initial shortcoming concerns clean disentanglement, which the marginal likelihood alone cannot provide because it collapses the four NIG parameters into a single Student- t density. Several divisions between the aleatoric and epistemic components can therefore fit any label equally well. Recovering a unique separation would require

assuming smoothness in the input and prior knowledge of at least one component, neither of which is supplied by the training loss alone. The second concerns the quantity represented by the epistemic estimate, since the regulariser’s gradient dynamics determine how quickly ν converges during training. The estimate consequently reflects the point-wise convergence rate of the model rather than a principled posterior, so its magnitude has no absolute meaning and supports only comparisons between inputs observed in the same pass. The third concerns calibration, which remains unattainable because epistemic uncertainty is a property of the model rather than the data, leaving no ground truth to calibrate it against. The same authors observe that the method remains useful when the response to a flagged instance depends only on the estimate being large rather than on its nominal value being correct. The approach then retains practical value, since its convergence-rate proxy is available for any input at inference time and flags slow-converging instances as uncertain without labelled out-of-distribution training examples. The evidential output is therefore treated as a single-pass, relative uncertainty signal rather than an exact Bayesian quantity, and the reliability of that relative information is tested in Section 4.3.1.

The second limitation is that the objective favours small predicted variance, and training drives the evidence parameters towards degenerate values that minimise it rather than reflect the data. Optimisers push ν to within machine precision of zero even under small λ_{reg} [22], rendering the epistemic proxy meaningless. The concern extends beyond regression, as the survey by Gao et al. [23] reports that subsequent Dirichlet-based formulations deprecate an analogous variance-minimising term to alleviate overconfidence. The consequences depend on the role the variance plays. In a supervised regressor, the damage is confined to the reported uncertainty, which understates the model’s ignorance while the predictions remain as accurate as before. Once the predicted variance also sets the exploration noise, as it will for the actor introduced in Section 3.5, the shrinking variance narrows exploration itself, a collapse mitigated by the adaptation set out in Section 3.5.2.

The final limitation is that the gradient vanishes at both boundaries of the evidential parameter space, preventing a network from being trained out of either extreme once there. The most severe case is the High Uncertainty Area (HUA) characterised by Ye et al. [24], in which the gradient of the loss with respect to the pre-activations vanishes as α approaches its lower bound of one. At this limit, the aleatoric estimate given by Eq. (2.7) also diverges, so training stalls exactly where the reported uncertainty is least trustworthy. Oh and Shin [25] identified a similar failure at the opposite extreme, where large evidence suppresses the gradient with respect to the predicted mean instead, and training ends without correcting the prediction. Their remedy alters the objective rather than the output range, penalising the region instead

of excluding it. Pandey and Yu [26] found the same failure in classification, among the activations that maintain non-negative evidence. Such activations return exactly zero over part of their input range, so samples mapped to that region contribute neither evidence nor gradient. Together, these findings mark the boundaries of the NIG parameter space as a region that any architecture producing these parameters must avoid. Two devices are introduced in the next chapter to enforce that, a head construction that bounds the outputs at source (Section 3.5.1) and a pair of anchoring and flooring terms added to the objective (Section 3.5.2).

Beyond these three limitations, disentangled uncertainty in RL has a precise definition. Charpentier et al. [27] propose four desiderata for uncertainty in a reinforcement-learning agent. Two of them directly govern the epistemic estimate, which should shrink as training accumulates experience of a fixed environment and rise on inputs from outside that training distribution. The remaining two concern the utility of the estimate, requiring that actions sampled from the epistemic distribution rather than the aleatoric one improve sample efficiency during training and generalisation at test time. Meeting all four is an architectural matter, and Charpentier et al. evaluate four uncertainty methods against the desiderata, of which only an evidential model satisfies the concentration criterion. Detection of departed states additionally requires the estimate to be tied to a density or a kernel over the encoded state. The estimate is attached to the predicted return in every variant they examine, whereas the head developed in Section 3.5 attaches it to the action distribution, so their conclusions carry over to the present design only as far as that change of target permits. These desiderata motivate the aleatoric-epistemic separation examined in Section 4.3.2, and the placements attempted against the four criteria are surveyed in Section 2.2.2.

2.1.4 Extended Kalman Filter and Satellite Positioning

State estimation provides a confidence metric alongside each pose estimate, so the structure of that signal determines what the proposed architecture can condition on. The Kalman filter is the optimal recursive estimator for linear-Gaussian systems. The EKF extends it to non-linear motion and measurement models by linearising both around the current state estimate [28]. Two phases alternate, the first being prediction, in which the motion model advances the state x_k while the process noise Q inflates the error covariance P :

$$P_k^- = F_k P_{k-1} F_k^\top + Q_k \quad (2.10)$$

where k denotes the filter timestep, F_k is the motion-model Jacobian, and the superscript minus marks a quantity not yet corrected by a measurement. The second phase weights a measurement with noise covariance R_k against the predicted

covariance through the Kalman gain K :

$$K_k = P_k^- H_k^\top (H_k P_k^- H_k^\top + R_k)^{-1} \quad (2.11)$$

with H_k representing the measurement Jacobian. A larger gain assigns greater weight to the measurement and therefore contracts the covariance more sharply:

$$P_k = (I - K_k H_k) P_k^- \quad (2.12)$$

The covariance therefore expands between position fixes under the prediction step and contracts at each accepted fix according to Eqs. (2.11) and (2.12), producing a sawtooth profile that is traced empirically for the configured filter in Section 3.4.2 and plotted in Fig. 3.5. The height of each tooth is set by the quality and recency of the measurements, so the profile records the fix state prevailing at every moment of a manoeuvre.

Such covariance is available to any policy at no additional inference cost, since it is computed as a by-product of estimation. Its informativeness depends on how the filter is configured. A study of a wheeled-robot EKF that varied the noise matrices and the parameters governing how uncertainty propagates found that a tuned filter substantially reduced position error relative to a fixed-parameter baseline [29]. That result motivates treating Q as a tuning variable for the present system in Section 3.4.2. Such tuning is exposed by the `robot_localization` package of Moore and Stouch [30, 31], the de facto standard EKF implementation in the ROS ecosystem.

GNSS supplies the position fixes that contract this covariance, so the size of each reduction follows the accuracy of the measurement producing it. A standalone receiver computes position from code-phase pseudoranges with metre-level errors, whereas RTK processing uses carrier-phase measurements relative to a nearby base station and achieves centimetre accuracy once the integer carrier ambiguities are resolved [32, 33]. The ambiguity-resolution status is reported alongside every fix, creating a discrete quality ladder: an ambiguity-fixed RTK solution at centimetre level, an ambiguity-float solution at decimetre level, and a standalone fallback at metre level, the three regimes into which measured highway error distributions separate [34]. Manufacturer datasheets specify representative accuracies for the fixed and standalone modes [1], quoted as Circular Error Probable (CEP), the radius within which half of the fixes fall. Recovering a per-axis standard deviation from that radius requires a circularly symmetric Gaussian assumption, a relation derived together with the margin adopted for each rung of the ladder in Appendix A.1. Each level can therefore parameterise a Gaussian noise model directly. This symmetry idealises the error of a real fix, which is elliptical rather than circular. The arrangement of the visible satellites

orients that ellipse, and the error is smallest along the direction the arrangement constrains most tightly [35, sec. 6.1.3]. The east and north components of a reported covariance therefore generally differ.

The accuracy a pose needs to be treated as reliable is set independently of the receiver, so the tiers of the ladder can be assessed against it. Reid et al. [36] derive localisation alert limits for automated vehicles from road geometry, obtaining bounds of 0.57 m laterally and 1.40 m longitudinally on freeways and a tighter 0.29 m in both directions on local streets, where the narrower lane leaves less room for error. Each bound is paired with an integrity risk, the permissible rate at which it may be exceeded. Commercial aviation suffers roughly one hundredth the fatalities per mile travelled that road transport does, and that rate is adopted as a reasonable safety target for a self-driving vehicle. It implies a fixed tolerance for failures per mile across the whole vehicle, and dividing that allowance among the subsystems leaves localisation a budget of 10^{-9} exceedances of the alert limit per mile. An ambiguity-fixed RTK solution errs by centimetres and so breaches the limit rarely enough to stay within that budget, whereas the lower two tiers breach it far more often. A float solution errs by a decimetre and a standalone fallback by a metre, both against limits of tens of centimetres. Descending the ladder therefore carries the pose beyond the alert limit on a growing share of fixes, and the consequences for a learned controller are taken up in Section 2.2.3.

A receiver ordinarily maintains the fixed tier of this ladder throughout a manoeuvre, but the centimetre accuracy required for parking is forfeited whenever that tier is lost. Reflections from structures and large vehicles are the principal cause of this degradation. Under multipath interference the directly received signal is distorted by a reflected copy, whereas under non-line-of-sight reception the blocked direct path is replaced by the reflection alone, and the latter produces large positive ranging errors that conventional multipath mitigation does not correct [32]. The same obstructions reduce the number of satellites tracked and may interrupt the correction stream on which RTK depends. Positioning does not stop outright, as the reported fix state instead descends the quality ladder while an obstruction persists and recovers once it clears, making the degradation observable to any system that consumes the fix. The appropriate response to a descent is for the vehicle to slow while its position is poorly known, or to stop until the fix is restored. An agent trained exclusively on a fixed solution would never encounter a state in which continuing at full speed was incorrect, and training therefore requires exposure to the entire ladder, which the Markov fix-state chain in Section 3.4.1 provides instead of a per-episode constant.

Reproducing that ladder in simulation does not require modelling the signal path of a GNSS receiver. Two bodies of work instead model the error statistically at successively later stages of the receiver chain. Rankin [37] focused on the pseudorange, representing the ephemeris, atmospheric and multipath contributions as

first-order Gauss-Markov terms, adding Gaussian white measurement noise, and then scaling the result to a position error using the dilution of precision of a simulated constellation. Karlsson and Mohammadiha [38] turned to the published fix, the final output of the receiver, fitting a state-based stochastic error model for automated driving. In their model, a discrete state selects the error statistics, and an autoregressive Gaussian-mixture model parameterises the error within each state. Both works establish statistical injection rather than signal-path modelling as an accepted simulation practice, and they differ in where the error is injected. Rankin's pseudorange-level treatment would require a simulated constellation that the driving environment does not expose, whereas the fix-level formulation of Karlsson and Mohammadiha acts on the quantity a vehicle actually consumes. The discrete-state formulation is adopted for that reason.

Position is not the only observable such a model must cover, as a receiver reports several. Velocity derived from Doppler measurements is considerably more precise than differencing successive position fixes, reaching centimetre-per-second accuracy even for a standalone receiver [39]. The Course Over Ground (COG) is the direction of that same Doppler velocity vector and inherits its precision, though it is defined only while the antenna is in motion. Position, velocity and COG are therefore available to a fusion filter as separate observables with distinct error budgets [32].

2.1.5 Curriculum Learning and Domain Randomisation

Exposing a policy to the full difficulty of a task from the outset can leave it without a viable path towards competence. Curriculum learning addresses that risk by ordering training experience from simpler to harder tasks, a principle formalised by Bengio et al. [40] and developed into a framework for RL by Narvekar et al. [41]. The same survey identifies the associated drawback, as each stage transition alters the training distribution and leaves a policy converged at one stage vulnerable to catastrophic forgetting [42, 43] once the meaning of its inputs changes at the next. The principal curriculum-related mitigation is to vary every input channel from the first step, ensuring that a stage transition changes how much a channel varies rather than whether it varies at all.

A second reason to vary every channel from the first step concerns plasticity, a network's capacity to keep adapting its weights as training progresses. Sokar et al. [44] show that neurons falling below an activation threshold become dormant and remain so for the rest of training, while Abbas et al. [45] demonstrate the resulting loss of plasticity across sequential tasks. In each case the damage is done by inputs that stop driving a unit's activation. The limiting case of such an input is a channel held constant across training samples, which never drives activation at all. The units reading it

therefore have nothing to fit, and their capacity to use the channel is diminished once it begins to vary after millions of training steps.

Domain randomisation addresses the same problem from the opposite perspective. Tobin et al. [46] vary every randomised quantity for each training sample across the full distribution of environment parameters, so no channel becomes degenerate and the plasticity loss described above cannot set in. The cost is that a policy must cope with the entire spread before it has learnt the task, a situation a curriculum is designed to avoid. Automatic Domain Randomisation (ADR), introduced for training in-hand manipulation at scale [47], reconciles the two approaches. The initial distribution is concentrated on a single calibrated environment, and each randomisation range is expanded independently as measured performance permits [47, 48]. Difficulty is therefore introduced progressively, although the narrow initial distribution reinstates the degeneracy avoided by the fixed scheme. The rule governing when each range expands is itself a pacing decision, and Narvekar et al. [41] treat pacing as a question distinct from task ordering, separating curricula advancing on a predetermined schedule from those advancing only once the agent meets a performance threshold. Florensa et al. [49] show that the latter scheme yields faster learning than the former, holding the agent at a stage until its success rate enters a target band.

The training protocol adopted in Section 3.7 draws on these findings and treats pacing as a decision separate from task ordering. The support-preserving property of full randomisation becomes the channel-liveness principle described in Section 3.7.1, applied to each observation channel rather than to environment parameters. Despite the faster convergence Florensa et al. report, a fixed schedule is adopted, since ablation arms governed by their own success would cross stage boundaries at different points. A difference at evaluation could then be attributed to the schedule as readily as to the head under test. Progression is instead tied to a per-stage budget, accepting weaker pacing in exchange for a comparison in which the head is the only term that varies.

2.1.6 Evaluation Methodology in Deep Reinforcement Learning

Empirical comparisons between Deep Reinforcement Learning (DRL) methods are prone to specific, well-documented failure modes, and avoiding them requires four conditions established across several studies of sound evaluation practice. Henderson et al. [50] demonstrated that outcomes in DRL are so sensitive to the random seed and hyperparameter configuration that two seeds of the same algorithm can differ by more than the gap between distinct algorithms. Single-seed comparisons are therefore unreliable, and multi-seed reporting with the variation shown is the minimum credible practice. Eimer et al. [51] extend this point to the tuning protocol, establishing that

hyperparameters must be tuned for each algorithm rather than transplanted between them. The tuning seeds must in turn differ from the evaluation seeds, since a configuration tuned on the evaluation seeds is fitted to the very runs meant to judge it. Results must also be reported as bootstrap intervals, since Agarwal et al. [52] show that a point estimate over few runs hides how widely the result varies between runs. A final condition concerns generalisation, for which Cobbe et al. [53] find that performance is overstated when training and testing occur under the same conditions. A credible assessment therefore requires evaluation under held-out conditions unseen in training.

All four requirements shape the experimental protocol that follows. The tension that tuning imposes between method-specific configuration and a controlled ablation is addressed when the ablation is designed in Section 3.8.1, and the held-out criterion sets a generalisation boundary tested last in Section 4.8.1.

2.2 Literature Review

The prior work most closely related to the proposed architecture falls into three threads, one for each subsection below: uncertainty-aware RL for driving, evidential methods in RL, and learned autonomous parking. Each thread is organised by where uncertainty is placed relative to the manoeuvre.

2.2.1 Uncertainty-Aware Reinforcement Learning for Driving

An uncertainty estimate can be added to the output of a driving policy, to its observation, or to its reward. Adding it at the output leaves the policy itself unchanged and gates the decisions it proposes. Hoel et al. [54] train an ensemble of value networks using randomised prior functions for tactical highway decisions. Epistemic confidence is derived from the spread of that ensemble, and the selected decision is replaced by a safe baseline action whenever its confidence falls too low. These randomised priors, described in Section 2.1.2, keep the ensemble members apart in states the agent has never visited, so such states produce a wide spread and low confidence. Each member, however, predicts only an expected return and has no variance of its own to capture aleatoric uncertainty. Once the members are trained on enough data, their spread therefore shrinks to zero even when the outcome itself remains highly variable. The later ensemble quantile networks of Hoel et al. [55] address that limitation by replacing each value network with one predicting a distribution over returns rather than an expected return. Aleatoric uncertainty is derived from the dispersion of that distribution, while epistemic uncertainty is obtained from disagreement between members. Any decision deemed unreliable on this basis is handed to a backup policy, which stops the vehicle if the remaining distance permits.

Despite the change of estimator, the value ensemble and its quantile successor share three characteristics. The quantity estimated is the value of candidate decisions rather than the state of the vehicle, and estimating it requires an ensemble costing M forward passes at every decision. The estimate is then applied as a veto over the discrete manoeuvres proposed by the policy. The veto is activated at a confidence threshold selected after training as a runtime hyperparameter, and the policy itself never observes the uncertainty. At runtime neither system scales its response to the size of the uncertainty. The quantile successor can produce caution proportional to it, though only after retraining with a risk-sensitive objective. These findings establish that a value-based estimate can withhold an unsafe manoeuvre, yet both quantities are read off an ensemble of value predictions rather than emitted by the policy that produces the action.

Moving the estimate into the observation gives the policy what the value-based systems withhold from it, an input read before the action is chosen. The signal therefore shapes the manoeuvre as it is formed, instead of correcting one already committed to. Puthumanaillam et al. [56] condition navigation policies on a Task-Specific Uncertainty Map (TSUM), an abstraction that divides the workspace into regions and records the acceptable level of state-estimation uncertainty in each. The policy reads the acceptable level at its location alongside a single scalar representing its current uncertainty. These two inputs enter a SAC learner, and the relevant property of that learner is the state-dependent action spread noted in Section 2.1.1. Adaptation to the context-dependent value of certainty is therefore divided between the map and the learner: the map retains the tolerance for each region, while the state-dependent spread converts it into the caution the vehicle applies. Kim et al. [57] retain the scalar form but remove the map, conditioning a policy solely on an epistemic uncertainty signal so that the vehicle drives differently in unfamiliar situations. The scalar alone has little effect in their ablation, so a mutual-information term linking uncertainty and action is added to the training objective. The manoeuvre must consequently be made to vary with the uncertainty input, rather than uncertainty being treated as a quantity to minimise. Both signals describe the vehicle’s own state, though a policy can be conditioned just as readily on how uncertain its surroundings are. Golchoubian et al. [58] illustrate this by predicting the motion of nearby pedestrians and supplying a crowd-navigation agent with the confidence of each prediction, training the agent to reduce its collision probability. A vehicle passing an unpredictable pedestrian consequently yields more room than one passing a steadily moving pedestrian, the closest any surveyed system comes to letting caution track the size of the estimate. The signal nevertheless reports the motion of other agents, so it is produced by a separate prediction module rather than the vehicle’s own state estimator. These systems establish that a policy conditioned on an uncertainty input can vary the

vehicle’s caution continuously with that input, instead of leaving it unchanged until a threshold is crossed.

A reward defined over the localiser’s covariance consumes it without a proxy, making uncertainty the target of optimisation instead of a condition on it. Placed and Castellanos [59] reward an exploration agent using the D-optimality of the pose covariance maintained by the particle filter of their Simultaneous Localisation and Mapping (SLAM) back end, so a discrete-action deep Q-learner learns to steer back towards previously mapped landmarks, where a repeat observation corrects the accumulated pose drift and tightens the filter’s covariance. Seeking uncertainty to reduce it is the appropriate response for exploration, in which information is the objective. In the forward perpendicular parking task defined in Section 1.1, the premise is reversed. Uncertainty arises exogenously from the satellite geometry, leaving the vehicle to manoeuvre under uncertainty it cannot lower. A reward term penalising it would therefore charge the vehicle for a condition that no trajectory within a bay-sized workspace can improve.

The remaining placement sits above the manoeuvre altogether, where the signal becomes a guarantee about the policy. Zhou et al. [60] estimate the performance uncertainty of a driving policy from the sparsity of its training data and the residual error of the fitted network, then derive a lower bound on its return. A rule-based planner operates by default, and the learned policy is engaged only when that bound exceeds the planner’s expected performance. The uncertain quantity is therefore the overall reliability of the policy rather than the ego state. The estimate gates exploration during training and selects between the planner and its learned counterpart at runtime, yet remains absent from the observation on which the policy conditions. Across this thread, localisation-related uncertainty becomes available to a driving policy only after an abstraction has converted it into another form. None of the reviewed ego-motion driving policies therefore includes the state estimator’s reported covariance among its observation features.

2.2.2 Evidential Methods in Reinforcement Learning

Evidential techniques are particularly attractive when ensembles are prohibitively expensive because of the single-pass property established in Section 2.1.3, although their adoption within RL remains limited to a handful of systems. Most apply the NIG prior given in Eq. (2.6) to the critic, expressing the value of a state as a distribution rather than a point estimate. The distribution serves three purposes across the systems surveyed. The first is to maintain the learner’s adaptability as the dynamics of its environment change. Akgül et al. [61] apply this construction to the value function of a PPO learner, the earliest evidential critic reported for an on-policy actor-critic. The

resulting head is claimed to address non-stationary conditions through two properties, one on each side of the learner. On the critic side, a change in dynamics makes the value target unfamiliar and so widens the spread. In turn, the update applied for a given error grows with it, so the critic continues to adapt long into training. On the actor side, the spread propagates into the advantage, with an optimistic bonus proportional to it added before the policy update. The complete method outperforms all on-policy baselines across continuous-control tasks perturbed at regular intervals. Both benefits nevertheless originate in a head placed on the value function and trained against the critic’s regression targets, so the advantage is the only route by which the estimate reaches the actor. The sampling distribution remains the conventional Gaussian head described in Section 2.1.1, so the spread of the manoeuvre during execution is left to a variance that carries no per-state information.

Delivering the estimate to the manoeuvre is only the first requirement, since a single combined quantity signals low confidence without indicating its cause. Separating the aleatoric and epistemic moments defined by Eq. (2.7) is therefore the second purpose. Stutts et al. [62] combine evidential estimates of the action values of a quantile-regression Deep Q-Network (DQN) with quantile calibration based on conformal inference, evaluated on miniaturised Atari games over a discrete action space. The evidential half of that construction has the network emit its parameters alongside the quantiles of the value distribution, so aleatoric and epistemic uncertainty are read off separately through the two moments without sampling. Both scores and learning speed improve, so the separation costs the agent no performance. An equivalent separation is reached from an identical value-side placement by the one agent that Charpentier et al. [27] find to meet all four desiderata set out in Section 2.1.3. The agent is a deep Q-learner of the same value-based family, drawing its action values from the Posterior Network [63]. A Gaussian likelihood over each return carries the aleatoric term, while the epistemic one shrinks as the evidence count rises. A normalising flow trained alongside the encoder supplies this count from its latent space, instead of an output head predicting the NIG parameters defined in Eq. (2.6). A sparsely visited region of the state space therefore attracts a low density and a correspondingly high epistemic score, which registers the familiarity of a state and not the confidence of the pose held within it. Both systems estimate the two components for each action in a finite set, so neither transfers directly to the continuous steering, throttle and braking commands of the parking manoeuvre assessed in this dissertation.

In the third application, the estimate is placed on a planning constraint rather than a value. Ham and Ahn [64] apply EDL to the position of a detected obstacle, and a distributionally robust Model Predictive Control (MPC) planner consumes the resulting distribution to define an ambiguity set that grows and shrinks with perception confidence. The planner therefore remains efficient when the perception network is

confident, while conservative margins are enforced when it is not. A response graded by confidence in the same way is sought from the evidential actor head developed in Section 3.5, where the spread of the action distribution widens as the evidence supporting the proposed action falls. The estimate is nevertheless derived from the position of another object rather than the ego pose, and it is produced by a planner downstream of any policy. The caution is therefore a property of the controller that overrides the manoeuvre rather than of the manoeuvre itself.

Beyond these three applications, one system moves the estimate off the learner entirely and onto the localisation stack that supplies its pose. Kalyanasundaram et al. [65] place a NIG head on a network that corrects the estimated velocity from onboard motion sensing alone, so the correction supplied to an error-state EKF arrives with an uncertainty estimate obtained in the same forward pass. The estimate weights the correction as a virtual measurement, which reduces position drift through prolonged GNSS outages on real vehicle data. The estimate is consumed by the filter upstream of any controller, leaving no learned policy to condition on it. In the architecture proposed in Section 3.5, the filter covariance is instead retained as the pose signal, and the evidential head is moved to the action selected under it.

The actor remains downstream of the estimate across all of these placements, and only Kanazawa et al. [66] close this distance, forgoing an evidential head for an ensemble of actor-critic learners in continuous control. They guide exploration with epistemic uncertainty, measured as the spread across the actors' outputs, and make the policy risk-sensitive with aleatoric uncertainty. The estimate is consequently available at the point of action, although reaching it costs the M passes established in Section 2.1.2 rather than the single pass that motivates evidential methods. The decomposition is therefore recovered across the ensemble members rather than from the quantity emitted by the actor, which draws the manoeuvre from a conventional Gaussian head.

Setting the ensemble aside, none of the five systems carrying an evidential head places an evidential distribution over the action, as all share a constraint on where such a head can be attached. Akgül et al., Stutts et al. and Charpentier et al. attach the head to the critic, where each prediction is regressed onto a bootstrapped return. Ham and Ahn instead attach it to a detected position and Kalyanasundaram et al. to a velocity, both regressed onto labelled ground truth. In each case, the authors take the return, position or velocity as the observed target y^* on which the regulariser in Eq. (2.8) depends. No comparable target exists for a policy-gradient agent, as its sampled command is never evaluated against a correct counterpart during training. Evidence over the action must therefore be accumulated through a route other than regression onto a target, and that route remains unexplored in the work surveyed.

2.2.3 Reinforcement Learning for Autonomous Parking

Production parking systems are organised as a modular pipeline, which fixes where a statement of confidence can enter. Heimberger et al. [67] describe the layered architecture of a camera-based assistant, in which the outputs of embedded vision algorithms are integrated at a stage that then informs motion planning. The precision of that vision stage is set in advance as a specification, and the planner thus receives a position presumed to meet it, never an estimate of the error actually incurred. The survey by Olmos Medina et al. [68] finds the same division across the field, where the perception stage is the only one to express confidence at runtime. The control stage is instead defined by tracking precision rather than by any runtime uncertainty estimate of its own. Confidence in both stacks is hence an attribute of the camera and fusion stages, resolved upstream of the controller and not transmitted into the manoeuvre. The deployed systems therefore share the shortfall of the evidential placements examined in Section 2.2.2.

Lazzaroni et al. [69] instead collapse planning and control into a single network, training a PPO agent to navigate to a randomly selected bay among sixty perpendicular parking spaces in a CARLA environment. Their task is the closest match among learned parking systems to the one adopted in Section 3.2. The observation combines a LiDAR scan, the position relative to the target bay, and the vehicle’s velocity, acceleration and yaw rate. The agent then issues continuous throttle, steering and brake commands together with a reverse-gear signal. Difficulty is raised through a three-stage curriculum, under which the agent attains a 97% success rate. Pacing of this kind is one mechanism governing how readily the manoeuvre is acquired, while the reward function is another. Suleman et al. [70] focus on the latter, keeping the target bay at a single location. In that setting, they find that a milestone-augmented reward, which issues a proximity signal once the vehicle enters an intermediate region of its approach, outperforms both a sparse goal-only reward and a fully dense proximity reward under on-policy and off-policy optimisation alike. The manoeuvre itself is thus treated as the difficulty to overcome in both studies, and training in each proceeds under nominal localisation, with no state estimator interposed between the simulator and the pose on which the policy operates.

Yang et al. [71], by contrast, account for uncertainty, applying it while fitting the policy. They distil demonstrations from a classical planner-controller stack into a goal-conditioned policy through end-to-end offline RL, with a conservative value regulariser pushing down the value of actions absent from the demonstrations. Such conservatism is an epistemic assessment of dataset coverage, resolved during the gradient updates. The deployed agent therefore inherits a value function already shaped away from the unsupported actions, and not a coverage score it can read at

decision time. The proposed evidential head instead yields the same epistemic quantity at the output, scoring the observation being acted on rather than the coverage of a fixed dataset. The estimate is consequently recomputed at every decision and remains available during deployment.

The assumption of an exact pose in the learned policies of Lazzaroni, Suleman and Yang et al. carries a quantitative threshold, since the alert limit established in Section 2.1.4 places the lower tiers of the fix ladder beyond the accuracy a pose must hold to be treated as reliable. Fix loss is intermittent, so a manoeuvre stays mostly within the bound and falls outside it when the solution degrades. Olmos Medina et al. [68] report that such errors are typically handled through probabilistic sensor fusion and perception-level confidence estimation. Neither is kept as a separate stage in an architecture that maps raw input directly to a control action, and the same survey lists uncertainty quantification within such architectures as an open research direction. The success rates reported for these three policies are therefore uninformative about parking under errors beyond the alert limit, which structures and large vehicles around a facility tend to induce. Degradation of the pose estimate is accordingly treated as a controlled variable in Section 3.4.1.

2.3 The Research Gap

The two absences previewed at the start of this chapter are stated below, each bounded by the scope of work it is asserted against.¹

The first of the two concerns what a learned controller is given. No continuous-control policy for ego-motion has been provided with covariance-derived features from its own state estimator, such as the standard deviations along each axis, as observation inputs, whether for parking or for any other driving task surveyed. Two threads of the review converge on that point from different directions. The estimate in every placement analysed in Section 2.2.1 reaches the policy only through an abstraction of the covariance, while the parking agents reviewed in Section 2.2.3 receive the pose with no statement of confidence in it at all. Availability does not account for that absence, since the navigation-grade EKF (Section 2.1.4) produces the signal continuously at no additional inference cost. The covariance tracks the trustworthiness of the pose only if the filter's noise models are tuned, and objective O1 accordingly requires that tracking to be confirmed before any policy is conditioned on it.

¹Both absence claims rest on searches of IEEE Xplore, Scopus, arXiv and Google Scholar, conducted in August 2026. The output side paired the evidential and normal-inverse-gamma vocabulary with policy-gradient and continuous-control terms. The input side likewise paired the localisation-uncertainty and pose-covariance vocabulary with observation-space and driving terms. No search establishes a universal negative, so both claims are limited to the work surveyed.

A corresponding absence on the output side governs what that controller emits. No published evidential RL system has attached the NIG distribution to the actor of a continuous-action policy-gradient agent. The restriction behind this absence was identified in Section 2.2.2, where every prior placement is anchored to the regression target presumed by the evidential loss, and no such target exists for an action. The only remaining alternative is the Gaussian head described in Section 2.1.1, which leaves per-state confidence inexpressible. Two questions are consequently left unexplored: whether an evidential action head can be trained without a ground-truth action target, and whether its uncertainty output carries a usable signal. Both are examined through objective O2, focusing on the NIG parameterisation and the single-pass property outlined in Section 2.1.3, with the pathologies catalogued alongside them bounding what such a head can deliver.

Both absences are visible at once in Table 2.2, where the surveyed systems are characterised along five axes. The first two concern the uncertainty estimate itself: the locus indicates where that estimate is placed relative to the manoeuvre, and the estimator is the mechanism that produces it. The remaining three bound what those systems can provide. The number of passes sets the per-decision cost, the action space the expressibility of a graded response, and the domain the proximity of the task to parking. A dash marks an axis that does not apply to a given system, either where no runtime uncertainty estimate is carried or where no action is emitted.

Across the rows, neither absence depends on the other, which is what makes their combination testable. An input-side mechanism can be integrated into a standard actor, an output-side mechanism can be trained independently of covariance observations, and both can be used together. This independence yields a factorial design in which each mechanism's contribution can be assessed separately. No cell of this design has been occupied by prior work under controlled localisation degradation, and in parking the omission has direct consequences, since an intermittent loss of the RTK fix carries the pose past the alert limit established in Section 2.1.4.

Objective O3 accordingly requires those contributions to be separated, under the multi-seed reporting and interval estimates established in Section 2.1.6. Objective O4 then demands evidence that any resulting benefit originates from uncertainty-conditioned behaviour rather than from the input alone, in the form of a vehicle that drives measurably more conservatively as the reported uncertainty increases. No reward term refers to uncertainty, so caution of that kind would emerge on its own, whereas Kim et al., reviewed in Section 2.2.1, had to add one to the training objective before the manoeuvre varied at all.

Table 2.2 Prior work across the three threads surveyed in Section 2.2.

Work	Uncertainty locus	Estimator	Passes	Actions	Domain
Hoel et al. (2020) [54]	Output (decision veto)	Bootstrapped ensemble with randomised priors	M	Discrete	Tactical driving
Hoel et al. (2023) [55]	Output (decision veto)	Ensemble of quantile networks	M	Discrete	Tactical driving
Puthumanaillam et al. (2026) [56]	Observation (abstracted map)	Task-specific uncertainty map over the workspace	1	Continuous	Robot navigation
Kim et al. (2022) [57]	Observation (scalar signal)	Epistemic scalar with a mutual-information term	1	Continuous	Automated driving
Placed and Castellanos (2020) [59]	Objective (reward)	Particle-filter pose covariance, D-optimality	1	Discrete	Active SLAM
Zhou et al. (2023) [60]	Guarantee (policy-level fallback)	Bootstrapped ensemble of value estimates	M	Discrete	Automated driving
Akgül et al. (2025) [61]	Output (critic)	NIG on the state value	1	Continuous	Locomotion
Stutts et al. (2025) [62]	Output (action values)	NIG with conformal calibration	1	Discrete	MinAtar games
Charpentier et al. (2022) [27]	Output (predicted return)	Posterior Network with a normalising-flow density	1	Discrete	Classic control
Ham and Ahn (2026) [64]	Constraint (perception into planner)	Evidential perception uncertainty	1	Continuous	Automated driving
Kalyanasundaram et al. (2026) [65]	Input (filter measurement)	NIG on a velocity correction	1	-	Vehicle localisation
Kanazawa et al. (2022) [66]	Output (actor and critic)	Distributional actor-critic ensemble	M	Continuous	Continuous control
Lazzaroni et al. (2024) [69]	None	-	-	Continuous	Parking (CARLA)
Suleman et al. (2026) [70]	None	-	-	Continuous	Parking
Yang et al. (2025) [71]	Objective (offline conservatism)	Conservative value regularisation	-	Discretised	Parking (CARLA)

3 Methodology

The experimental system and the design decisions taken to answer the research objectives are documented in this chapter in the order of construction.

3.1 System Overview and Infrastructure

The two-layer uncertainty architecture introduced in Section 1.4 is realised by a three-container simulation, localisation and training stack. Every signal crossing those containers is traceable to a named source, from the raw sensor streams to the rate-limited vehicle command.

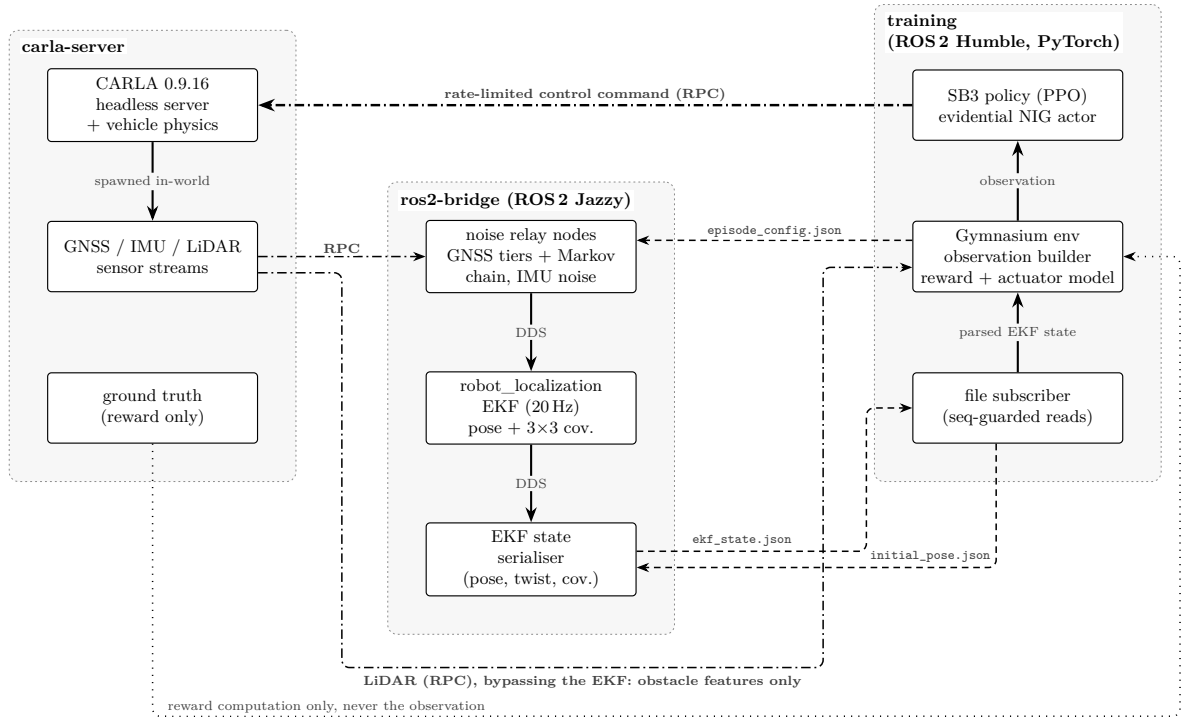


Figure 3.1 The three-container system and its data flow.

The trajectory traced in Fig. 3.1 begins with the raw GNSS, IMU and LiDAR streams published by CARLA. Datasheet-derived noise is added to the GNSS and IMU measurements by relay nodes that republish them with stamped covariances. The `robot_localization` EKF node [30, 31] fuses those two republished streams into a pose estimate with a live 3×3 covariance, and a state node serialises both outputs to a shared file. The LiDAR stream takes no part in that fusion, serving the obstacle-clearance features alone, so its degradation leaves the localisation uncertainty signal untouched. Ground-truth pose is routed separately into the reward calculation and never enters the observation, as it has no counterpart on a real vehicle, a point

detailed further in Appendix B.1. The observation is then built from the serialised state by the Gymnasium environment in the training container and passed to the SB3 policy. The resulting action is rate-limited by the actuator model and returned to CARLA as a vehicle command.

The simulator, the ROS 2 Jazzy bridge and the CUDA-compatible training stack rely on different Ubuntu releases, so each runs in its own Docker image. No common message format survives that split, as Jazzy and Humble serialise messages differently, and the state therefore crosses the boundary as three JavaScript Object Notation (JSON) files on a shared volume, each written atomically and stamped with a sequence number. The bridge remains passive, and the simulator advances only when the CARLA client within the training container requests a tick. Each step is therefore executed at a fixed point in the training loop, irrespective of the processing time of each container, so a rerun from the same seed reproduces an identical sequence of steps. Both properties, together with the curriculum and ablation assumptions set out later in Sections 3.7.1 and 3.8, are verified by more than 470 unit and integration tests. The container stack, file bridge and supporting tooling are documented in full in Appendices D.1 to D.3, together with the staged inspection that isolated faults in the lot geometry, sensors and observation before training began.

3.2 The Parking Environment

The world is a flat OpenDRIVE surface built through the programmatic world generator of CARLA, leaving the layout as the sole source of geometric variation. Two lots, shown in Fig. 3.2, are produced for this purpose by a bespoke layout generator and verified in simulation prior to training, as described in Appendix D.3.

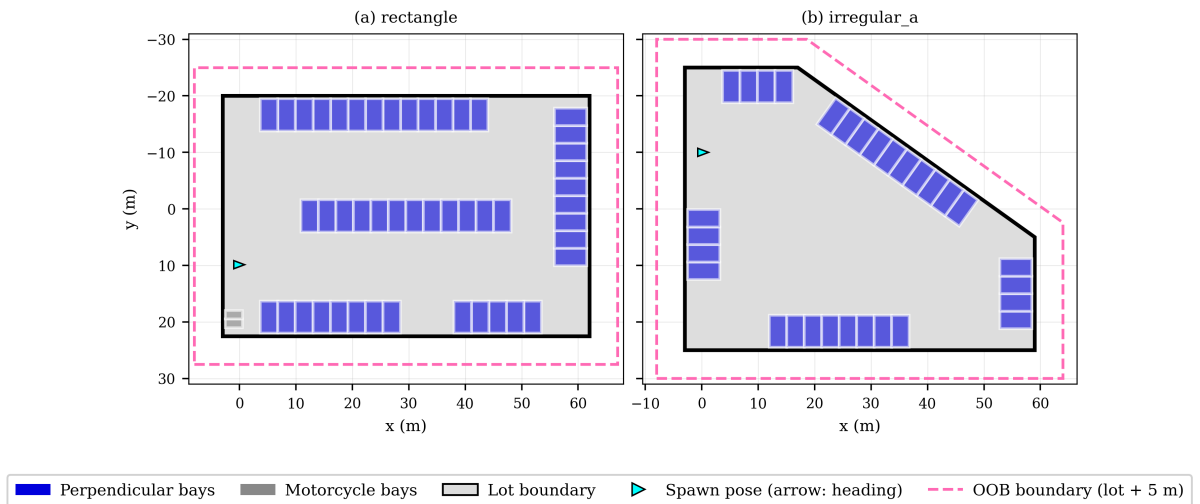


Figure 3.2 Bird's-eye plots of both lot layouts as generated, (a) rectangle and (b) irregular_a.

Training uses only `rectangle`, an axis-aligned 65×42.5 m lot with 47 targetable perpendicular bays in five clusters across three bay orientations. Its counterpart `irregular_a`, by contrast, has five sides, one corner cut away by a diagonal wall, and is reserved for the out-of-distribution evaluation in Chapter 4. In both lots, each approach starts at the cyan spawn marker and ends in one of the blue bays within the shaded lot, while the dashed line marks the out-of-bounds limit 5 m beyond the lot boundary. Their aisles are held to the 6.0 m minimum that published parking-design guidance stipulates for 90° bays [72], so the manoeuvre is performed under the spatial constraint of a compliant real facility.

3.2.1 Scope Decisions

Beyond the layouts, the only obstacles in the lot are stationary parked vehicles, and the reverse gear is disabled because of the single-antenna sensor configuration specified later in Section 3.4. One GNSS antenna cannot measure absolute heading, so the heading is inferred from the COG of successive fixes and fused by the EKF as its sole yaw correction. The substitution holds only under forward motion, since COG reports the direction of travel rather than the orientation of the vehicle. In reverse the two diverge by 180° , and an error of that size would propagate into the covariance feature σ_ψ given to the policy in Section 3.3.

Rotating the COG by 180° whenever reverse is engaged could restore a heading in principle, but not an uncertainty signal suitable for the study. The rotation would need a direction-of-travel input such as the gear state, tying the localisation to one drivetrain and to platforms that expose it, the objection that also rules out wheel odometry in Section 3.4. Even a sound rotation would not help, as the vehicle must stop before reverse is engaged. The heading is suppressed whenever consecutive fixes are too close to define a direction, a rule given later in Section 3.4, so the yaw correction is lost for the duration of the pause. Repeated shunts would accumulate those gaps, raising σ_ψ through the manoeuvre itself rather than through the fix state and confounding the manipulated variable. Bridging the gaps with the IMU gyroscope would confound it in the same way, as the gyroscope measures rotation rate and leaves the accumulated heading error unbounded between fixes.

Limiting the vehicle to forward motion therefore preserves the signal the study manipulates. The task remains achievable, as the success criterion defined later in Section 3.6 requires only the lateral clearance that a forward approach into a perpendicular bay provides. The vehicle cannot withdraw from a misaligned approach, however, so the alignment must be established within a single trajectory, a cost shared by all four arms that leaves the contrasts between them unaffected.

Stationary parked vehicles likewise suffice as the sole obstacles, since each is both a LiDAR landmark for the clearance features and a collision hazard that a mistimed or badly angled approach can reach under an accurate fix. Occupancy is thereby a genuine axis of difficulty, and failure is not confined to what the localisation estimate reports.

3.2.2 Episode Mechanics

Each episode begins with the ego vehicle at the spawn pose of the lot. A target bay is then drawn uniformly from those available at the current stage of the curriculum, four bays in the first stage, 15 in the second and all 47 from the third onwards. A single occupancy rate $q \sim \mathcal{U}(q_{\min}, q_{\max})$ is drawn for the episode, between bounds set by the same stage. Every bay other than the target is then occupied, each by its own draw, with probability q by a randomly selected vehicle blueprint. Each parked vehicle is rotated by 180° in half of all cases, so a target bay is approached with nose-in and nose-out neighbours alike.

The spawn pose is fixed, so the distribution of approaches is identical across the arms. The variation encountered during training therefore arises from the target bay, the occupancy draw, the orientation of neighbouring parked vehicles and the fix-state chain rather than from the starting point, so memorisation of the approach from that pose cannot be ruled out by the design alone. The question is settled against the evaluation results later, in Section 4.8.1.

An episode that neither succeeds nor collides runs until the clock expires at 1750 simulation ticks, which caps it at 437 policy decisions at the 5 Hz decision rate set out in Section 3.3. Two mechanisms can end it earlier, the first applied softly at the lot boundary rather than through a collidable wall. The lot polygon is expanded outward by 5 m into a run-off skirt, and each decision taken with the vehicle centre beyond it earns -0.5 until the accumulated penalty reaches 10.0 and ends the episode. Twenty decisions outside the skirt are therefore tolerated, so a single erratic deviation under a degraded fix cannot end an otherwise salvageable rollout. The tolerance matters mainly in early exploration, since the reward defined later in Section 3.6 pays nothing for movement outwards and the perimeter is rarely approached after convergence.

The second mechanism addresses the opposite failure, an episode that makes no progress. A rollout is deemed stalled after 50 consecutive decisions, or ten seconds, spent outside the target bay below 0.1 m/s, the speed the success criterion requires (Section 3.6). The rest of the episode is thereby reclaimed, as a motionless vehicle would otherwise be carried to the clock ceiling, flooding the buffer with repetitive zero-advantage frames. Deliberate waiting must be exempt, since a pause under a degraded fix is the behaviour the uncertainty inputs defined next in Section 3.3 are

intended to elicit. The stall counter is therefore suspended whenever the live EKF position standard deviation exceeds 0.4 m, the boundary between the float and standalone tiers listed later in Table 3.4, just above the float-tier figure of 0.36 m. The threshold of 50 decisions follows from those same tiers, under which the longest degraded excursion lasts approximately seven seconds on average, within the ten the counter permits, and any longer dwell falls under the suspension, so a justified wait for recovery is never cut short.

Stalling offers no cheaper exit either, since a stalled episode and one that reaches the clock ceiling receive nominally the same graded timeout penalty set out in Section 3.6.3. Discounting weakens a penalty the later it is received, so stalling during a failing episode costs more than seeing it through to the ceiling.

3.3 Observation and Action Spaces

The policy observes the 13-dimensional vector defined in Table 3.1, grouped into four semantic blocks: vehicle state, localisation uncertainty, relative target pose and obstacle clearance. The EKF estimate is thereby supplied to the policy together with its reported uncertainty (Section 2.1.4), addressing the input-side absence identified in Section 2.3.

Table 3.1 The 13-dimensional observation vector.

Index	Feature	Source	Units
0	speed (body-frame, ≥ 0)	EKF filtered twist	m/s
1	yaw rate $\dot{\psi}$	EKF filtered twist	rad/s
2–4	$\sigma_x, \sigma_y, \sigma_\psi$ (yaw)	EKF covariance diagonal	m, m, rad
5–7	$\Delta x, \Delta y, \Delta\psi$ (target, ego frame)	EKF pose estimate	m, m, rad
8–9	left distance, left bearing	LiDAR (bearing $> +15^\circ$)	m, rad
10–11	right distance, right bearing	LiDAR (bearing $< -15^\circ$)	m, rad
12	forward distance	LiDAR ($ \text{bearing} \leq 15^\circ$)	m

3.3.1 Justification of Each Block

These blocks occupy contiguous indices, as Fig. 3.3 shows, with the localisation-uncertainty dimensions at indices 2–4 subject to the covariance gate. This gate constitutes the input-uncertainty factor of the 2×2 study, and disabling it yields the 10-dimensional observation of the covariance-blind arms.

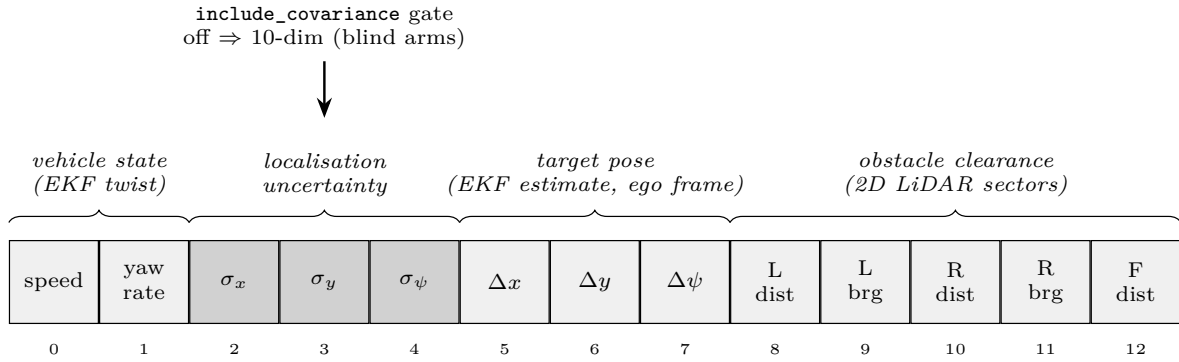


Figure 3.3 Block structure of the observation vector.

The vehicle-state block holds the two components of the motion of the vehicle, its speed along the body axis and its yaw rate. The speed enters the success criterion defined later in Section 3.6 directly, since one of its two conditions requires the speed to stay below 0.1 m/s for five consecutive decisions before a rollout counts as a park. The yaw rate carries no such threshold, being the rate at which the vehicle rotates under the steering command delivered on the previous decision. Both are derived from the filtered twist of the EKF, so the policy is trained against the Doppler and gyroscope error characteristics described in Sections 2.1.4 and 3.4.

The uncertainty block consists of the three standard deviations on the diagonal of the $[x, y, \psi]$ covariance submatrix of the EKF. The off-diagonal correlations could also be supplied, on the premise that the shape of the uncertainty is characterised by more than the variances alone, yet they are deliberately left out. Their contribution in simulation is negligible, since each fix is generated with a diagonal covariance, the error on each axis being scaled independently by the degradation model introduced later in Section 3.4.1. A change in fix quality therefore leaves the correlations almost untouched, while the three standard deviations already separate the localisation regimes to which the vehicle must respond. Supplying the correlations would expand the observation without adding information, attenuating the three features in which the uncertainty signal resides.

Those deviations describe the reliability of the third block, the target pose, which locates the assigned bay in the ego body frame through a forward offset, a lateral offset and a heading discrepancy derived from the EKF pose estimate. The target pose degrades with the localisation while the covariance features register the decline, so the appropriate degree of restraint can be learned from them. The body frame serves a second purpose, as each lot has its own origin and bay positions, so world coordinates supplied as observations would bind a policy to the survey on which it was trained. An offset to the target carries no such dependence, since the same

approach geometry yields the same observation whatever frame the bay was surveyed in, and absolute position is consequently absent from the observation altogether.

The target pose gives the direction to the bay without revealing what stands in the way, so the vehicles parked in neighbouring bays are reported alongside it by the obstacle block. Its features come from the LiDAR scan, split at bearings of $\pm 15^\circ$ into a narrow forward cone and a sector to either side, each reduced to its nearest return. The side sectors keep both distance and bearing, as an obstruction to the side must be located as well as detected, whereas the forward cone is narrow enough for the distance alone to serve that purpose. Returns behind the front-mounted two-dimensional scanner [73] are excluded beforehand, so only the forward 180° of its 270° aperture is used and every bearing lies within $\pm 90^\circ$.

3.3.2 Fixed Physical-Range Normalisation

The four blocks are measured in metres, radians, metres per second and radians per second, so a raw 40 m target offset is orders of magnitude larger than a heading standard deviation of a few tenths of a radian. Such differences can leave the gradient dominated by scale, so some rescaling is necessary, the prevalent choice being a running normaliser that divides each dimension by statistics gathered throughout training. That normaliser was used in an earlier design of this environment, where an input was amplified into saturation at a boundary between curriculum stages. A fixed physical scale is therefore adopted for each dimension, listed in Table 3.2 and applied through the division and clip given by Eq. (3.1):

$$\tilde{o}_i = \text{clip}\left(\frac{o_i}{\kappa_i}, -5, 5\right) \quad (3.1)$$

where o_i is the raw value of dimension i in the physical units of Table 3.2, κ_i the fixed scale by which it is divided, and $\tilde{o}_i$ the normalised input supplied to the policy after the clip. A value at its scale, $o_i = \kappa_i$, thus reaches the network as $\tilde{o}_i = 1$, while any measurement of five scales or more saturates the clip. Each κ_i is set to the typical physical range of its dimension, so the magnitude of a feature at the network input represents the fraction of that range it occupies rather than the unit in which it is measured. A dimension confined to a narrow part of that range is consequently compressed towards zero, at a cost in resolution that a running normaliser would have recovered. The scales are structural constants of the environment rather than tuned parameters, identical across every curriculum stage and layout, and that constancy is worth the resolution forgone. Adaptive scaling is accordingly confined to the discounted returns maximised by the clipped surrogate objective of PPO, leaving the observations untouched.

Table 3.2 Fixed normalisation scales κ_i by observation block.

Block	Dimension	Scale κ_i	Rationale
Vehicle state	speed	5 m/s	forward parking-speed scale
	yaw rate	π rad/s	clipped to $[-\pi, \pi]$
Uncertainty	σ_x, σ_y	1 m	EKF position-std ceiling
	σ_ψ	0.5 rad	heading-std ceiling ($\sim 29^\circ$)
Target pose	$\Delta x, \Delta y$	40 m	offset over the 65×42.5 m lot
	$\Delta\psi$	$\pi/2$	wrapped to $[-\pi/2, \pi/2]$ (180° symmetry)
Obstacle	distances	25 m	LiDAR maximum range
	bearings	$\pi/2$	forward-hemisphere bound

3.3.3 Action Space and Actuator Model

The simulator runs in synchronous mode at a fixed 20 Hz, updating physics and sensors on every tick, and an action is issued once every four ticks, a decision rate of 5 Hz. A faster reaction brings no benefit at parking speeds, so the command persists across the three intervening ticks while the physics continues to integrate it. A single call to the step function of the environment therefore represents one policy decision and one SB3 timestep spanning 0.2 s, and every budget and metric in this study is expressed in decisions.

Each decision is a continuous action over the three axes that control the vehicle, [steer, throttle, brake], each emitted on $[-1, 1]$ by the tanh-squashed policy constructed in Section 3.5. Steering uses that interval in full, a negative command steering left and a positive one right. Throttle and brake carry magnitude only, so their negative halves are folded to zero and each pedal is driven on $[0, 1]$ by the positive half alone. The fold also withholds the reverse gear from the interface, under the single-antenna heading constraint explained in Section 3.2.1.

Requests from the policy pass through an actuator model before reaching CARLA. A steering wheel or pedal in a real vehicle moves at a finite slew rate, which bounds the change attainable within one decision interval. A raw request is subject to no such bound and may demand full left lock on one decision and full right on the next. Every axis is therefore confined to a band centred on its previous command:

$$a_t^{\text{cmd}} = \text{clip}(a_t, a_{t-1}^{\text{cmd}} - \Delta_{\text{max}}, a_{t-1}^{\text{cmd}} + \Delta_{\text{max}}) \quad (3.2)$$

where a_t is the raw action requested on that axis at decision t , a_{t-1}^{cmd} the command actually delivered on the previous decision, and Δ_{max} the per-axis rate limit.

Each $\Delta_{\max}$ matches the rate of a real actuator rather than being optimised for training, and Table 3.3 compiles the values adopted from the published measurements that anchor them. Steering is the most constrained axis, at 0.20 normalised units per decision, a road-wheel rate of approximately 70° per second. That rate lies below both the 140° per second ceiling of a driverless electric power-steering design [74] and the 115° per second MPC comfort envelope [75]. The pedals are limited less stringently, at 0.50 normalised units per decision, reaching full application in 0.4 s, which is still slower than the sub-200 ms response of production drive-by-wire and electro-hydraulic hardware [76, 77]. The action space does not prevent both pedals being demanded at once, so the brake is given priority on safety grounds: a delivered brake above 0.3 sets the throttle to zero immediately rather than following the throttle rate limit.

Table 3.3 Actuator rate limits.

Axis	Limit / decision	Physical rate
Steering	0.20	$70^\circ/\text{s}$ road wheel
Throttle	0.50	0 to full in 0.4 s
Brake	0.50	0 to full in 0.4 s
Priority	brake > 0.3	throttle cut to 0

A governor independent of the rate limits caps the speed of the vehicle at 6 m/s by cutting the throttle above it. The ceiling sits well above parking speed, as a limit close to it would suppress the deceleration behaviour the evaluation is designed to detect.

3.4 Localisation Pipeline and Sensor Noise Models

The noise behind the covariance features introduced in Section 3.3 comes from published hardware specifications rather than being tuned for the task. The simulated suite is accordingly built around a named target platform, a VectorNav VN-100 IMU [78], a u-blox ZED-F9P RTK GNSS receiver [1] and a SICK TiM571 two-dimensional LiDAR [73]. Every value in Table 3.4 traces to one of those three datasheets except the two marked †, each resting on a different substitute, and the per-tier velocity scaling that follows the position ladder. No manufacturer figure exists for the ambiguity-float tier, as its accuracy varies with both atmospheric conditions and baseline length. The tier is therefore assigned the decimetre regime that Reid et al. [34] report for float solutions over 30 000 km of highway driving, where the observed errors separate into centimetre, decimetre and metre regimes. The degraded tier rests on no measurement at all, being a stress condition rather than a documented operating mode. Its 5.0 m error exceeds the worst case the datasheet describes, establishing how

the vehicle behaves once the fix is too poor for the manoeuvre to be attempted. The derivation of every row from its source specification, the two exceptions included, is given in Appendix A.

Table 3.4 Sensor noise parameters and their provenance.

Sensor & quantity	Parameter	Value (1σ)	Provenance / derivation
ZED-F9P GNSS	rtk_fixed position	0.020 m	datasheet accuracy table [1]
	rtk_float position	0.360 m	measured decimetre float regime [34], no datasheet value [†]
	standalone position	1.802 m	datasheet accuracy table [1]
	degraded position	5.0 m	stress condition beyond the datasheet [†]
ZED-F9P velocity	Doppler speed (rtk_fixed)	0.05 m/s	datasheet velocity accuracy [1]
	per-tier scaling	$\times 1 / 18 / 90 / 250$	matched to the position-noise ladder
VN-100 IMU	gyro variance / sample	7.46×10^{-8}	noise density $0.0035^\circ/\text{s}/\sqrt{\text{Hz}}$ at 20 Hz [78]
	accel variance / sample	3.77×10^{-5}	noise density $0.14 \text{ mg}/\sqrt{\text{Hz}}$ at 20 Hz [78]
	bias (per episode)	gyro $\pm 5^\circ/\text{h}$ accel $< 0.04 \text{ mg}$	resampled per episode within datasheet in-run bias stability [78]
TiM571 LiDAR	systematic range bias	$\pm 60 \text{ mm}$	datasheet systematic error, per episode [73]
	statistical range noise	20 mm	datasheet statistical error, per point [73]

Variance units: gyro $(\text{rad}/\text{s})^2$, accel $(\text{m}/\text{s}^2)^2$, per sample.

3.4.1 The Global Navigation Satellite System Fix-State Markov Chain

In a parking lot, the deterioration mechanisms described in Section 2.1.4 depend on the surrounding obstructions, including the vehicles parked in neighbouring bays. Fix quality is therefore governed by conditions external to the manoeuvre rather than by the location of the vehicle.

The fix state is consequently modelled as a discrete-time Markov chain over the four receiver tiers rather than as a constant sampled once per episode. Each tier adds zero-mean Gaussian error at the per-axis standard deviation (1σ) listed in Table 3.4, obtained from the CEP figures of the manufacturer, where they exist, under the circularly symmetric Gaussian assumption discussed in Section 2.1.4. Those figures were measured under static open-sky conditions, without the reflectors of a parking lot, and the per-axis standard deviation is therefore raised by a factor of $\sqrt{2}$ over the strict conversion. The arithmetic behind each tier, including any further adjustments, is

set out in Appendix A.1. Transitions between the tiers are evaluated at every GNSS callback (20 Hz), so the quality of the solution can change within a single run, exposing the vehicle to deterioration and recovery alike during a manoeuvre.

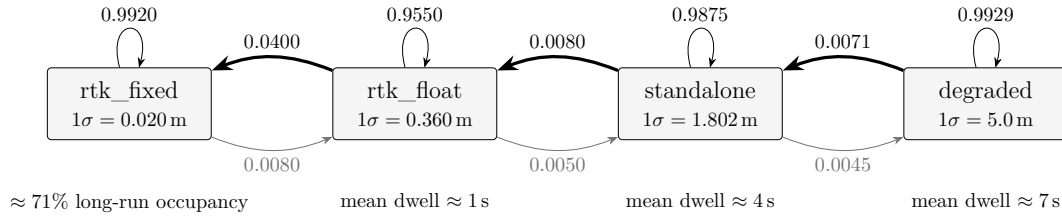


Figure 3.4 The four-state GNSS fix-state Markov chain.

The design depends on three attributes of the chain depicted in Fig. 3.4, with its probabilities taken directly from the runtime configuration laid out in Appendix D.6. Transitions are permitted only between adjacent tiers, as a receiver cannot move from centimetre to metre accuracy without passing through the float solution, the intermediate stage of carrier-phase ambiguity resolution detailed in Section 2.1.4. Movement through the tiers is then biased upward, as recovery is more likely than further deterioration at every level below `rtk_fixed`. The chain is therefore fixed-dominant, with around 71% of its long-run occupancy in `rtk_fixed`. The dwell times of the lost-fix tiers lengthen down the hierarchy, from approximately one second in `rtk_float` to four seconds in `standalone` and seven seconds in `degraded`. Severity and duration consequently rise together, and the longest exposure falls in the tier where waiting for recovery is the only appropriate response.

The initial tier of each episode is drawn from a fixed distribution placing probability 0.40 on `rtk_fixed` and 0.20 on each of `rtk_float`, `standalone` and `degraded`. The split is deliberate, as a substantial share of episodes is seeded into a diminished fix from the outset rather than left to deteriorate during the run. Competent action is therefore required on a fix that was already poor when the episode began, with no preceding deterioration to signal it.

The upward-biased chain replaced an earlier, free-diffusing one that spent approximately 30% of its long-run time in `rtk_fixed` and recovered over intervals measured in minutes. A degraded episode was consequently unwinnable, as the fix would not return before the episode ended. Under the adopted chain, waiting for recovery is both physically plausible and learnable within a single episode, and that wait is the behavioural mechanism the evaluation is designed to detect. The chain can also be modified for evaluation to isolate a particular failure pattern. Its one-way drift mode begins at `rtk_fixed` and deteriorates monotonically without any possibility of recovery, and serves as a stress condition in the evaluation set up later in Section 4.1.

The chain governs the magnitude of the GNSS position error and nothing else, so five secondary effects are added for the error characteristics it leaves unmodelled, extending the noise model to the rest of the sensor suite.

The first effect concerns the direction of the positional error. Each tier in Table 3.4 carries a single per-axis figure, reproducing the circular idealisation contrasted in Section 2.1.4 with the elliptical error of a real fix. Direction is restored by drawing an east/north ratio of at most 1.5, which multiplies the 1σ value of the tier on a randomly chosen axis and divides it on the other. The eastward and northward components thereby diverge while their geometric mean stays at the tabulated figure. Satellite geometry orients the ellipse [32] and holds throughout a manoeuvre, shifting only between one arrival and the next, so the ratio is drawn once per episode. The ratio acts on the two diagonal components of the reported covariance as well as the position, since altering the position alone would leave the eastward and northward uncertainties indistinguishable, so the anisotropy would never reach the covariance features observed by the policy.

Four further effects are distributed across the receiver, the IMU and the LiDAR. Two originate in the receiver, one affecting the reported position and the other the heading. The positional mode consists of cycle slips and transient satellite obstructions [32], reproduced by discarding each fix with a probability of 0.02, after which the EKF estimate advances on the prediction step alone until the next fix arrives. The COG heading is instead conditioned on speed, through a variance obtained by analytic error propagation that scales as the Doppler noise divided by speed. That variance is published with the COG measurement and used as the measurement noise R_k of the yaw channel in the Kalman gain given by Eq. (2.11), which downweights a heading obtained at low speed. On the same reasoning, the heading is suppressed outright in two cases, a displacement between consecutive fixes too small to define a direction and an IMU reading of standstill. The second is handled on the inertial side by a Zero-Velocity Update (ZUPT), the standard inertial-navigation correction applied over a detected stationary interval to bound drift [32], during which the gyro and planar accelerations are clamped to zero. Sensor noise from a motionless vehicle is thereby kept from being integrated as motion.

The remaining two effects introduce no failure, instead randomising characteristics that a physical unit would hold fixed. The IMU bias and scale-factor errors are resampled every episode within the datasheet bounds listed in Table 3.4 and derived in Appendix A.2, so no single calibration becomes ingrained during training. The LiDAR noise in that table follows the same principle, applied in polar coordinates as a systematic bias drawn once per episode alongside statistical noise drawn for each point.

3.4.2 The Extended Kalman Filter

The `robot_localization` EKF node described in Section 2.1.4 runs at 20 Hz in planar mode. Three aspects of its configuration bear on the study: the inputs fused, those deliberately withheld, and the tuning of the process noise.

Two of the three filter inputs come from the receiver, while the third, the IMU yaw rate, drives the prediction step. The first receiver input combines the flat-earth projected position with the Doppler-derived longitudinal velocity in a single fused measurement, and the second is the gated COG heading, which corrects the yaw. Linear acceleration is left unintegrated, as the velocity states are already constrained by the 20 Hz velocity measurement, so an accelerometer channel would add integration drift with no benefit at low speeds. The LiDAR is withheld on different grounds, serving obstacle clearance alone, so the covariance features carry no trace of the obstacle channel.

On the evidence of Kovacs et al. [29] reviewed in Section 2.1.4, the process-noise covariance Q is treated as a tuning variable rather than left at its defaults, and is configured so that the posterior grows appreciably whenever the corrections weaken. A degraded fix carries large measurement noise R , which makes the gain defined in Eq. (2.11) small. Each update consequently removes less uncertainty than the preceding prediction added, and the reported covariance climbs for as long as the fix remains poor. The active diagonal of Q sets the steepness of that climb, and Appendix A.4 gives the diagonal entries of the estimated states with the reasoning behind each value, identifying the two entries that would need retuning on a physical vehicle.

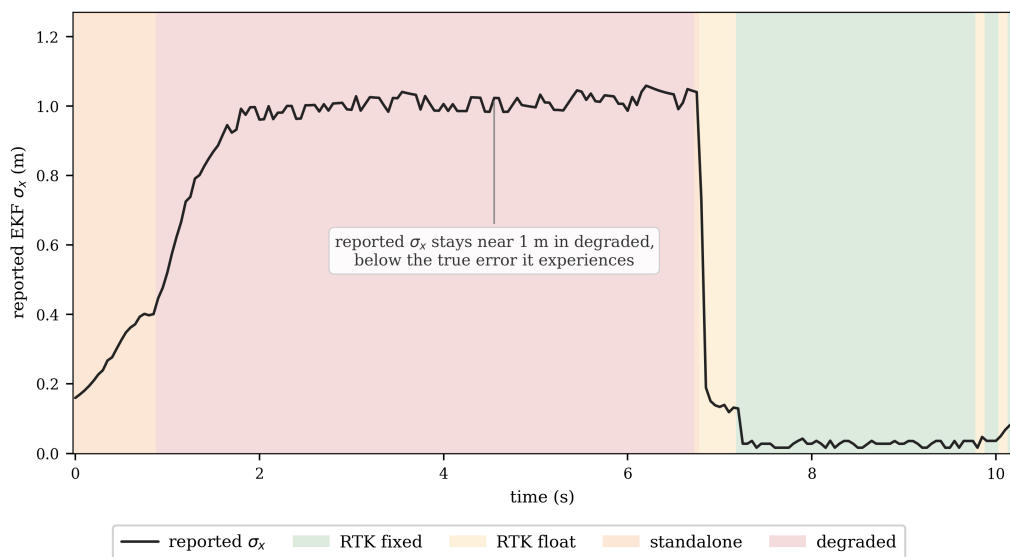


Figure 3.5 Reported EKF σ_x over one training episode.

A representative trace of σ_x over one training episode is shown in Fig. 3.5, with the fix-state tiers shaded behind it, and presents the empirical sawtooth anticipated in Section 2.1.4. The reported σ_x follows those tiers closely, rising as the solution deteriorates into standalone and degraded and holding while the vehicle remains within them. Its decline is far steeper than its rise, as a reaccepted clean fix restores a small R and therefore a large gain, so a handful of updates recover the confidence that many predictions had eroded.

The same trace shows the limit of the reported σ_x as an uncertainty signal. During the dwell in the degraded tier, where the injected position noise reaches 5 m (Table 3.4), the reported σ_x stays near 1 m and so understates the true localisation error several times over. The measurements received during the dwell are too noisy to constrain the estimate, and the posterior is therefore inflated at the bounded rate set by the tuned Q instead of being measured against the error actually incurred. That bound keeps the full injected error out of reach within a single dwell, so the reported standard deviation indicates the fix state and the direction in which it is moving without estimating the instantaneous error.

Any localisation approach for this task must produce an uncertainty that varies within an episode and is not tied to one vehicle. Two alternatives were implemented and rejected before those requirements became clear. The first, scan-matching against perimeter cones with Cartographer [79], was attractive because match quality fell as parked vehicles occluded the cones, coupling uncertainty directly to occupancy. The coupling was also its defect, as occupancy is fixed throughout an episode, leaving a rollout well or poorly localised from beginning to end. The reported uncertainty would therefore never rise and fall within an episode, making the appropriate response to a given level unlearnable. The second, wheel odometry fused with the other sensors, was briefly attempted and then reverted. It would have sufficed on this lot, but it ties the signal to a particular drivetrain, restricting the setting to platforms that expose clean wheel encoders.

Neither objection applies to RTK GNSS with IMU fusion, as the fixed and standalone rungs of its degradation ladder are documented by the manufacturer and its fix state varies within an episode rather than being set at the outset. The choice constrains the experiment but not the method, since the policy conditions on a faithful covariance rather than on the sensor producing it.

3.5 The Evidential Actor

The covariance described in Section 3.4 measures the uncertainty in the estimated pose but conveys no confidence in the selected action. That confidence comes from

the evidential NIG actor head, which produces an action distribution and a decomposition of its uncertainty in a single forward pass.

The head uses the DER construction described in Section 2.1.3 without structural alteration, differing only in the regression target. The original formulation was designed for a supervised label y^* , whereas here the Gaussian-NIG hierarchy (Eq. (2.6)) is applied to a sampled action a on one axis of the continuous action space defined in Section 3.3.3, with the substitute for the absent label established in Section 3.5.2. Four parameters (μ_0 , ν , α and β) are generated for each action dimension, and the virtual-observation interpretation set out in Section 2.1.3 transfers with them, so ν reflects the evidence supporting a proposed action rather than a predicted label. The two uncertainty channels then follow from the NIG moments stated in Eq. (2.7):

$$\sigma_{\text{alea}}^2 = \frac{\beta}{\alpha - 1} \quad \sigma_{\text{epi}}^2 = \frac{\beta}{\nu(\alpha - 1)} = \frac{\sigma_{\text{alea}}^2}{\nu} \quad (3.3)$$

A single NIG head therefore obtains σ_{epi}^2 by dividing σ_{alea}^2 by ν , tying the two channels together through the evidence count alone, and the consequences of that coupling are examined in Section 4.3.2. Neither estimate is claimed as an exact Bayesian posterior, both being treated as the heuristic single-pass signal argued for in Section 2.1.3.

3.5.1 The Normal-Inverse-Gamma Actor Head

The two uncertainty channels are meaningful only while the four parameters remain well defined, which requires three properties of the head. A constraint on the outputs must bound the region the parameters can occupy, an initialisation must then fix their starting point within it, and a sampling rule must finally convert them into an action. The constraint on the outputs arises from the HUA described in Section 2.1.3, the region around $\alpha = 1$ in which the evidential gradient vanishes and training stalls. Entering that region costs a supervised regressor only accuracy, whereas the cost to an actor is considerably greater. The aleatoric variance defined by Eq. (3.3) sets the action sampling variance and becomes unbounded as $\alpha \rightarrow 1$, signalling policy failure.

The head employs the generic softplus-with-offsets constraint discussed in Section 2.1.3, with an offset on every activation rather than on α alone, configured so that the HUA lies beyond the reachable range:

$$\nu = \text{softplus}(\xi_\nu) + 10^{-6} \quad \alpha = \text{softplus}(\xi_\alpha) + 1.5 \quad \beta = \text{softplus}(\xi_\beta) + 10^{-6} \quad (3.4)$$

The offsets on ν and β are negligible floors that keep the values strictly positive, with each activation clamped at 100 at the opposite end as a numerical safeguard. The critical choice is the offset on α , set to 1.5 rather than the usual 1.0, which holds the denominator $\alpha - 1$ shared by both channels at 0.5 or above by construction. The

pathological region is therefore structurally unreachable rather than merely regularised against.

The offsets imposed by Eq. (3.4) bound the region in which the head can settle but leave its starting point unconstrained, so a deliberately selected initial state, established in three parts, replaces the default.

Control over the outputs is first moved from the weights to the biases by scaling all head weights by a factor of 0.01, so each output is dominated by its bias at the first step. Those biases are then set to the targets in Table 3.5, where one group determines the initial action means. The throttle bias is positive whereas the brake bias is firmly negative, so under the freshly initialised policy the vehicle creeps forward rather than engaging both pedals at once. The remaining group starts the evidence count ν at approximately 0.313, and since Eq. (3.3) divides σ_{alea}^2 by ν , a count below one makes the starting prior epistemic-dominant, the regime intended for a novelty-reading gate in unfamiliar states. These targets survive integration only if they are re-applied, as the SB3 orthogonal-initialisation pass zeroes all biases after the head is built.

Table 3.5 Bias-driven NIG initialisation targets.

Parameter	Bias	Activation	Value at init	Interpretation
μ_0 steer	0.0	linear	0.0	straight ahead
μ_0 throttle	+0.5	linear	+0.5	pedal default-on during approach
μ_0 brake	−1.0	linear	−1.0	pedal default-off (folded to zero)
ν	−1.0	softplus + 10^{-6}	≈ 0.313	sub-one evidence: epistemic-dominant prior
α	0.9	softplus + 1.5	≈ 2.741	well-defined finite variance, clear of the HUA
β	0.0	softplus + 10^{-6}	≈ 0.693	moderate scale

A head constrained by the softplus offsets in Eq. (3.4) and initialised to the targets listed in Table 3.5 yields its actions through a tanh-squashed Gaussian. An unbounded latent $\tilde{a}$, the action before squashing, is sampled from the Gaussian likelihood (Eq. (2.6)), centred on μ_0 and scaled by the square root of the aleatoric estimate given by Eq. (2.7). The epistemic estimate is excluded from that scale, as it quantifies uncertainty over the mean μ_0 rather than further noise on the action itself. The latent is drawn under a clamp on the sampling variance, motivated in Section 3.5.2, and is transformed through tanh into the bounded action a , with the throttle and brake axes folded to their non-negative halves as described in Section 3.3.3:

$$a = \tanh(\tilde{a}) \quad \tilde{a} \sim \mathcal{N}(\mu_0, \text{clamp}(\frac{\beta}{\alpha-1}, \sigma_{\min}^2, 1.0)) \quad (3.5)$$

3.5.2 Adapting Evidential Learning to Reinforcement Learning

Placing an evidential head on an actor instead of the supervised regressor considered in Section 2.1.3 introduces two opposite pathologies. The first is a divergence, in which the regulariser increases indefinitely and the uncertainty estimates rise with it. The second is the inverse, a collapse in which the predicted variance approaches zero and exploration halts. Both trace to the same origin, the absence of a fixed regression target under policy sampling, and each is identified and addressed in turn.

The divergence stems from the loss itself, which must be modified before it can be applied to RL. The missing target identified in Section 2.2.2 bears directly on the evidence regulariser (Eq. (2.8)), as the sampled action must stand in for the label. The residual $|a - \mu_0|$ then measures the inherent exploration noise of the policy rather than a prediction error, so the regulariser rewards reduced evidence on exploratory actions, penalising the very variability on which exploration relies.

The effect of that penalty appeared in an early training run conducted with the unmodified loss and none of the constraints imposed in Section 3.5.1. Its traces, which record the uncertainty estimates and task metrics at each step, show the regulariser loss increasing indefinitely and driving the aleatoric estimate from approximately 2 to above 11 000. Exploratory actions were suppressed under that penalty, so the policy entropy collapsed alongside it and success stayed at zero throughout. Two faults combine to produce the divergence, the first being a residual-weighted regulariser poorly specified under policy sampling. That condition is absent from every supervised placement surveyed in Section 2.2.2, and the penalty consequently grows with the exploration it suppresses. The second is that the NIG parameters are left unconstrained, so their growth under the penalty has no upper limit.

In the final design, the runaway parameters are contained by the output clamping and initialisation set out in Section 3.5.1, which restrict the NIG to a bounded operating range, and the poorly defined residual-weighted penalty is replaced by the prior-anchoring regulariser below.

$$\mathcal{L}_{\text{reg}} = \mathbb{E} \left[\left(\frac{\alpha}{\alpha_0} - 1 \right)^2 + \left(\frac{\beta}{\beta_0} - 1 \right)^2 \right] \quad \alpha_0 = 2.741 \quad \beta_0 = 0.693 \quad (3.6)$$

This raw-ratio quadratic needs no target, as each factor compares a live parameter with a fixed reference rather than a data label, and the expectation $\mathbb{E}$ is taken over a batch of collected experience in which both parameters vary from state to state. The reference is the initialisation prior chosen in Section 3.5.1, so the loss falls to zero exactly at $\alpha = \alpha_0$ and $\beta = \beta_0$, the post-softplus outputs of the initial biases in Table 3.5.

The head is therefore drawn back towards the configuration set by those biases without ever being fixed to it, since the restoring term counteracts the PPO gradient

rather than overriding it. The parameters settle at the balance between the two, tracking the task while their deviation stays constrained. Normalising by α_0 and β_0 before squaring makes the penalty scale-free, as the restoring gradient then reflects the relative deviation from the prior rather than its absolute magnitude. Under the quadratic form that gradient rises linearly with the relative distance, so a parameter far from the prior is drawn back more firmly than one close to it. Both are thereby anchored with comparable force, even though α and β differ by a factor of four in magnitude. The references stay fixed throughout training, so the resistance of the anchor does not diminish as the policy improves, whereas a residual-weighted term would have relaxed as the residuals shrank.

The anchor is deliberately restricted to those two parameters, as extending it to ν would bind the epistemic evidence to its initial value and defeat the purpose of the head, under which epistemic uncertainty rises in unfamiliar states and falls as familiarity grows. The anchor is itself phased in, the regulariser coefficient λ_{reg} (Eq. (2.9)) rising gradually from zero to 0.02 over the first 50 000 decisions of each stage, so that the averages stabilise under PPO before it takes effect.

A restoring term at 0.005, the coefficient in use before the final 0.02, leaves β substantial freedom, so once training was otherwise stable a second pathology emerged, in which the sampling variance collapsed towards zero. The mechanism lies in the sampling standard deviation given by Eq. (3.5), a quantity derived from the loss in which β , unlike α , is floored only at 10^{-6} and can therefore approach zero. As some episodes begin to yield returns, β can be reduced profitably by the policy gradient, diminishing the sampling noise and shifting the policy from exploration towards exploitation. The conventional Gaussian PPO head (Section 2.1.1) is not exposed in the same way. Its log standard deviation is a free parameter, independent of the input and acted on directly by the closed-form entropy bonus, whereas the evidential head derives that quantity from β and α . The collapse is therefore a cost of the substitution, to be addressed within the evidential design itself.

The severity of that cost was measured on a run after the divergent one, carrying the anchor and constraints adopted in response to it, in which the shrinkage in β ran its full course. Its aleatoric estimate fell from 0.40 to 0.05 and its epistemic estimate from 0.32 to 0.04, reducing the sampling variance by approximately an order of magnitude. Driving performance declined with the narrowing spread, the success rate falling from a peak of 0.74 to 0.52 and the mean final position error rising from approximately 3 to more than 9 metres. The decline is early convergence towards the suboptimal deterministic policy that the entropy term of Eq. (2.4) exists to prevent.

A spread this narrow confines the vehicle to manoeuvres that have already earned reward, so alternatives are rarely sampled and a better approach to the bay is never found. The traces thus document the variance-minimisation bias identified in

Section 2.1.3 in its actor-side form, in which the evidential objective drives the sampling distribution towards a degenerate density. No prior instance of this failure has been reported, and the placements surveyed in Section 2.2.2 account for its absence. Each attaches the evidential head to a critic, a detected position or a velocity correction, none of which sets the spread from which an action is drawn. A variance driven towards zero therefore costs them nothing beyond an overconfident report.

Eliminating the drift requires three mitigations, the first being an aleatoric floor. The sampling variance given by Eq. (3.5) is held at or above $\sigma_{\min}^2 = 0.04$, so the sampling standard deviation cannot fall below 0.2 whatever value β attains. The exploration noise is thereby bounded from below by construction, a bound the policy gradient cannot overcome however profitable the shrinkage. The floor is the operational form of the depreciation of the variance-minimising regularisation term documented in Section 2.1.3, applied to the sampling spread rather than to the objective. Its reach is nonetheless confined to the sampling trajectory and never extends to the value passed to the uncertainty consumers, which would distort the signal under study.

Holding the sampling spread above 0.2 conceals the shrinkage in β without opposing it, so a second mitigation is directed at the shrinkage itself, raising the coefficient λ_{reg} that weights the anchor (Eq. (3.6)) in the loss from 0.005 to 0.02. The increase is deliberately restrained, as a stronger anchor would hold α and β near the prior and flatten the epistemic channel with them, inflating the reported uncertainty beyond what the data warrants. The value adopted is accordingly the smallest that counters the drift while leaving the channel responsive.

Neither the bound nor the anchor incentivises exploration, which remains the function of the entropy term $\mathcal{H}$ of the PPO objective in Eq. (2.4). The third mitigation concerns how that term is computed, as no closed-form entropy exists for the tanh-squashed distribution defined by Eq. (3.5). For such a distribution SB3 reports no entropy at all, falling back on the negative log-probabilities of the squashed samples. Saturating the squashed action raises that estimate more cheaply than widening the standard deviation, so the bonus would reward saturation in place of the exploration it was meant to sustain. The head therefore overrides the fallback with the differential entropy of the pre-squash Gaussian:

$$\mathcal{H}[\pi_{\theta}](s_t) = \sum_{i=1}^3 \frac{1}{2} \ln(2\pi e \sigma_i^2) \quad (3.7)$$

The index i runs over the steer, throttle and brake axes, and each σ_i is the square root of the clamped variance from which the action on axis i is sampled. The only route to a higher bonus is now to widen those standard deviations, so the floor and the bonus act

on the same quantity, the first holding it at 0.2 from below and the second rewarding every value above.

3.5.3 Integration

The gated observation described in Section 3.3 enters the actor-critic architecture depicted in Fig. 3.6.

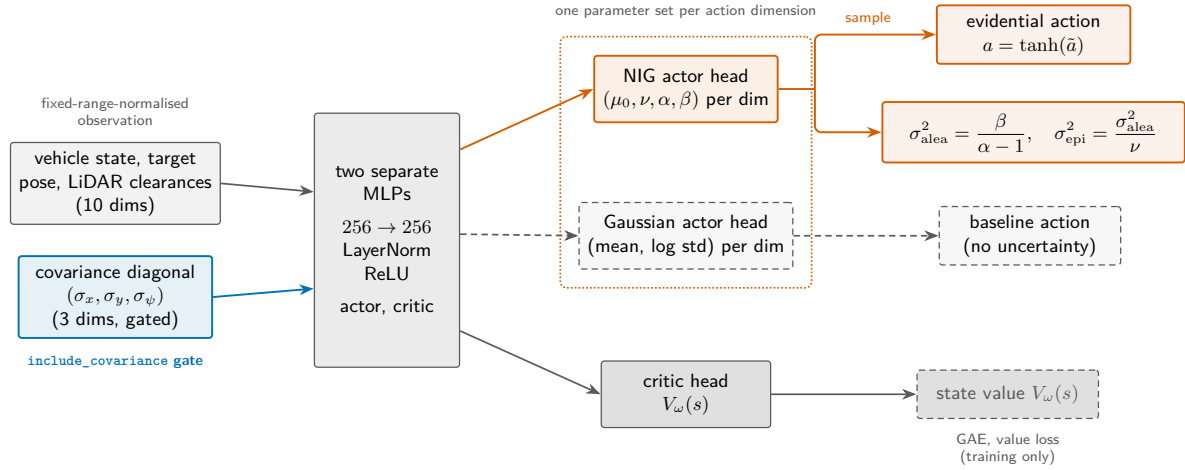


Figure 3.6 The actor-critic architecture.

The observation is processed by two separate networks of identical shape, one for the actor and one for the critic, each of two 256-unit layers in which layer normalisation precedes the Rectified Linear Unit (ReLU) activation to guard against the loss of plasticity described in Section 2.1.5.

The actor network passes its features to one of two interchangeable heads, selected by the arm under training. The NIG head (Section 3.5.1), drawn solid, supplies the sampled action with the aleatoric and epistemic estimates defined by Eq. (3.3) attached, whereas the standard Gaussian head, drawn dashed, emits only a mean and a log standard deviation, leaving the action without any uncertainty estimate. The critic network is not swapped, staying the same across all arms and providing the scalar state-value estimate V_ω on which the GAE advantages and the value loss depend. That estimate is consumed within the update rather than issued as a command, so it shapes the policy only through the gradient and never reaches the vehicle.

The interchangeable pair forms the output-uncertainty axis of the ablation designed later in Section 3.8. Swapping one head for the other changes nothing upstream of the head, leaving the input axis free to vary without an architectural confound between the two.

The policy is trained under SB3, and using the evidential head within it means modifying its stock action distribution, policy class and PPO algorithm. Even with these

modifications, SB3 returns the action without its uncertainty estimate, so the estimate reaches the safety wrapper described in Section 3.9 through a dedicated interface that supplies the action and the complete uncertainty dictionary in one call. Every command is consequently gated on the uncertainty from the same forward pass that generated it.

Algorithm 1 Evidential PPO training iteration

- 1: collect n_{steps} decisions, sampling a via Eq. (3.5) from the normalised observation (Eq. (3.1))
 - 2: compute returns and advantages $\hat{A}$ with GAE
 - 3: **for** each of n_{epochs} epochs over minibatches **do**
 - 4: evaluate $(\mu_0, \nu, \alpha, \beta)$, values, log probabilities and entropy
 - 5: $\mathcal{L} \leftarrow -L^{\text{CLIP}} + c_1 L^V - c_2 \mathcal{H} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}$
 - 6: clip gradients and take an Adam step, halting the epochs early on the KL target
 - 7: **end for**
 - 8: advance the learning rate, entropy coefficient and λ_{reg} per the stage schedule
-

The update departs from stock PPO at four separate points, located in the single training iteration presented in Algorithm 1, namely the floored sampling rule during collection, the pre-squash entropy and the anchoring regulariser within the epoch loop, and the λ_{reg} schedule at the close. Advantage estimation and the clipped update between them are left unmodified, so the PPO update is identical across all four arms and, like the architecture, held constant between them.

3.6 Reward Design

The advantage estimates incorporated into the PPO update (Section 3.5) are derived from a scalar reward, structured so that its components characterise the parking task. Its five components, four dense and one sparse, are tabulated with their magnitudes and roles at the end of the section in Table 3.6, and summed at every decision:

$$r_t = r_{\text{prog}} + r_{\text{end}} + r_{\text{clear}} + r_{\text{oob}} + r_{\text{term}} \quad (3.8)$$

Three of the dense terms shape the approach itself, in that r_{prog} rewards progress towards the bay, r_{end} rewards a clean endgame, and r_{clear} penalises encroachment on a neighbouring vehicle. The fourth, r_{oob} , charges for crossing the out-of-bounds line defined in Section 3.2.2, a charge that accumulates across the episode rather than clearing on re-entry. Finally, the sparse term r_{term} is the payment issued once the episode ends.

None of the five draws on the localisation covariance or the two uncertainty estimates, so the reward contains no uncertainty term, scoring the parking manoeuvre

alone. Any uncertainty-conditioned behaviour measured later is emergent under such a reward rather than incentivised.

3.6.1 Progress: the Bay-Frame Shaping Potential

The progress term r_{prog} is built on a shaping potential expressed in the frame of the target bay, with the parked pose as its origin and the depth direction of the bay as its axis. Measuring the ego pose against that axis separates the manoeuvre into three components a potential can weight independently, the along-track distance d_{along} down the axis, the cross-track distance d_{cross} across it and the symmetry-wrapped heading error ψ_{err} . The three combine into the potential Φ :

$$\Phi = -(w_{\text{along}} |d_{\text{along}}| + w_{\text{cross}} |d_{\text{cross}}| + w_{\text{head}} \psi_{\text{err}}) \quad (w_{\text{along}}, w_{\text{cross}}, w_{\text{head}}) = (1, 2, 3) \quad (3.9)$$

The potential reaches zero only at the perfectly parked pose and is negative everywhere else. Being monotonic in each coordinate, it rises with every reduction in depth, offset or heading error, so the direction of an approach governs its reward as much as the ground it covers.

The coefficients w_{along} , w_{cross} and w_{head} weight each coordinate within Φ but cannot be compared directly, as the two distances are measured in metres and the heading error in radians. The heading error carries the heaviest coefficient, yet at the magnitudes met in the task a 20° error and a metre of depth incur roughly equal penalties, while a metre of lateral offset incurs double the penalty of either. Those penalties set the priority among the three corrections, so reaching the bay centreline pays most, while straightening onto the bay axis and closing the remaining depth pay less. Advancing into the bay is therefore never the best-paid correction, so the vehicle aligns outside it rather than approaching at an angle.

A ceiling nonetheless applies to w_{cross} and w_{head} , though not to w_{along} , since closing depth earns its reward immediately while the other two corrections do not. In an S-shaped correction the vehicle turns away from the bay axis before gaining any ground towards the centreline, so the heading penalty is incurred ahead of the cross-track reward. Raising either weight beyond its value in Eq. (3.9) would deepen that intervening loss until a forward-only vehicle stalls in aligned-but-offset poses.

The three coordinates and the axis against which they are measured are illustrated in Fig. 3.7. Since the potential characterises the quality of a pose, r_{prog} converts it into a reward as the change in Φ across a single decision, normalised per episode and scaled:

$$r_{\text{prog}} = \frac{\Phi_t - \Phi_{t-1}}{\max(|\Phi_{\text{start}}|, 5)} \times \mathcal{R}_{\text{target}} \quad \mathcal{R}_{\text{target}} = 39 \quad (3.10)$$

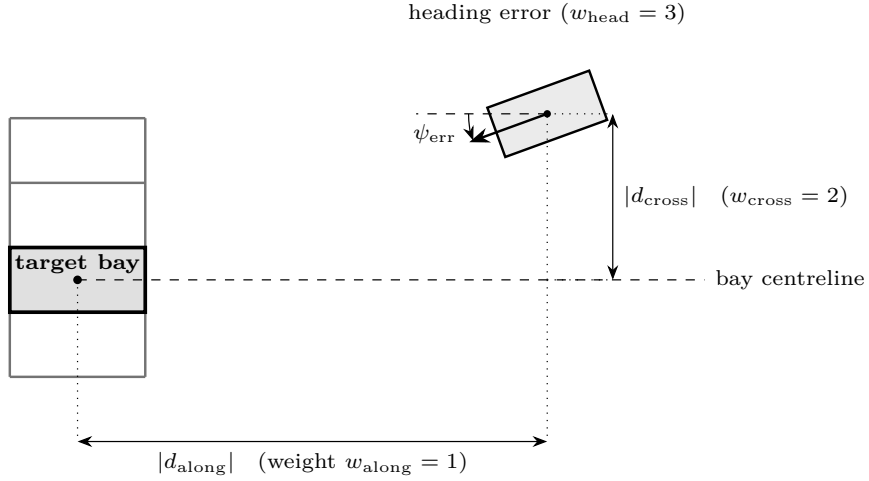


Figure 3.7 The bay-frame coordinates of the shaping potential.

This follows the potential-based formulation discussed in Section 2.1.1, without the discount on which its guarantee of optimal-policy invariance rests. The calculation employs $\Phi_t - \Phi_{t-1}$ instead of discounting Φ_t , so successive terms cancel exactly and the sum telescopes to the change from the initial pose. Cyclic accumulation is closed off as a result, since a stationary vehicle earns nothing, a loiter sums to zero, and a full, clean approach earns the multiplier whatever route it takes. That multiplier, $\mathcal{R}_{\text{target}}$, stays below the +50 paid for success, a terminal figure given later in Table 3.6, so approaching a bay perfectly remains less valuable than parking in it.

The denominator in Eq. (3.10) holds the episode total at $\mathcal{R}_{\text{target}}$. The total would otherwise scale with the commanded distance, so a far bay would pay more dense reward than a near one purely for having further to cover, and for the most distant targets would exceed the terminal payment itself.

3.6.2 Endgame Bonus and Clearance Nudge

Two further dense terms complement the progress term, both acting in the final stages of an episode. The first is an endgame bonus, awarded for holding a pose near the parked target and graded by the offset, heading and depth remaining against it:

$$r_{\text{end}} = (0.008 + 0.006 \cdot f_{\text{stop}}) \cdot f_{\text{line}} \cdot f_{\text{align}} \cdot f_{\text{depth}} \quad (3.11)$$

The factors f_{line} , f_{align} and f_{depth} rescale the three bay-frame coordinates onto $[0, 1]$, each reaching unity at the parked value and decaying to zero over a cutoff of its own, 2 m of offset for f_{line} , 45° of heading for f_{align} and 6 m of depth for f_{depth} . The factors are multiplied rather than summed, so a deficiency in any one suppresses the bonus and a vehicle that stops short and crooked collects almost nothing.

The two leading coefficients set the rate at which the bonus accrues, a base rate of 0.008 while the vehicle is moving and a further 0.006 scaled by the slowness factor f_{stop} , which reaches unity only at a held stop. Even that settled rate sums over an episode to less than the discounted terminal payment, so completing the park is worth more than hovering beside it.

The second is the clearance nudge r_{clear} , a penalty for nearing an adjacent parked vehicle, which acts on the closest clearance d across the LiDAR sectors described in Section 3.3:

$$d = \min(d_{\text{fwd}}, \min(d_{\text{left}}, d_{\text{right}}) + 0.5) \quad (3.12)$$

The side returns are discounted by 0.5 m before the minimum so that the forward cone dominates, since a return directly ahead of a moving vehicle is the real collision risk.

That clearance then drives a ramp gated on the pose:

$$r_{\text{clear}} = -w_{\text{clear}} \underbrace{\min\left(\max\left(\frac{0.8 - d}{0.5}, 0\right), 1\right)}_{\text{ramp}} \underbrace{(1 - f_{\text{line}} f_{\text{align}})}_{\text{gate}} \quad (3.13)$$

The gate reuses the corridor factors from Eq. (3.11) and falls to zero as the vehicle squares onto the centreline, while the 0.8 m onset lies below the approximately 0.98 m side gap left beside an occupied neighbour. Parking next to another vehicle is therefore not penalised, and the penalty falls only on a vehicle that drifts towards a neighbour while still manoeuvring.

3.6.3 Terminal Payments and the Success Criterion

Unlike the dense terms, which are paid at every decision, the terminal payment r_{term} is issued once at the end of an episode, fixed for success and collision but graded by the final error when the budget expires:

$$r_{\text{term}} = \begin{cases} +50 & \text{success} \\ -25 & \text{ego-fault collision} \\ \max\left(-\frac{1.5 e_{\text{pos}} + 2.5 e_{\text{yaw}}}{\max(|\Phi_{\text{start}}|, 5)} \times \mathcal{R}_{\text{target}}, -24\right) & \text{timeout} \end{cases} \quad (3.14)$$

The timeout branch assesses the final pose, weighting the orientation error e_{yaw} above the position error e_{pos} . A vehicle ending square with the bay but slightly out of position is therefore charged less than one ending closer but badly angled. The branch also reuses the normaliser from Eq. (3.10), which places the penalty on the dense progress scale so that a distant bay is not penalised for its distance. That scaled penalty is

bounded below at -24 , one point above the -25 charged for an ego-fault collision, so ending a failing episode in a collision is never cheaper than letting the clock expire. Payment at the opposite end of that scale, the $+50$ for success, requires the vehicle to be both inside the bay and stationary. Every corner of the ego bounding box must lie within the target bay polygon and the speed must remain below 0.1 m/s, both for five consecutive decisions, so that a fast drive-through cannot register as a park.

Before the corners are tested, the polygon is adjusted by a dimensional tolerance, the margin, which is the only quantity in the criterion that varies. A negative value expands the polygon outwards and permits a looser park, while one closer to zero contracts it and requires a tighter fit. The margin starts loose in the early curriculum stages and is stepped down as training advances, on a schedule Section 3.7 defines in full. It ends at the tightest setting, which is the one the evaluation in Chapter 4 applies.

Table 3.6 The additive reward defined by Eq. (3.8).

Term	Type	Magnitude	Role
Progress r_{prog}	dense	sums to $\mathcal{R}_{\text{target}}$ per episode	potential difference given by Eq. (3.9), normalised so every bay offers the same dense pool
Endgame bonus r_{end}	dense	below discounted terminal	aligned, near-depth, stopped factors that settle the vehicle in
Clearance nudge r_{clear}	dense	small ramped penalty	discourages closing on a parked neighbour, off once square
Out-of-bounds r_{oob}	dense	-0.5 per decision outside	the accumulating drift-out cost described in Section 3.2.2
Success	terminal	$+50$	bounding box inside the bay, held stopped
Ego-fault collision	terminal	-25	striking a vehicle
Timeout	terminal	floored at -24	graded on final pose error (Eq. (3.14)), above the ego-fault floor

3.7 Training Protocol: the Curriculum

The reward in Section 3.6 is sparse in its crucial component, as the $+50$ terminal payment is awarded only once an entire manoeuvre has been completed under drifting localisation. Such a payment would rarely be reached by a policy trained from scratch on the full task across all 47 bays, so the difficulty is organised into stages.

3.7.1 The Channel-Liveness Principle

The staging is constrained throughout by the liveness of the observation channels, so every channel varies from the first stage rather than being introduced once the geometry has been mastered. Each later stage then widens only the extent or severity of one difficulty axis. The support-preserving property carried over from fixed domain randomisation in Section 2.1.5 is thus combined with the gradual widening of a curriculum, so no channel stays degenerate long enough for its input weights to decay. The principle is recorded for each channel in Table 3.7, together with the stage at which its range is opened up.

Table 3.7 The channel-liveness principle.

Channel	Kept live from stage 1 by	Ramped later by
$\sigma_x, \sigma_y, \sigma_\psi$	the always-on GNSS fix-state Markov chain (mid-episode drift)	not ramped: the chain is stage-invariant
$\Delta x, \Delta y, \Delta \psi$	four bays spanning all three bay orientations	more bays, reaching all 47
LiDAR clearance	parked neighbours at up to 10% occupancy	denser occupancy, reaching 80%

The liveness requirement follows from the shortcomings of an initial prototype for the curriculum, built on the more intuitive assumption that geometry and noise are separate skills to be taught in sequence. Training began with one bay in an empty lot, with the fix-state chain from Section 3.4.1 pinned to its best tier through a clean opening phase. Those three simplifications were meant to ease the task, but each flattened a group of observation dimensions instead. Holding the chain at one tier fixes the covariance it reports, pinning those three dimensions to the floor of their range, and a single bay does the same to the target pose and an empty lot to the five clearance readings. Only two of the thirteen dimensions consequently varied, too few to compel a target-conditioned controller, so a single trajectory was fitted and the weights on the other eleven decayed towards zero. Those inputs were therefore unusable by the time later stages began to vary them.

This is the dormant-neurone and plasticity-loss failure described in Section 2.1.5, under which an input that has fallen out of use is not readily re-engaged once it carries information again. The sequential ordering was therefore abandoned, and the liveness requirement that replaced it grades the task without silencing any channel.

The same collapse arose through a second fault, left untouched by the liveness requirement and remedied separately. A revised prototype kept every channel live but

still normalised observations with the running statistics identified as a liability in Section 3.3.2, and those statistics crossed each stage boundary with the weights. A dimension held narrow in one stage therefore entered the next carrying the variance measured under that setting. The target offset was affected most, as the bay set expands at each of the first two transitions. Offsets drawn from the larger set of bays, divided by a variance gathered over the smaller one, fall outside the clip bounds set by Eq. (3.1), so the network received a saturated constant on that dimension precisely as it began to carry information.

The principal implication of the liveness requirement is that the GNSS degradation process is not part of the curriculum at all. The single Markov chain described in Section 3.4.1 runs unchanged throughout, neither gated nor scaled by stage, so localisation uncertainty is present from the first training decision.

3.7.2 The Six Stages

The six stages are defined in Table 3.8, with budgets counted in policy decisions and the learning rate decaying linearly within each stage rather than across the whole run. The occupancy column sets only the upper bound $q_{\max}$ of the per-episode draw defined in Section 3.2, with $q_{\min}$ held at zero throughout, so every density from an empty lot up to that stage's maximum remains reachable.

Table 3.8 The six curriculum stages as committed.

Stage	Bays	Occ. max	Margin	Budget	LR	Primary axis
1	4	0.1	-0.75	1.5×10^6	$3.0 \rightarrow 0.5 \times 10^{-4}$	bootstrap manoeuvre
2	15	0.1	-0.55	1.5×10^6	$1.5 \rightarrow 0.5 \times 10^{-4}$	bay variety
3	47	0.1	-0.40	2.0×10^6	$1.5 \rightarrow 0.4 \times 10^{-4}$	bays to full
4	47	0.1	-0.25	1.5×10^6	$1.2 \rightarrow 0.4 \times 10^{-4}$	margin to strict
5	47	0.5	-0.25	2.0×10^6	$1.2 \rightarrow 0.3 \times 10^{-4}$	occupancy
6	47	0.8	-0.25	1.5×10^6	$1.0 \rightarrow 0.2 \times 10^{-4}$	occupancy to final

The order of the stages is not arbitrary, as each widens an axis that becomes tractable only once the skill beneath it has been consolidated in the stage before. The hardest skill, parking at all three bay orientations with both turn directions, is concentrated in stage one while every other axis remains forgiving. The loosest acceptance margin, light occupancy and the dense per-bay-normalised shaping from Section 3.6 combine so that the +50 terminal payment is received often enough to bootstrap PPO from scratch. Once that manoeuvre is reliable at four bays, it is generalised across the lot in stages two and three, the bay set expanding through

supersets from 4 to 15 and finally all 47 targets. The acceptance margin is tightened to the strict evaluation criterion in stage four, only after that generalisation is complete and while conditions are otherwise clean, so precision is acquired before the lot is densely occupied. Occupancy is raised to the operating point last, in stages five and six, since a crowded lot obstructs a manoeuvre that can by then be executed precisely at any bay.

That ordering fixes what each stage contains, while a separate choice governs when one gives way to the next. Progression follows the difficulty setting rather than the outcome achieved within a stage, consistent with the fixed pacing argued for in Section 2.1.5. The acceptance margin, bay set and occupancy ceiling therefore advance to the values in Table 3.8 once each stage's decision budget is spent, and the run is never stopped early or held back for missing a success rate. That uniformity keeps the stage boundaries identical across the arms compared later in Section 3.8.

A schedule that advances on budget alone is only as sound as the training state it hands to the next stage. The learning rate is the parameter most at risk across a boundary, since the finding in Section 2.1.1 places its value among the most influential in on-policy training while leaving its decay schedule far less consequential. Two choices therefore govern what crosses that boundary, the first being that every stage runs its own linear learning-rate and entropy curves over its budget, so no schedule state is inherited from the stage before. The second sets the starting learning rate of the restarted curves to roughly three times the final value of the preceding stage rather than to the from-scratch rate, so a nearly converged policy is nudged rather than destabilised.

Those two choices apply to every arm, and the schedules they carry differ between the two head types in a single respect. The two per-policy blocks, one for the standard-head arms and one for the evidential arms, are initialised identically except for a raised entropy floor in the evidential case and, in the later stages, a higher starting value. The asymmetry follows from where the entropy pressure lands. On the evidential head the action standard deviation is the square root of the aleatoric estimate (Eq. (3.5)), so the pressure acts directly on that estimate. The log standard deviation of the standard head is, by contrast, a single parameter, independent of the state. This difference is a further planned response to the collapse tendency described in Section 3.5.2.

3.7.3 The Learnability of Waiting

The exploration those schedules protect must reach the behaviour the uncertainty inputs exist to elicit, a pause under a degraded fix. No reward term pays for that behaviour, yet four choices made independently across this chapter combine to make it

learnable. Target-pose features are derived from the EKF estimate, while success is assessed on ground truth unavailable to the policy (Section 3.3.1). The perceived target is therefore corrupted under a poor fix at the moment the covariance features signal the degradation, and a vehicle that commits to the manoeuvre in that state misses the bay and earns no terminal payment. The tier is nonetheless restored within seconds by the upward-biased chain in Section 3.4.1, while the wait is tolerated in the meantime by the stall counter in Section 3.2.2. A pause is consequently reachable without ever being paid for.

3.8 Ablation Design and Hyperparameters

Uncertainty enters the agent through two pathways: the covariance observation on the input side and the evidential head on the output side. Their contributions are evaluated by crossing the two in a 2×2 factorial ablation. The same hidden layers in every arm and the stock SB3 optimiser from Section 3.5 hold learning constant, while the stages advance on a fixed budget rather than on measured performance (Section 3.7). The four arms listed in Table 3.9, each a complete training configuration, therefore differ only in the actor head and in whether the three covariance features from Section 3.3 are admitted through the gate. The one exception is the entropy floor, which is tied to the head rather than held common (Section 3.7.2).

Table 3.9 The 2×2 ablation arms.

Arm	Covariance obs.	Actor head	Obs. dim
vanilla_ppo	no	standard Gaussian	10
input_uncertainty	yes	standard Gaussian	13
output_uncertainty	no	evidential NIG	10
full_method	yes	evidential NIG	13

Each axis carries two settings, the mechanism under test and the component it replaces, so the evidential head is compared against the standard Gaussian head and the covariance observation against its own absence. A further setting on either axis would divide a contribution the design is meant to attribute whole, so the finer-grained arms that would apportion it are left to the future work proposed in Section 5.2.

Every arm draws its hyperparameters from Table 3.10, which summarises the committed configuration files reproduced in full in Appendix C. The discount of 0.997 gives an effective horizon of approximately 333 decisions, or around 66 seconds, so the terminal payment remains visible from the spawn pose throughout the approach.

Neither the entropy coefficient nor the learning rate holds the tabulated value during training, as both are starting points that the per-stage schedules in Table 3.8 supersede.

Table 3.10 Committed hyperparameters.

Parameter	Value	Rationale
rollout length n_{steps}	2048	~ 4.7 episodes per update at ~ 437 decisions each
batch size / epochs	256 / 5	standard PPO regime for this rollout size
discount γ	0.997	effective horizon $1/(1 - \gamma) \approx 333$ decisions
GAE λ	0.95	standard bias-variance point
clip range	0.2	PPO trust region; primary stability control
target KL	0.04	early-stops update epochs; backup to the clip
value coefficient	0.5	equal-footing value regression
gradient-norm clip	0.5	caps the joint gradient of actor and critic
learning rate (base)	$3 \times 10^{-4} \rightarrow 1 \times 10^{-5}$	linear decay; stage schedules override (Table 3.8)
entropy coefficient (base)	$0.02 \rightarrow 0.006$	decaying pressure with a floor; evidential floor higher (Section 3.7.2)
log-std init (standard head)	-0.6	initial exploration standard deviation ≈ 0.55 , close to the evidential prior scale
λ_{reg}	$0 \rightarrow 0.02$ over 5×10^4	prior anchor defined by Eq. (3.6); brittle, raised cautiously (Section 3.5.2)
aleatoric floor	0.04	sampling-variance floor applied to Eq. (3.5); exploration guard

3.8.1 Fixed Rather Than Tuned Hyperparameters

No Hyperparameter Optimisation (HPO) was conducted for any of the experiments, the values in Table 3.10 being predetermined defaults applied identically across every arm and seed except for the entropy coefficient, which the evidential arms hold above a higher floor and, in the later stages, start from a higher value. The covariance axis of the ablation contrasts two arms of matched head type and is therefore unaffected by this exception, whereas the head axis carries it alongside the head itself. That decision resolves a conflict between two requirements. Established practice, recorded in Section 2.1.6, is to optimise each method under comparison separately, as the values that suit one method rarely suit another [51]. In a 2×2 ablation, by contrast, the actor head and the covariance gate are isolated by holding all other variables constant. A separate search for each arm would make the configuration a third variable, and any measured difference would then be partly attributable to the four searches converging unevenly. Priority is therefore given to the controlled comparison, since the contribution rests on the ablation.

That priority leaves a residual weakness, as a set selected once may suit the optimiser dynamics of one arm better than another, and no claim is made that it suits all four equally. No such preference is introduced by construction, however, since every value is derived from the task or from established PPO practice rather than from any arm’s measured performance, and the complete set was committed before the ablation began, so the results could not influence it. A search would also have consumed Graphics Processing Unit (GPU) time on the scale of a training chain, time that was allocated instead to the seed matrix described next in Section 3.8.2. An advantage arising from the suite rather than from the architecture would be expected to vary across seeds, so averaging over three reduces its influence on the reported contrasts. That reduction falls short of elimination, and the possibility of such an imbalance is carried forward as a limitation in Section 4.8.3.

3.8.2 Seeds and Compute

Every arm is trained as a complete six-stage chain of ten million policy decisions under each of the three evaluation seeds 42, 123 and 7, giving twelve chains in total. The per-stage budgets were set on the first seed alone and reused unchanged for seeds 123 and 7, with training traces inspected at each stage and investigated only upon divergence, so no per-seed tuning could leak into the comparison.

3.8.3 Training Trajectories

The training trajectories are plotted in Fig. 3.8, averaged across the three seeds and marked at the stage boundaries. Success and collisions in its two panels are recorded on the training distribution to track convergence, while performance on the same metrics is measured under the evaluation protocol defined later in Section 4.1.3.

The highest success rate belongs first to the two arms observing the covariance and later to the full method alone. Occupancy is capped at 10% in stage one (Table 3.8), leaving localisation error as the most likely source of uncertainty. These two arms accordingly reach approximately 75%, against around 60% for the output-uncertainty arm and 50% for vanilla PPO. The input-uncertainty arm falls behind the full method from the third stage and reaches the level of the output arm once the lot fills in stage five. By the end of the final stage the full method attains approximately 50%, against around 27% for the input- and output-uncertainty arms and approximately 23% for vanilla PPO. The three lower arms therefore finish within six percentage points of one another, so neither the covariance observation nor the evidential head lifts an arm far above vanilla PPO on its own. The gain appears only where the two are combined, indicating an interaction between the axes.

The lowest collision rate, by contrast, belongs to the two evidential arms, though only once obstacles become common, since all four remain below 10% and within a few percentage points of one another through the sparsely occupied stages. The rise in occupancy at stage five lifts every arm to between roughly 25% and 30%, the intended consequence of admitting more parked-vehicle obstacles rather than a loss of competence, and success recovers over stage six. The two evidential arms then settle near 20% while vanilla PPO and the input-uncertainty arm remain around 30%. The lower collision band is consistent with the state-dependent sampling variance given by Eq. (3.5), an adjustment unavailable to the standard head. The two panels are therefore consistent with parked vehicles joining localisation error as a source of uncertainty in the final two stages, which would account for the lower collision rates of the evidential arms and motivates the uncertainty-triggered handover specified next in Section 3.9.

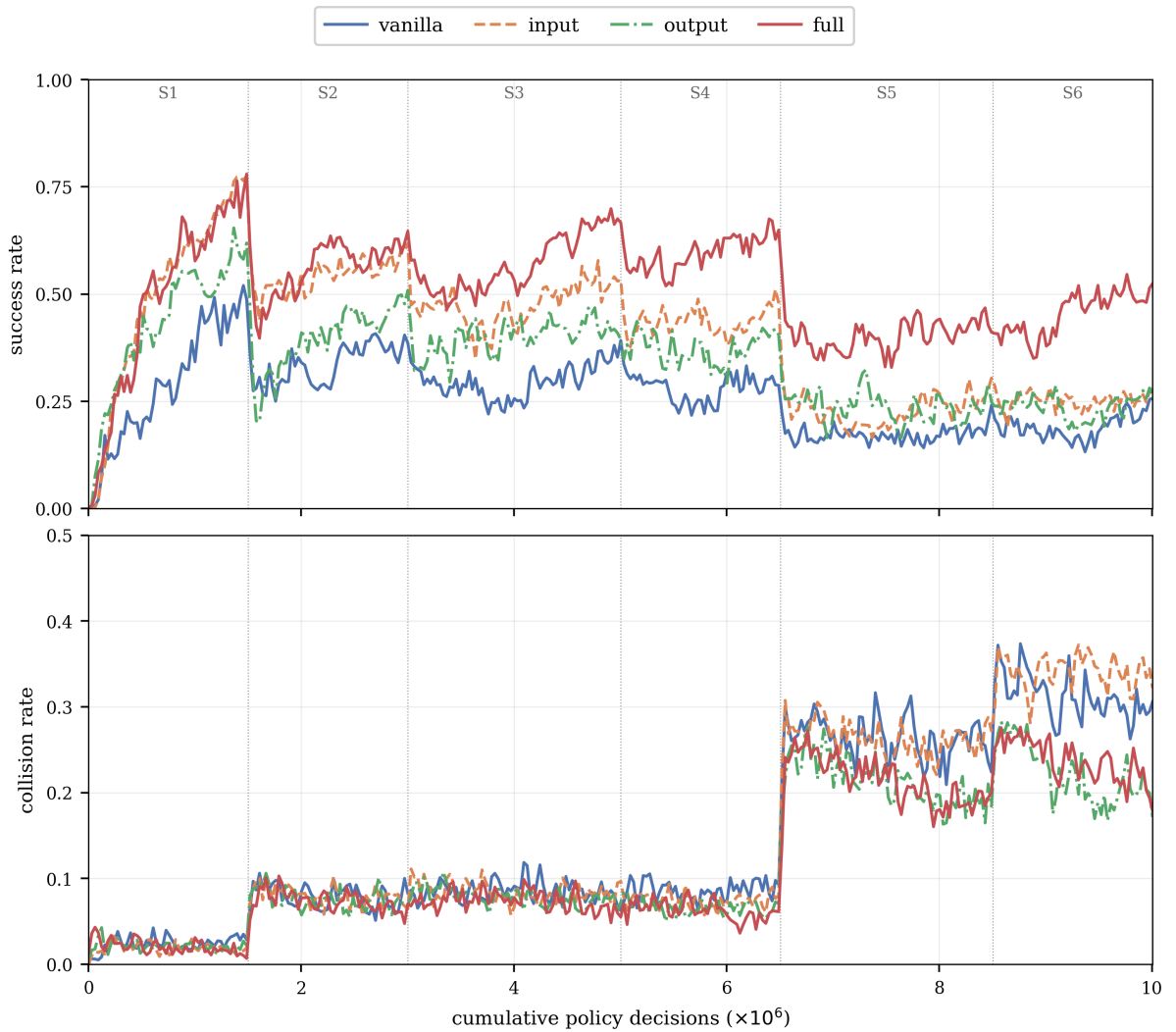


Figure 3.8 Training success rate (top) and collision rate (bottom).

3.9 Safety Wrapper and Deployment Path

A runtime safety layer surrounds the environment during deployment but not during training, so learning is unrestricted and supervision applies only where actions reach the vehicle. Separate thresholds on the two uncertainty channels would not distinguish an unfamiliar state from intrinsic task noise, since the channels are coupled in a single-head NIG actor (Section 4.3.2). The aleatoric and epistemic estimates defined in Eq. (3.3) nonetheless remain variances of a single predictive distribution and sum to the total variance given by Eq. (2.5), which therefore rises whenever either channel does. The wrapper is accordingly triggered on that sum, which spans a wider range than either channel alone and so permits a more precisely positioned threshold.

Once the total reaches 1.2, the commanded action is overridden: steering is zeroed and full braking is applied until the vehicle stops. The handover is then flagged and the episode ends, so no post-intervention behaviour is attributed to the policy.

Runtime arbitration between a learned policy and a fallback places this wrapper within the guarantee-and-fallback family reviewed in Section 2.2.1. The surveyed systems, however, condition a stop on a verdict about policy reliability, whereas the total predictive uncertainty originates in the actor and rises with the current state.

A rule based on magnitude alone is blind to duration, so a momentary excursion above 1.2 and a sustained one both stop the vehicle. A persistence variant is consequently examined alongside the magnitude rule, requiring the total to remain above the same 1.2 for ten consecutive decisions, or approximately 2 s, while any reading at or below the threshold resets the count. The delay rests on the expectation that a sustained rise signifies a genuine loss of confidence whereas a brief one reflects noise, so the stop is reserved for cases that warrant a handover. Neither configuration is assumed superior, and both are therefore replayed against the outcome reached by each episode, and Section 4.8.2 reports what the wrapper can and cannot discriminate.

3.10 Summary

The resulting system couples a PPO policy to an EKF localisation stack, with the filter's posterior covariance entering the observation and the action distribution carried on an evidential NIG head. Training and evaluation share a single code path, a unified configuration source and consistent observation scaling, so the covariance input and the evidential head, together with its entropy floor, remain the only designed differences between the four arms set out in Section 3.8. This system is assessed in the next chapter against the objectives stated in Section 1.3.

4 Evaluation and Results

The framework established in Chapter 3 is evaluated in this chapter against the four objectives set out in Section 1.3, using the trained policies generated by the factorial ablation specified in Section 3.8.

4.1 Experimental Setup

No established benchmark is available for this evaluation to adopt, as no prior system combines closed-loop parking control with controlled localisation degradation (Section 2.3). The absence extends to the available environments, since the public parking environment closest to this task supplies neither the state estimator observed here nor an axis along which its sensing can be degraded [80]. The protocol is therefore assembled from the practices of reinforcement learning and of parking control, the two fields the task spans. Reporting practice is drawn from the former, under the multi-seed requirement established in Section 2.1.6, so every headline result is pooled across three training seeds rather than taken from a single run. The two headline metrics are drawn from the latter, namely the task success rate reported as the primary outcome by Lazzaroni et al. [69] and the final position error reported alongside it in end-to-end benchmarks [68]. Both are supplemented by a five-class outcome taxonomy and by per-step behavioural proxies, each defined in Section 4.1.4.

4.1.1 Arms and Controlled Contrasts

The four arms (Table 3.9) each serve as the controlled comparator for their adjacent counterparts on either axis, a role that no external trained system could provide.

Any two arms adjacent on one axis differ by a single factor, giving four contrasts. Along the covariance axis, the full method against the output-uncertainty arm shows the effect of the covariance observation on the evidential head. The same effect on the standard Gaussian head appears in the input-uncertainty arm against vanilla PPO. Along the head axis, the full method against the input-uncertainty arm isolates the evidential head where the covariance observation is available, and the output-uncertainty arm against vanilla PPO does so where it is withheld. A pair on either axis is analysed together rather than separately, since a factorial design determines whether the effect of one factor depends on the setting of the other.

4.1.2 Evaluation Conditions

Every arm is evaluated under the same four conditions, which address distinct properties of the deployment environment. The conditions are defined in Table 4.1 and grouped into a reference pair, a generalisation probe and a within-episode drift case. Each isolates one factor against the deployment anchor, so any performance difference can be attributed to that factor. Closed-loop control in simulation is not a supervised prediction task, so no held-out labelled test set can be constructed, and a fixed, seeded battery of conditions takes its place.

Table 4.1 The four evaluation conditions.

Group	Condition	Factor varied	Purpose
Reference pair	anchor_deployment	none (occupancy 0.5, training noise process live)	in-distribution reference
	anchor_empty	occupancy $\rightarrow$ 0.0	no-obstacle control
Generalisation probe	ood_irregular_rtk_fixed	unseen irregular layout, clean GNSS	geometry as the only OOD factor
Within-episode drift	gnss_degrade_one_way	mid-episode one-way drift to degraded	ecologically valid degradation

The reference pair provides the baseline, holding conditions as close to the training settings as the deployment target allows. The anchor condition (anchor_deployment) departs from those settings only in bay occupancy, which is fixed at the deployment value of 0.5, whereas the curriculum described in Section 3.7 resamples it for each episode. Fixing it prevents a GNSS-tier effect from being confounded with how busy a particular episode’s lot happened to be. Localisation is unchanged, so a fix-state tier is drawn for each episode and remains free to drift through the Markov transition defined in Section 3.4.1. The condition is therefore reported in its own right, as the performance obtainable on the instrumented lot, and the three remaining conditions are each evaluated against it.

The second member of the pair (anchor_empty) is identical to the anchor except that its occupancy is reduced to 0.0. No parked vehicles remain to return a LiDAR echo or present a collision hazard, so the filter estimate is the only signal by which a bay can be reached. Any difference between the two conditions consequently measures the extent to which task performance depends on the presence of parked vehicles.

The remaining two conditions extend the evaluation beyond the anchor, one through the lot geometry and the other through localisation quality. In the generalisation probe (ood_irregular_rtk_fixed), the task is moved to the held-out lot

established in Section 3.2, with localisation held clean rather than following the Markov chain. The unfamiliar geometry is therefore the sole OOD factor, so the cost of the layout reported in Section 4.8.1 is not confounded with GNSS degradation.

The second condition (`gnss_degrade_one_way`) guarantees a transition from confident to degraded localisation. An anchor episode follows the Markov chain wherever it leads and may never degrade, whereas the replay of the handover threshold reported later in Section 4.8.2 requires a degradation that is certain. Localisation therefore starts at RTK fixed and proceeds one way down the ladder without recovery.

4.1.3 Protocol

Every arm is assessed over 200 episodes under all four conditions, and the whole battery is repeated for the three training seeds fixed in Section 3.8.2, so each condition-arm combination rests on $n = 600$ episodes. The policy is run deterministically at the fixed 5 Hz decision rate adopted in Section 3.3.3, where the 0.2 s spanned by a decision is established as the unit of every budget. Quantities reported in decisions are converted to time on that basis, and the complete protocol can be reproduced step by step from the guide in Appendix D.

The safety mechanism described in Section 3.9 remains disabled throughout every episode. An activation would replace the selected action with a full stop and truncate the run, so the behavioural signal would be corrupted and the outcome the episode would otherwise have reached withheld. Each episode therefore runs to its natural end, and the implemented rule is assessed in Section 4.8.2 by replaying its threshold over these records rather than intervening within them.

4.1.4 Metrics

Each numerical value in this chapter is built from four types of measurement: a pair of primary results for each episode, a categorical label recording how the episode ended, four per-step behavioural proxies, and two grouping quantities that sort episodes into comparable bands rather than report an outcome.

The first of those types, the primary pair, records both the outcome of an episode and its margin. Success is scored categorically, awarded once the vehicle comes to rest fully inside the target bay on the criteria fixed in Section 3.6.3, evaluated at the tightest bay margin of the curriculum. A binary label of that kind records nothing of how an unsuccessful episode ended, so it is paired with the final position error. That error is the Euclidean distance in metres from the ego to the target bay centre when the episode terminates, so a failed attempt is measured by how close it came to the target. Episodes ending tens of metres away pull the mean of this quantity several times above

the median. The mean is therefore the informative measure where the composition of outcomes is itself the subject, and the median where the concern is how precisely the vehicle parks. Both are reported, and the caption of every results table in this chapter states which of the two it carries. Aggregating the binary outcome over a set of episodes yields a success rate, and two such sets are compared by the difference between their rates. That difference is expressed in percentage points (pp), denoting an absolute change in the proportion of episodes parked rather than a relative one.

The manner in which an episode ended is captured instead by the outcome label, which assigns each episode exactly one of five mutually exclusive categories: success, collision, out-of-bounds, near-miss and stuck. The first three are determined by the event that concludes the episode, success being the criterion defined above, collision a registered contact, and out-of-bounds the ego leaving the drivable area. The last two cover episodes in which no such event occurs before the step limit is reached or the stall counter ends the episode, and are distinguished by the final position error, a near-miss ending within 1.5 m of the target bay centre and a stuck episode beyond it. The finer split matters because a bare success-or-failure label would score an attempt that ran out of time short of the bay and one that halted far from it as the same failure.

A hierarchy resolves the cases in which more than one label could apply, with the physical outcome taking precedence over any other event at the same step. A collision rate reported in this chapter is therefore the proportion of episodes labelled collision once the hierarchy has been applied, not a count of contact events.

The outcomes and labels above characterise an episode by its conclusion. The conduct of the vehicle within it is characterised instead by four per-step behavioural proxies, on which the account of the driving given later in Section 4.6 is built. The first two record the commands issued to the vehicle, read from the vector delivered to the simulator once the rate limits described in Section 3.3.3 have been applied. The brake command is the brake axis of that vector, on $[0, 1]$, whereas action jerk draws on every axis of it, being the Euclidean norm of the difference between the full steering, throttle and brake vectors of successive decisions. A large jerk value therefore marks a decision that departed sharply from the one before it. Jerk in this sense describes the commanded signal rather than the resulting motion, and is distinct from mechanical jerk, the rate of change of the vehicle's acceleration.

The remaining two record the resulting motion rather than the commands producing it, each under a restriction that keeps the measurement interpretable. Speed while moving is the vehicle's speed sampled only on steps at or above the 0.1 m/s threshold that the success criterion set out in Section 3.6.3 treats as stationary. Omitting slower steps prevents the stationary hold at the end of a completed park, and any deliberate wait, from pulling the average towards zero. The yaw rate is recorded under the same moving-only restriction and as a magnitude, so that opposing turns do

not cancel across an episode. All four proxies are averaged over the steps of an episode, so each episode contributes a single value per proxy.

Every measurement to this point describes a single episode, whereas the last two, realised difficulty and reported uncertainty, group episodes for comparison and are never reported as outcomes. Each condenses an episode into a single value used to sort it into a band, the term this chapter assigns to a group of complete episodes. Realised difficulty is the mean per-step distance between the EKF position estimate and the simulator's ground-truth pose across the episode. It therefore quantifies the localisation difficulty actually encountered in each episode and forms the basis for the banding presented later in Section 4.5. Reported uncertainty is instead the episode-mean position standard deviation reported by the filter itself, so it groups episodes by the signal available at the policy's input at the time, as the behavioural comparison requires. The two cannot substitute for one another, since the reported posterior inflates only at the bounded rate its process noise allows and so understates the true error, as established in Section 3.4. Both are divided into equal-population bands, so the same number of episodes supports every value reported from a band.

4.1.5 Reporting Statistics

The claims of this chapter rest on three statistics computed from those metrics, each answering a distinct question. An interval estimate indicates how precisely a contrast is known, a rank correlation measures how strongly two quantities track one another [81], and the area under the Receiver Operating Characteristic (ROC) curve [82] assesses how effectively a scalar signal separates two outcomes.

The pooled episodes of an arm are a finite sample of its performance. A success rate calculated from them is therefore only an estimate of the arm's true rate, and a different sample of episodes would give a different figure. Quantifying that variability by running further episode pools is prohibitively expensive, so the existing episodes are instead resampled with replacement and the contrast recomputed for each resample. The spread of those recomputed values indicates how far the estimate could shift if the evaluation were repeated, while the 2.5th and 97.5th percentiles of their distribution provide the stated Confidence Interval (CI) on the small-sample grounds established in Section 2.1.6.

4.2 The Precondition: Covariance Honesty

Each claim made in this chapter assumes that the standard deviations of the EKF provide a meaningful conditioning signal, such that higher reported uncertainty predicts greater true error. A policy conditioned on a signal that fails this test would

instead be responding to noise, rendering any resulting behavioural outcome uninterpretable. This prerequisite is therefore established before any policy comparison. The standard deviation reported at each decision step is paired with the true error of the estimate it describes, and the covariance ranks that error to the extent that the two rise together. The EKF implemented in Section 3.4 is identical across the four arms, so what is measured is the localisation pipeline rather than any policy. A single arm is consequently sufficient to characterise it, and every decision step analysed in this section is thus drawn from the vanilla PPO runs.

The relationship is illustrated in Fig. 4.1 and quantified by a rank correlation and a binned mean error. Assessing the covariance requires a fix state that varies, a criterion the unseen irregular layout does not meet, since its fix is held at RTK fixed throughout (Section 4.1.2). Its exclusion leaves three conditions, pooled into the single population from which every figure reported in this section is drawn. The two panels cover position and heading and plot the same decision steps at two granularities. The blue dots are the individual steps, each placed by the standard deviation the filter reported against the true error incurred. The joined red markers build on the individual dots to give the mean absolute error of each of five equal-count bins, formed by ordering the steps by reported standard deviation. Both axes of the position panel are logarithmic, since around 48% of them cluster within a few centimetres of the RTK-fixed floor of the reported standard deviation, while those taken under a degraded fix reach values beyond 1 m.

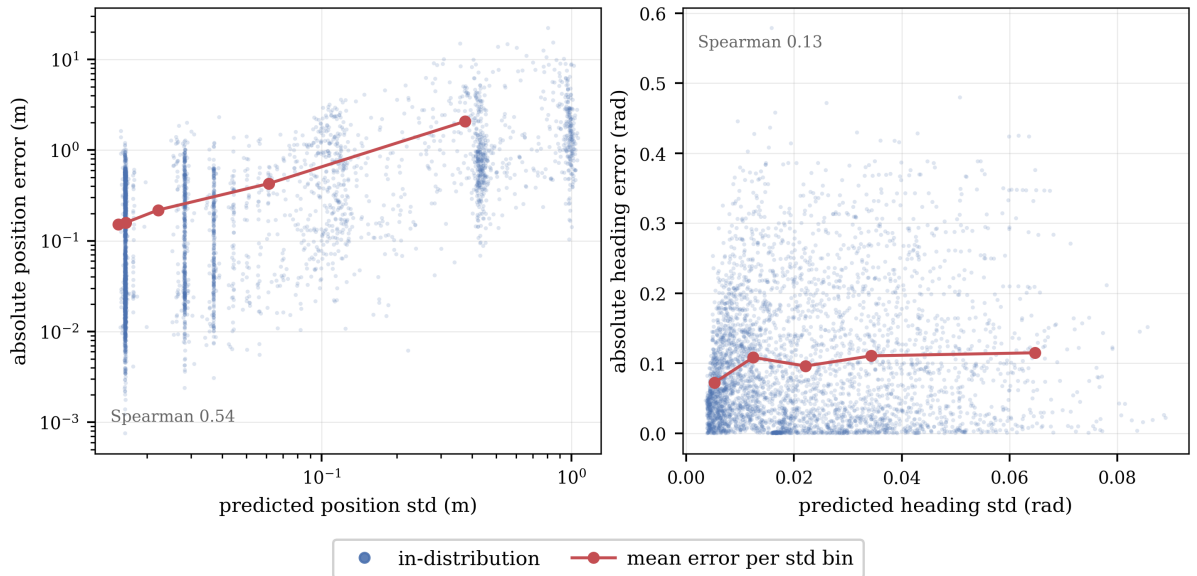


Figure 4.1 Reported EKF standard deviation against true error, pooled over the three conditions in which the fix state varies.

Quantifying the rise in the position channel begins with the ordering shared by the reported position standard deviation and the true position error. Their Spearman

rank correlation is 0.54 over that population, against a value of one for a perfect ordering. Part of the distance from that ceiling is attributable to ties in the data, since a large share of the steps run on a stable RTK fix, for which the filter reports nearly the same value every time. A ranking cannot separate steps that share a value, so such pairs contribute no agreement to the coefficient despite those estimates erring by only about 0.15 m on average. Recomputed above that floor, where the reported figure genuinely moves, the coefficient rises to 0.63, so the ties explain only part of the shortfall. The remainder is expected, since a standard deviation gives the typical size of the error and not the error itself, so even a perfectly calibrated filter could not rank every step correctly. A population-level ordering is what a conditioning signal requires, since the response to a reported level is learned from the average outcome over the many steps that share it. The same ranking could, however, be an artefact of pooling. If one evaluation condition (Section 4.1.2) had both higher reported uncertainty and larger errors than another, the correlation across all steps would be positive even if, within each condition, the reported uncertainty of a step bore no relation to its error, so the coefficient is recomputed for each condition separately. It remains above zero in all three, at 0.41 for the one-way drift, 0.42 for the obstacle-free control and 0.63 for the deployment anchor. Two of the three nonetheless fall below the pooled 0.54, so part of that figure reflects differences between conditions and not the ordering within each one.

The bins address a question the coefficient leaves open, since a rank correlation gives the direction of the relationship without showing whether the error grows enough to justify slowing the vehicle. The mean absolute position error rises at every step of the sweep across five equal-count uncertainty bins, taking the values 0.15, 0.16, 0.22, 0.43 and 2.07 m. The highest-uncertainty bin therefore averages roughly fourteen times the error of the lowest, a gap wide enough to warrant action rather than one detectable only through a statistic.

Two limits govern how far that climb can be read. The first concerns how the bins were formed, as holding equal numbers of steps rather than equal ranges of reported uncertainty makes their widths differ by orders of magnitude. The two lowest bins sit at the RTK-fixed floor of the reported standard deviation because part of every condition runs on a stable solution. A large mass of steps therefore accumulates at a single floor value, which equal-count binning must split. The second-lowest bin is narrower than 0.0001 m, whereas the widest admits 0.98 m. The second limit concerns magnitude, as the bounded inflation rate of the posterior caps how high the reported figure can climb within a single dwell. The consequence is the understatement of the true error identified in Section 3.4.2, where the tuned inflation rate is shown to indicate the fix state without tracking the error incurred under it.

The heading panel permits no such interpretation, as its reported standard deviation has a pooled Spearman correlation of 0.13 with the true heading error. The estimate itself remains sound, with a median heading error of 0.07 rad and a 99th percentile of 0.38 rad, since yaw is propagated from the IMU yaw rate at 20 Hz and corrected by the gated COG estimate, as specified in Section 3.4. This dead reckoning does not depend on the GNSS tier, so heading error stays small even under the degraded tiers in which position error grows to metre scale. The 99th percentile of heading error is roughly five times its median, compared with a factor of 49 for position. High position error therefore occurs without a corresponding rise in heading error, which remains uniformly small from step to step. A rank correlation needs variation in the quantity being ranked, so the low coefficient reflects that absence rather than a failure of the filter. The weak heading correlation nonetheless leaves every result in this chapter intact, since the response graded in Section 4.6.1 is measured against the position standard deviation and no claim conditions on the heading figure. A policy responding to the position standard deviation is therefore responding to a signal that ranks its own error, and the behavioural contrasts that follow can be read as responses to uncertainty rather than to noise.

4.3 The Evidential Actor's Uncertainty Channels

The evidential head presented in Section 3.5 reports a second uncertainty, the policy's confidence in its own actions. It warrants examination now that the covariance entering the policy as an observation has been shown to rank the error it describes. Under the NIG parameterisation a single forward pass should produce both per-action epistemic and aleatoric uncertainty (Table 2.1), and their sum is the total predictive uncertainty defined in Section 3.9.

4.3.1 Failure Prediction from the Head's Reported Uncertainty

The claim under test is that the head's reported uncertainty warns of failure, which holds only if that signal separates failing episodes from successful ones. The quantity on trial is therefore the head's total predictive uncertainty, the sum of the epistemic and aleatoric estimates given by Eq. (3.3) at each decision step. The EKF standard deviation for the two position axes, averaged into a single planar figure, is scored beside it as a reference measure rather than a candidate replacement, since it reports on the state estimate alone. It is therefore insensitive to the collisions and stalls arising from the manoeuvre itself, both of which remain possible under an accurate fix (Section 3.2.1), leaving the head's per-action uncertainty as the sole quantity able to anticipate them.

Both signals are condensed into a single score per episode, defined as the peak value the signal attains during that episode. These scores are computed over runs that follow the protocol set out in Section 4.1.3 with no threshold active, so every run terminates naturally with a definitive outcome label. For this analysis, the four categories short of success defined in Section 4.1.4 are collapsed into a single failure class. Separating failures from successes requires a mixture of the two, which restricts the analysis to the three conditions described in Section 4.1.2 where the GNSS tier varies within the episode. A candidate threshold is then swept over the scores, and each setting yields the two rates that characterise a live gate, the fraction of failing episodes caught and the fraction of successful parks needlessly aborted. Plotting one against the other gives Fig. 4.2, with one curve per arm and signal. The area under each curve (AUC, Section 4.1.5) then summarises it, so the reading rests on the whole sweep and not on a single operating point [82].

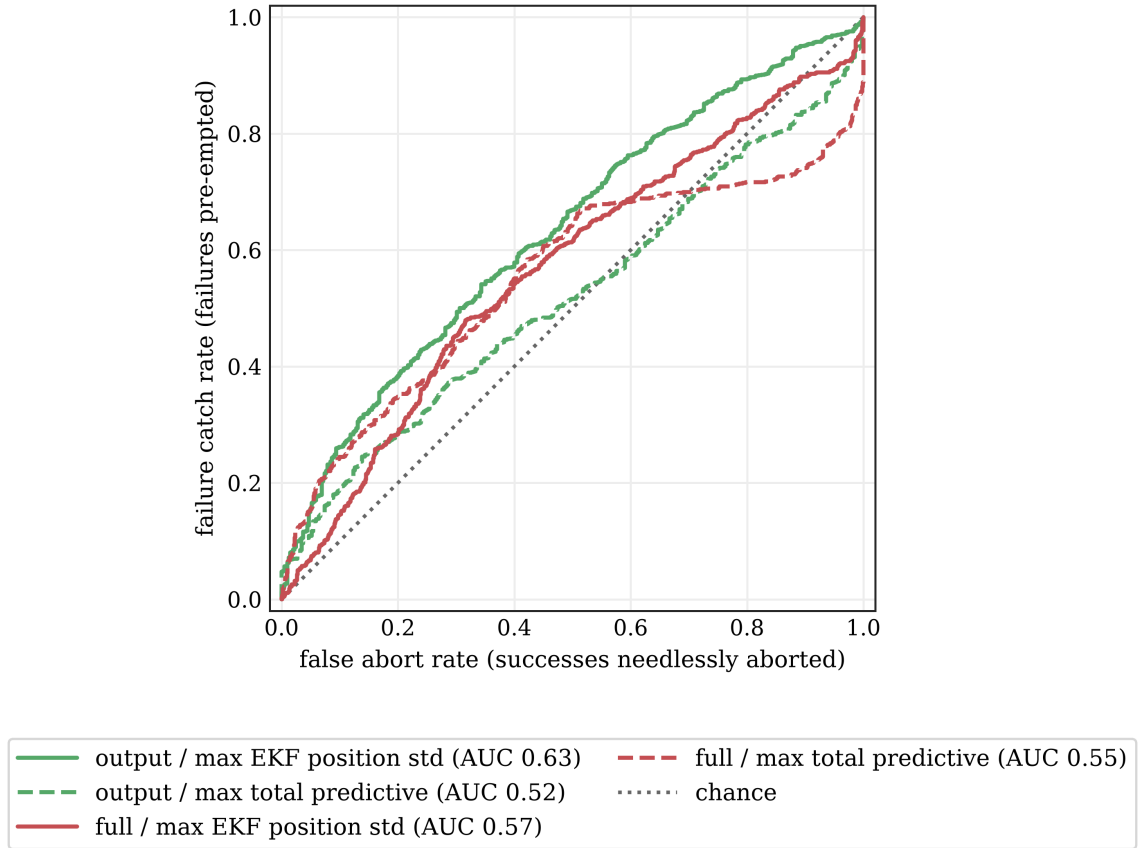


Figure 4.2 Pooled failure-prediction ROC curves over the three varying-GNSS conditions.

Pooled over the three seeds and conditions, the head's total predictive uncertainty is close to chance as a predictor of failure on both evidential arms. Its AUC is 0.55 for the full method (95% bootstrap interval $[0.52, 0.58]$) and 0.52 for the covariance-blind evidential arm ($[0.49, 0.55]$). The EKF position standard deviation, scored on the same episodes as a control, attains a higher value on both arms, at 0.57

for the full method ($[0.54, 0.60]$) and 0.63 for the covariance-blind evidential arm ($[0.60, 0.66]$), the highest recorded. Three of these four intervals lie wholly above the 0.5 baseline, so a per-step signal carries some information about failure and the prediction task is attainable in principle. The exception is the covariance-blind arm scored on its own uncertainty, which is indistinguishable from chance. For the full method, the interval on its own uncertainty overlaps that of its EKF control over most of its width, leaving no detectable difference between the head and the filter. None of the four approaches the level a usable warning would require, so the claim that the head's reported uncertainty warns of failure is unsupported.

A per-condition analysis of the same scores sharpens that reading. In the obstacle-free control, the maximum score of the full method separates failing from successful episodes with an AUC of 0.88 ($[0.86, 0.91]$). It falls, however, to 0.44 in both obstacle-bearing conditions, on $[0.393, 0.487]$ at the deployment anchor and $[0.394, 0.495]$ under the one-way drift. Both intervals lie below 0.5, so in these conditions the ordering is inverted and not merely absent, with successful episodes ranked above failing ones. The inversion arises because manoeuvring near stationary cars raises the head's reported uncertainty on successful episodes to, and slightly beyond, the level recorded on failing ones. The near-random pooled result therefore reflects a signal tracking obstacles, the same sensitivity that separates the evidential arms by collision rate in training (Section 3.8.3), and not an absent one.

4.3.2 The Coupling of the Two Channels

The failure of the two channels to separate is accounted for by the evidence parameter ν . That parameter remains inert for the same reason as the divergence and the collapse addressed in Section 3.5.2, all three tracing to the absence of a fixed regression target under policy sampling. The epistemic estimate is the aleatoric one divided by ν (Eq. (3.3)), so the two channels separate only when ν varies across states, and nothing supervises ν towards doing so.

Separation therefore demands a state-dependent ν , as epistemic uncertainty is the channel expected in Section 2.1.2 to rise with novelty and fall with familiarity, and the head was constructed to accommodate precisely that behaviour. The initialisation protocol specified in Section 3.5.1 establishes ν below unity, the regime in which the epistemic estimate exceeds the aleatoric one. The anchor (Eq. (3.6)) is then imposed on the remaining parameters alone, leaving ν unconstrained as training proceeds. The parameter nonetheless holds at a state-invariant value under evaluation, as an unconstrained parameter acquires state dependence only under a gradient that rewards it, and the actor-side objective provides none.

Such a gradient arises under the two placements surveyed in Section 2.2.2, supervised regression against a labelled target and value-side RL against the TD error. Neither is available on the actor, where a sampled action does not serve as its own label and no action-value is regressed from which a TD error could be formed. Either would resolve the collapse at the cost of abandoning the actor-side placement examined in response to the research gap identified in Section 2.3, and both are accordingly returned to as future work in Section 5.2. The deficiency is therefore one of placement and not of evidential methods in general, although a single signal reported under two names is the most the present head can offer. That signal meets none of the four desiderata for uncertainty in a reinforcement-learning agent established in Section 2.1.3, since each is defined over an epistemic estimate that moves independently of the aleatoric one.

4.3.3 The Head Against a Heteroscedastic Baseline

The collapse of the epistemic channel leaves the evidential head close to a heteroscedastic Gaussian head, emitting a per-state variance over the action distribution. The conventional SB3 PPO head (Section 2.1.1) structurally lacks that property, so the operating mechanism of the evidential head is best described as heteroscedastic. The two differ in what each can represent out of distribution. The NIG head has an evidence mechanism under which uncertainty can rise in new states, whereas the Gaussian standard deviation is a reward- and entropy-adjusted exploration parameter. The distinction remains a property of the method rather than a measured outcome, since no arm in the factorial disentangles the NIG parameterisation from state-conditioned variance. The control needed to separate the two mechanisms therefore falls outside the factorial and is proposed as future work in Section 5.2. Under deterministic evaluation (Section 4.1.3), the state-conditioned variance that survives the collapse bears on the driving solely through the mean action learned under it, a further reason the factorial cannot isolate its effect. The sole capability forfeited is therefore the independent failure warning tested in Section 4.3.1.

4.4 Task Performance Across Conditions

The pooled success rate and mean final position error for the four arms under the four conditions are shown in Fig. 4.3, with the success rates given numerically in Table 4.2 and the mean error alongside the median in Table 4.4. The full method achieves a success rate of 45.8% to 64.7% across the three trainable conditions, against 19.0% to 33.3% for the other three arms. The unseen irregular layout is the exception: every arm registers a 0% success rate, a finding examined among the limitations in Section 4.8.

All four arms are reported separately, so the contribution of each factor can be isolated by comparing two arms that differ only in that factor. Two of these pairings concern the covariance input and are quantified with bootstrap intervals. The first, set out next in Section 4.4.1, contrasts the full method with the output-uncertainty arm, thereby measuring the effect of supplying the covariance to the evidential head. The second, in Section 4.4.2, repeats that measurement on a standard Gaussian head by setting the input-uncertainty arm against vanilla PPO, thus establishing how far the gain depends on the architecture.

Table 4.2 Pooled success rate by condition and arm ($n = 600$ episodes per arm and condition).

Condition	Vanilla	Input	Output	Full
anchor_deployment	23.8	25.2	19.0	45.8
anchor_empty	33.3	30.7	28.5	64.7
gnss_degrade_one_way	27.3	28.5	20.0	52.5
ood_irregular	0.0	0.0	0.0	0.0

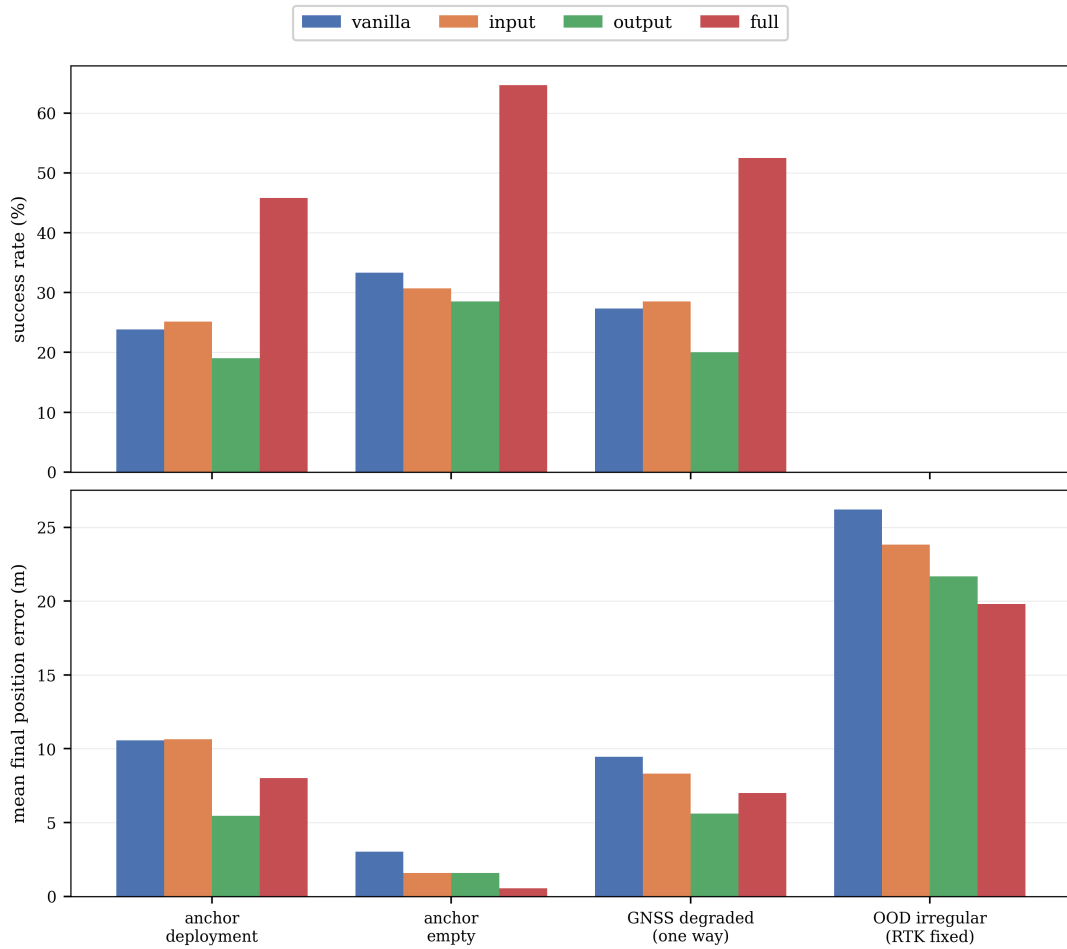


Figure 4.3 Pooled task performance ($n = 600$ episodes per bar).

4.4.1 The Covariance Contrast

The contrast examined below sets the evidential head trained with the covariance against the same head trained without it, under an identical curriculum, reward and set of seeds. Any difference in outcome is thereby attributable to that input alone.

The covariance raises parking success by 26.8 pp at the deployment anchor (95% CI [21.7, 32.0]) and by 26.8 to 36.2 pp across all trainable conditions. Those figures are drawn from Table 4.3, which reports every contrast under the success rate defined in Section 4.1.4, with the evidential head on the left and the standard Gaussian one on the right. Exposing the localisation covariance is therefore the sole modification that more than doubles the success rate of the evidential arm in every trainable condition, which constitutes the primary quantitative finding of the evaluation.

Table 4.3 Covariance contrasts in success rate (treatment minus control) with 95% bootstrap confidence intervals, pooled over $n = 600$ episodes per arm and condition.

Condition	Evidential head		Standard head	
	Δ success (pp)	95% CI	Δ success (pp)	95% CI
anchor_deployment	+26.8	[21.7, 32.0]	+1.3	[−3.5, 6.2]
anchor_empty	+36.2	[30.8, 41.3]	−2.7	[−8.0, 2.7]
gnss_degrade_one_way	+32.5	[27.3, 37.7]	+1.2	[−3.8, 6.3]
ood_irregular	0.0	[0.0, 0.0]	0.0	[0.0, 0.0]

The magnitude of the controlled effect is best assessed beside the end-to-end comparison. Vanilla PPO and the full method differ in both the covariance input and the actor head, so their success-rate gap confounds the two changes. That gap spans 22.0 to 31.3 pp across the same three trainable conditions. The margin of the full method over vanilla PPO is thus smaller than its margin over the output-uncertainty arm, which lacks only the covariance and trails the full method by 26.8 to 36.2 pp. The evidential head used without the covariance is therefore a liability, leaving that arm below the unmodified baseline. The covariance is the only indication available to that head of a deteriorating position estimate. In its absence the vehicle is near-uniformly cautious rather than cautious in proportion to the circumstances, an account elaborated in Section 4.6.1.

An arm parking more often should finish closer to the bay on average, yet the mean and median columns of Table 4.4 disagree on that point at the deployment anchor and under the one-way drift alike. At the anchor the full method records a higher mean than the output-uncertainty arm, 8.02 m against 5.45 m, but a lower median, 0.76 m against 1.23 m. The parked vehicles obstruct the approach, so a failed attempt halts metres from the bay rather than beside it. An arm that parks more often

is therefore left with failures further out that dominate its mean. The reversal is confined to that statistic, since the full method holds the lowest median under all three trainable conditions, the ordering that a higher success rate would predict.

Table 4.4 Pooled final position error over all episodes, successful and failed alike ($n = 600$ episodes per arm and condition).

Condition	Mean (m)				Median (m)			
	van.	in.	out.	full	van.	in.	out.	full
anchor_deployment	10.58	10.65	5.45	8.02	2.83	2.22	1.23	0.76
anchor_empty	3.02	1.58	1.59	0.54	0.99	0.80	0.93	0.41
gnss_degrade_one_way	9.45	8.33	5.61	6.99	1.70	1.46	1.18	0.64
ood_irregular	26.22	23.84	21.69	19.82	20.49	19.62	19.31	19.60

4.4.2 The Interaction: The Covariance Depends on the Evidential Head

The standard-head columns of Table 4.3 show the same covariance input conferring no benefit once the evidential head is removed, with success-rate contrasts spanning -2.7 to $+1.3$ pp and no interval excluding zero. The variance of that head is a single learned parameter, so the covariance influences only the action mean and leaves exploration unaltered. The evidential head instead conditions the variance on the input, allowing the same dimensions to yield the gain in the corresponding columns. The null result is therefore specific to this control and does not extend to Gaussian policies in general, since a heteroscedastic baseline could exploit the covariance, a possibility examined in Section 4.3.3.

The benefit of the covariance is thus contingent on an architecture capable of representing uncertainty-conditioned behaviour, and the central claim stated in Chapter 1 consequently rests on a measured interaction rather than a presumption.

4.5 Graceful Degradation

Graceful degradation denotes the vehicle operating more conservatively as localisation uncertainty rises. Observing such behaviour requires a fix state that deteriorates mid-manoeuve, which the live Markov chain described in Section 3.4.1 supplies. The transitions of that chain determine how long an episode remains poorly localised. The realised difficulty, the first of the two measures set out in Section 4.1.4, is therefore a property of the episode itself rather than of the condition from which it was drawn.

The 2400 anchor episodes, 600 per arm, are ranked by this realised difficulty, defined as the mean true position error over the episode, and partitioned into terciles

to form the three bands plotted in Fig. 4.4. Episodes with a mean error of at most 0.26 m are assigned to the low band, those between 0.26 m and 0.62 m to the mid band, and those above 0.62 m to the high band, in which the most severe error recorded is 5.4 m. Each band is thus populated by 800 episodes, between 163 and 263 per arm, and the proportion of these ending in a successful park is the quantity plotted.

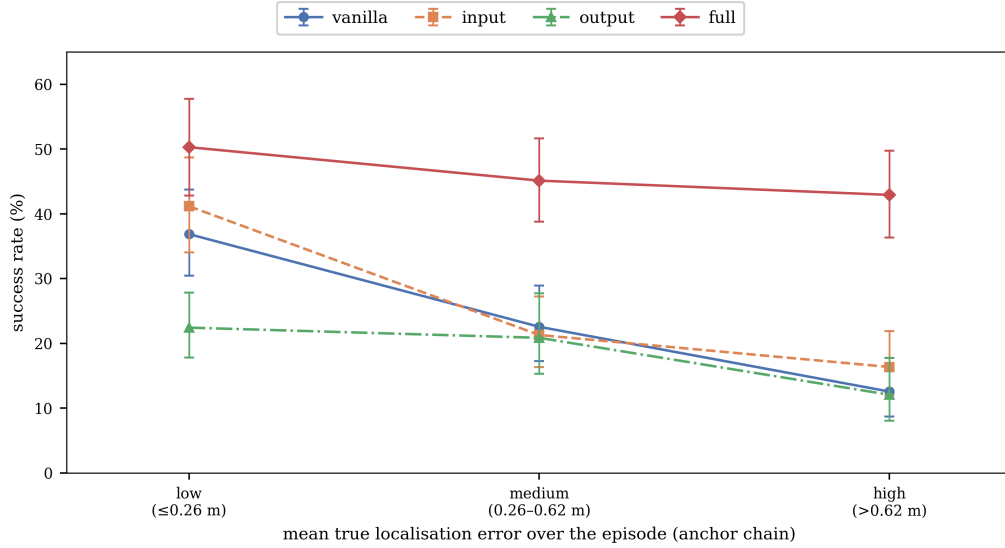


Figure 4.4 Pooled success rate under the live anchor GNSS chain (~ 800 episodes per band).

Across that range, the full method's success rate remains almost level, falling by 7.4 pp from 50.3% to 42.9%. The other three arms instead lose roughly half their success or more over the same bands. Vanilla PPO falls from 36.9% to 12.6%, the covariance-fed standard head from 41.2% to 16.4%, and the covariance-blind evidential arm from 22.4% to 12.1%. The widening gap also exceeds sampling uncertainty in the mid and high bands, where the CI of the full method, plotted as error bars in Fig. 4.4, lies clear of that of every other arm. In the low band, by contrast, it overlaps those of vanilla PPO and the covariance-fed standard head.

The full method alone holds its performance as localisation error increases. The two single-factor arms, one carrying the covariance without the evidential head and the other the head without the covariance, both regress to the level of vanilla PPO in the high band. Robustness to a deteriorating estimate is thus a property of the pairing, which is the same interaction Section 4.4.2 reports on the pooled outcomes. The mechanism behind that robustness remains open, as an outcome records whether an episode ended in a park and nothing of the conduct that led there. The driving itself is accordingly examined decision by decision in Section 4.6.

4.6 The Timing and Effort of the Response

An arm that conditions on localisation uncertainty should produce more conservative driving when the filter reports a poor estimate. The episodes are therefore grouped by reported uncertainty, the second of the two measures set out in Section 4.1.4. The covariance, as the sole indication of localisation quality available to the policy, is the appropriate benchmark for assessing a response. That response is quantified through the four per-step behavioural proxies defined alongside the measure.

A condition can support such a comparison only if its reported uncertainty varies, since a covariance that remains constant provides no basis against which behaviour can be graded. The irregular layout evaluated later in Section 4.8.1 is therefore excluded, because it holds RTK fixed throughout the episode. The three conditions in which the GNSS tier changes mid-episode are consequently pooled, giving $n = 1800$ episodes per arm.

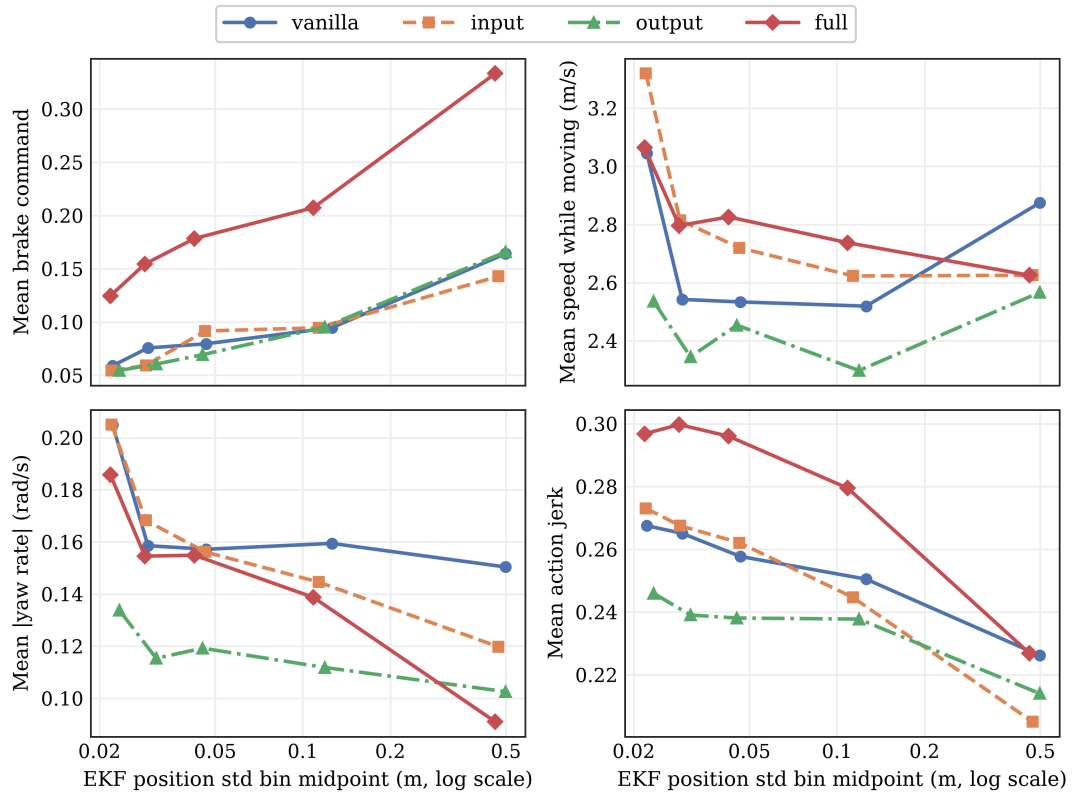


Figure 4.5 Per-episode behavioural proxies, pooled over the three conditions in which the GNSS tier changes mid-episode.

The magnitude of the behavioural response is read from Fig. 4.5, where each of the 1800 episodes of an arm is summarised by five averages taken across its steps. Those are the reported standard deviation and the four behavioural proxies defined in Section 4.1.4. The episodes are arranged by their reported uncertainty and divided into

five bands of equal population, separately within each condition to keep the mix of conditions constant from one band to the next. Dividing the pooled 1800 instead would concentrate the obstacle-free episodes in the highest band, where the absence of parked vehicles elevates the mean speed on its own, as the band figures reported later in Section 4.6.1 will show. Every proxy is then averaged a second time over the episodes each band contains, weighting every attempt equally regardless of its duration, to give the markers plotted. An average of this kind is raised as readily by a proxy that moves moderately in every episode of a band as by one that moves sharply in only a few. The consistency of the response is therefore established separately in Table 4.5. Its upper block correlates each proxy with the reported uncertainty through the rank statistic discussed in Section 4.1.5, yielding one value per proxy and arm. Its lower block averages the operating quantities those correlations leave unrecorded.

Table 4.5 Uncertainty-conditioning analysis pooled over the three conditions in which the GNSS tier changes mid-episode ($n = 1800$ episodes per arm).

Metric	van.	in.	out.	full
<i>Rank correlation (Spearman vs episode-mean position std, within condition)</i>				
Brake command	+0.265	+0.133	+0.223	+0.406
Speed while moving	−0.039	−0.209	+0.010	−0.127
Absolute yaw rate	−0.207	−0.353	−0.187	−0.446
Action jerk	−0.257	−0.437	−0.165	−0.432
<i>Operating level (pooled)</i>				
Mean speed while moving (m/s)	2.70	2.82	2.44	2.81
Mean brake command	0.09	0.09	0.09	0.20
Median final error (m)	1.46	1.10	1.09	0.54
Successful parking (%)	28.2	28.1	22.5	54.3

Pairing the two blocks matters because a coefficient conveys no information about the magnitude of the response. An arm may grade the driving by the reported uncertainty more sharply than a covariance-blind control and gain nothing in success for it, a case examined in Section 4.6.2. The analysis therefore begins by comparing the full method with the output-uncertainty arm, since that pair differs only in the covariance input and thus isolates its effect on the proposed architecture.

4.6.1 The Effort and Consistency of the Response

A vehicle driven more conservatively as uncertainty rises should brake harder and manoeuvre more gently, which across the bands means an increase in the brake

command and a decrease in speed, yaw rate and action jerk. Those trends are read against bands plotted at their midpoints on a logarithmic axis, since the reported standard deviation clusters near its RTK-fixed floor, as observed in Section 4.2.

Every one of those expectations is met by the full method. Its mean brake command, delivered on $[0, 1]$ by the action interface (Section 3.3.3), increases by a factor of 2.7 across the sweep, rising at every band from 0.125 in the lowest to 0.333 in the highest. The mean speed in motion falls over the same range from 3.06 m/s to 2.63 m/s, so the vehicle is both slowing and braking as the estimate deteriorates. Its manoeuvring softens alongside that response, as the mean absolute yaw rate more than halves, from 0.186 rad/s to 0.091 rad/s, ending below every other arm, while the mean action jerk falls from 0.297 to 0.227. The vehicle consequently approaches the bay more slowly and turns half as much once its own position can no longer be pinpointed.

The output-uncertainty arm shows almost none of that response over the same sweep. Three of its four proxies move in the conservative direction: the brake command rises across the bands from 0.054 to 0.166, the yaw rate declines from 0.134 rad/s to 0.103 rad/s, and the jerk falls from 0.246 to 0.214. Speed alone follows no such ordering, with its means confined between 2.30 m/s and 2.57 m/s and oscillating between bands rather than declining as the filter deteriorates. Braking, yaw rate and jerk therefore follow the trend expected of a conservative policy, yet none of the three reaches the magnitude at which the full method operates. The yaw rate and jerk of the output-uncertainty arm begin at the lowest levels of any arm and move little across the sweep, the former falling by a third as much as that of the full method.

The weakness of this response is a consequence of the ablation itself. Covariance is withheld from the output-uncertainty arm, so a degraded fix reaches its ten features as a shifted target offset, indistinguishable from a genuine one, and nothing marks the episode at the point where a slower approach must be selected. Since no episode can be singled out as warranting caution, the arm hedges across all of them, holding the lowest mean speed of any arm in Table 4.5 at 2.44 m/s. The safety of that hedge is not repaid at the bay, where the vehicle comes to rest a median 1.09 m from the centre against 0.54 m under the full method.

The response the full method shows across the bands counts for little if it appears in only some episodes. The rank correlations reported in the upper block of Table 4.5 therefore provide the second half of the evidence. The sign a coefficient must carry to qualify as conservative differs by proxy. A positive one on the brake command indicates firmer stopping effort as the filter deteriorates, while a negative one on the remaining three marks the slowing, straightening and smoothing that accompany it. Each coefficient is computed within a condition and averaged across the three. The full method carries the conservative sign throughout, with +0.406 on braking, -0.127 on

speed, -0.446 on yaw rate and -0.432 on jerk. The ordering it imposes is therefore consistent across every axis rather than the product of one proxy alone.

The coefficients of the output-uncertainty arm fall below those of the full method on all four proxies, by between 0.14 and 0.27 . The widest gaps are on jerk and yaw rate, at 0.27 and 0.26 , whereas speed carries the narrowest at 0.14 and is distinguished instead by its sign, its $+0.010$ running opposite to the full method's -0.127 . Differences of that size are comparable to the output-uncertainty coefficients themselves. The benefit conferred by the covariance is therefore distributed across all four proxies rather than concentrated in any one of them.

4.6.2 The Limits of Timing Alone

Under the full method the driving is both graded by uncertainty and more often successful than under any other arm. The two effects cannot be separated in an architecture that carries the covariance input and evidential head together. Establishing the value of grading alone therefore falls to the input-uncertainty arm described in Section 4.1.1, which retains the covariance without the evidential head. Its standard head lets the covariance reach the commanded action and not the variance around it, and its success rate is accordingly level with vanilla PPO in Section 4.4. The evidence available from the arm is thus confined to the grading of the driving rather than the outcome of the episode.

Three of the four proxies confirm that expectation. In Fig. 4.5, the input-uncertainty arm opens the sweep level with vanilla PPO in yaw rate and ahead of every other arm in speed (3.32 m/s). Its yaw rate and jerk decline consistently from those values, reaching 0.120 rad/s and 0.205 , whereas speed descends unevenly, flattening at the end of the sweep before finishing at 2.63 m/s. Those trends yield -0.209 on speed against vanilla PPO's -0.039 , -0.353 against -0.207 on yaw rate and -0.437 against -0.257 on jerk, the strongest speed and jerk correlations of any arm and the second strongest on yaw rate. None of that ordering reaches the outcome. The input-uncertainty arm parks on 28.1% of episodes and vanilla PPO on 28.2% , both far short of the full method's 54.3% (Table 4.5).

Braking, the fourth proxy, partly accounts for that result, since all three baselines fall well short on its correlation. Vanilla PPO reaches $+0.265$, the output-uncertainty arm $+0.223$ and the input-uncertainty arm $+0.133$, the lowest of the four, whereas the full method stands clear of them all at $+0.406$. The operating levels give the same picture: the input-uncertainty arm produces a mean brake command of 0.089 against 0.095 for vanilla PPO, 0.089 for the output-uncertainty arm and 0.200 for the full method. Under the input-uncertainty arm, the two readings describe a vehicle braking

more firmly on uncertain episodes than on clean ones, yet too gently throughout for that ordering to register in the parking outcome.

Vanilla PPO, the reference for that grading, receives no uncertainty signal at all, and its speed fluctuates between 2.52 m/s and 2.87 m/s over the upper four bands without settling into a decline. The vehicle is consequently as likely to be travelling quickly on a poorly localised episode as on a well localised one. A coefficient of -0.039 records that absence of ordering, and the vehicle comes to rest a median 1.46 m from the bay centre, the worst of the four arms by roughly a third of a metre.

Setting its mean speed beside that of the full method across the bands shows what the signal provides. The two begin the sweep at almost the same speed, 3.06 m/s for the full method and 3.04 m/s for vanilla PPO, and move from there in opposite directions. The full method sheds speed as the estimate deteriorates, finishing at 2.63 m/s, whereas vanilla PPO runs slower than it across the lower four bands and then accelerates in the highest to 2.87 m/s. The two therefore cross, and the uninformed arm carries most speed precisely where the estimate is least trustworthy.

A crossing of that kind establishes whether a cautious response is timed to the uncertainty without indicating how cautiously the vehicle drives, a distinction preserved in Table 4.5, the correlations addressing the former and the operating levels the latter. The correlations identify the covariance as the prerequisite for that timing. The output-uncertainty arm, despite its evidential head, grades speed, yaw rate and jerk no more sharply than vanilla PPO, which is equipped with neither component, whereas the input-uncertainty arm grades all three on a standard head. The head therefore contributes nothing to the timing once the input is withheld.

The operating levels divide the arms on a different principle. Three axes are graded by uncertainty under the input-uncertainty arm, yet the vehicle brakes no harder than under the arm with no uncertainty input at all, so timing on its own leaves the outcome unaltered. The effort is supplied by the evidential head once the covariance reaches it, and the two components are combined only in the full method, which constitutes the mechanism behind the benefits reported in Section 4.4.

4.6.3 The Origin of the Behaviour

The uncertainty-conditioned behaviour reported above was never specified in the objective against which the policy was trained. The reward formulation set out in Section 3.6 contains no uncertainty component, certainty incentive or covariance-weighted penalty, and the standard deviations enter only through the observation. The vehicle nevertheless adjusts all four proxies with the reported position uncertainty, establishing the behaviour as emergent. That adjustment earns the terminal payment (Section 3.6.3) more often, so it was learned for the outcome it

produces rather than for any intrinsic merit. Its discovery depended in turn on the curriculum (Section 3.7), which kept the uncertainty channel active from the first stage and thereby left the strategy within reach throughout training.

Slowing while the estimate is poor and resuming once it recovers requires more than an ordering of speed against the reported uncertainty, which the covariance-fed standard head supplies through the commanded mean alone. That ordering is not held where it matters, since the speed oscillates across the bands under the covariance-blind evidential arm and, in the highest band, rises under vanilla PPO and stops falling under the covariance-fed standard head, whereas the speed under the full method falls to its lowest there. Sustaining the reduction across that band calls for an action distribution able to widen with the reported uncertainty during training, a capacity available only to the evidential head and exercised only when the covariance reaches it. The finding accords with that reported in Section 4.4.2, where the covariance raises success on that head alone.

4.7 Robustness to Seed Selection

Training each arm under three seeds, as established in Section 4.1, produces a second body of evidence that the pooled means leave unread. Every CI reported so far is obtained by resampling episodes, as described in Section 4.1.5, while the three policies trained for that arm remain fixed. Such an interval therefore bounds the precision of the arm’s estimate without revealing the variability produced by the same training procedure across seeds. Separating the three runs recovers that second spread.

Table 4.6 Per-seed success ranges for the evidential-head pair and the matched-seed covariance contrast ($n = 3$ seeds; 200 episodes per seed).

Condition	Success [min, max] (%)		Matched-seed Δ (pp)	
	full	output	min	max
anchor_deployment	[39.5, 51.0]	[17.0, 22.0]	+21.5	+34.0
anchor_empty	[48.0, 78.5]	[19.5, 34.0]	+16.0	+59.0
gnss_degrade_one_way	[40.0, 62.5]	[16.5, 23.0]	+17.0	+46.0

At $n = 3$ seeds, the strongest claim available is unanimity, which the per-seed record provides for the covariance contrast measured in Section 4.4.1. Each full-method policy is compared with the output-uncertainty policy trained from the same seed, forming a matched pair. The three seeds combined with the three conditions in which any arm achieves a successful park give nine such pairs. The contrast is positive in all nine, with margins ranging from +16.0 to +59.0 pp (Table 4.6). Discarding the seed matching leaves the advantage intact, since the weakest

full-method policy surpasses the strongest output-uncertainty policy in every condition, by +14.0 pp at the narrowest. No ordering of the six policies therefore leaves the advantage attributable to a fortunate draw, the failure mode the multi-seed protocol is designed to rule out.

The unanimity of the matched-seed contrast does not imply narrow per-seed ranges within the individual arms. The full method’s range overlaps those of both standard-head arms on the one-way drift, between 40.0% and 41.5%, as shown by the whiskers in Fig. 4.6 and the per-seed scores in Appendix E. The overlap is a property of individual runs, not of the pooled estimate on which the comparison rests. Every finding in this chapter is reported on that basis, as a pooled average over all episodes from all three seeds (Section 4.1) rather than as the result of any single run.

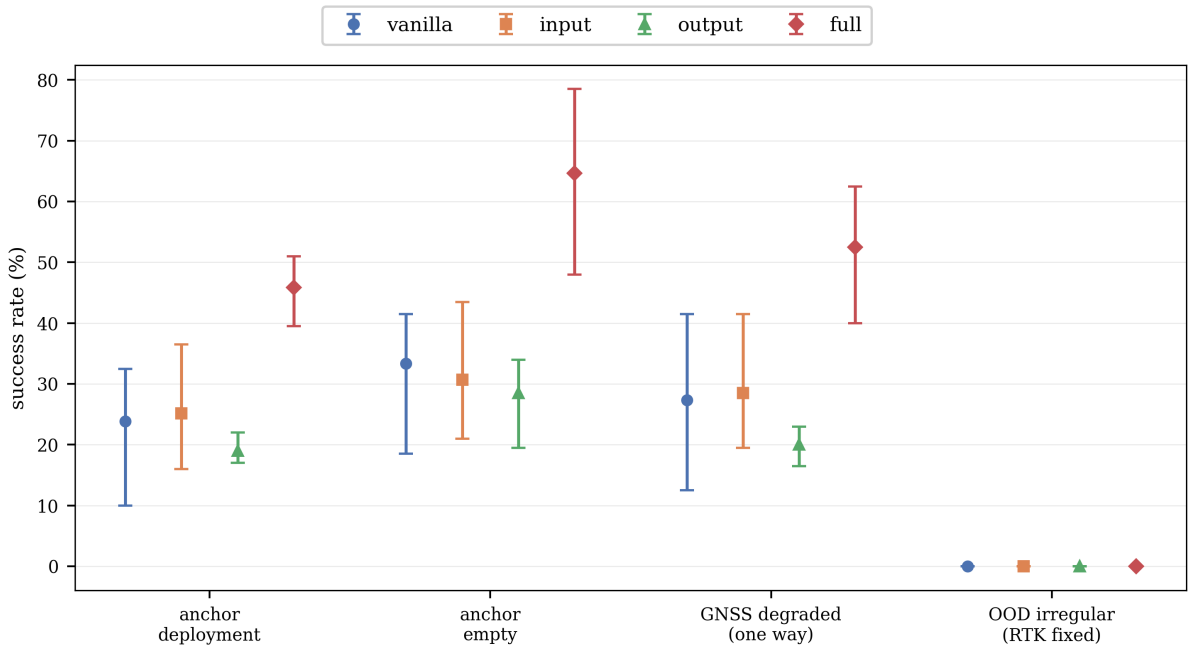


Figure 4.6 Pooled success rate, with whiskers spanning the per-seed minimum and maximum.

The standard Gaussian head yields no comparable consistency. The contrast between the input-uncertainty arm and vanilla PPO varies in sign across seeds in all three trainable conditions, spanning -22.0 to $+29.0$ pp on the one-way drift. The covariance observation therefore raises the success rate on one seed and lowers it on another. The per-seed record thus reproduces the pooled inertness reported in Section 4.4.2. It is consistent with the asymmetry lying in the head rather than in the seeds, since the identical covariance input is decisive for the evidential head while remaining directionless for the standard one.

The seed evidence accordingly strengthens the architectural claim rather than passing a robustness check. The covariance observation is of value only in a head able to represent what it encodes, and that dependence is reproduced on every seed.

4.8 Limitations

Three limitations bound the results reported above. The mechanism behind each is set out below, with a precise statement of what it does and does not invalidate.

4.8.1 No Successful Park on Out-of-Distribution Geometry

No arm records a success on the unseen irregular layout across 600 pooled episodes apiece (Table 4.2). Each phase of the curriculum described in Section 3.7 is conducted on the rectangular lot, producing a policy linked to a single geometry, while the irregular lot is held out as the measure of generalisation required in Section 2.1.6.

A complete failure on unseen geometry raises the possibility that no transferable capability was acquired and that the policy merely memorised the lot on which it was trained. Every training episode commences at a single spawn pose, as discussed in Section 3.2.2, so every trajectory originates from that point, which leaves the environment design unable to rule memorisation out. Two findings on the trained lot nonetheless weigh against memorisation of the lot and its bays, although neither addresses memorisation of the routes from that pose. The observation defined in Section 3.3 is egocentric and lacks absolute position, so no representation of the entire lot can be retained. The second concerns how often each target is practised. The deployment anchor draws from 47 bays, of which the early phases described in Section 3.7 use only 15 before the remainder become eligible. Memorisation would leave that heavily practised group markedly the more reliable of the two, whereas it attains a success rate of only 0.49 against 0.44 for the bays introduced later.

A skill that falters in unfamiliar geometry can fail at more than one stage, and a success rate of zero provides no indication of which. The vehicle may reach the bay and execute the manoeuvre poorly, or it may never arrive at all. Under the full method at the deployment anchor, the median final distance from the target is 0.76 m and 73% of episodes end within 5 m, whereas across the remaining 27% that median is 24.2 m. Those proportions reverse on the irregular layout, where the median over all episodes extends to 19.6 m and fewer than 8% end within 10 m, at step counts and speeds matching the anchor. A failed manoeuvre would leave the vehicle a few metres from a bay it had reached, yet hardly any episode ends at such a distance. The closest approach to any bay is also confined to the clusters nearest the spawn, so the vehicle stops early, before covering much of the new layout. The failure therefore lies in traversing the lot rather than in the final manoeuvre. Every trained route starts from the rectangular lot's single spawn pose, and none exists on the irregular layout, which is consistent with that failure. All four arms fail in this manner, so the comparison among them remains unaffected.

The training set contained a single layout, so the result reported above cannot speak to training on a distribution of layouts, which might yield a policy that does transfer. The layout generator that produced this probe could supply that varied training set, an experiment deferred to Section 5.2 because it calls for a fresh training campaign on every arm. The contribution of this work is therefore uncertainty-conditioned control on the trained geometry class, not zero-shot generalisation to unseen layouts.

4.8.2 Total Uncertainty Does Not Separate Outcomes

The safety layer specified in Section 3.9 brings the vehicle to a stop and flags a handover once the total predictive uncertainty crosses a fixed threshold. Its activations follow that signal alone and are therefore blind to the outcome an episode is heading for. All rates in this subsection are derived from replaying the threshold over the one-way drift episodes run under the protocol established in Section 4.1.3. The rule was inactive over those episodes, and a degradation is guaranteed in them rather than left to the Markov chain (Section 4.1.2). Each episode consequently carries both the moment the rule would have fired and the outcome it actually reached, a combination that a live intervention disrupts by terminating the run.

The total predictive uncertainty rises over the course of an episode under every assessed condition, including those without degradation, and it rises for the same reason in each. The signal tracks the signature of a slow, heavily braked final approach to the bay rather than any degradation. The one-way drift adds a degradation-driven rise on top of that approach signature, as the localisation estimate moves from confident to uncertain within a single episode.

A useful handover should fire on the episodes heading for failure and spare the rest. The threshold is instead crossed in 56.2% of the episodes that ended in a successful park, marginally more often than the 50.5% reached on those that did not. The two rates stand 5.7 points apart, so the rule surrenders roughly one successful park for every failure it averts, which is the 0.44 gate AUC this condition records in Section 4.3.1 restated as a firing rate. Moving the threshold cannot repair this, since the ordering is inverted and the evidence parameter ν holds at a constant rather than tracking familiarity (Section 4.3.2). The obstacle sensitivity behind the lower training collision rate of the evidential arms (Section 3.8.3) thus raises the signal on successful manoeuvres as readily as on failing ones.

A rule that waits for the threshold to be exceeded on several consecutive decisions widens that separation without resolving it. The 600 one-way drift episodes of the full method were replayed at the threshold fixed in Section 3.9, now demanding ten successive decisions above it, approximately 2 s, before a handover is issued. The

firing rate falls from 56.2% to 10.5% on episodes that would have parked and from 50.5% to 26.7% on those that would have failed. The larger fall is the desirable one only if a surrendered park costs more than a collision, which it does not, so the delay improves the rule on the side that matters less. Selectivity must therefore come from a signal that separates the two outcomes, or from a criterion reading more than just the uncertainty level, and both routes are set out as future work in Section 5.2.

4.8.3 Scope Limitations

Every result in this chapter was obtained in simulation. The real-vehicle pipeline is implemented, tested and structurally identical to the simulation path, and it covers the safety wrapper specified in Section 3.9 along with the calibration and pre-deployment sequence set out in Appendix B. None of it has ever been executed on hardware. The correspondence between the two paths is nonetheless substantial, since the observation is built by the same code from the same message types on both, as recorded in the interface listing given in Appendix B.2. The trained weights are therefore compatible with the vehicle interface by construction, although whether they would perform without retraining remains untested.

A second bound arises from the hyperparameter suite, which is held fixed across all four arms rather than tuned for each, as argued in Section 3.8.1. The arms are therefore ranked at a single operating point rather than at the best one available to each, and any arm poorly served by the fixed values is understated. Averaging over three seeds reduces that effect without removing it.

A third bound follows from the forward-only interface adopted in Section 3.2.1. The vehicle cannot withdraw from a misaligned approach, so the alignment must be established in a single trajectory under every arm. The absolute success rates are consequently likely to understate what a vehicle able to reverse would achieve, although the cost is shared by all four arms and leaves their contrasts unaffected.

A fourth bound concerns what the uncertainty-conditioned behaviour measured in Section 4.6 achieves. The gain is one of accuracy rather than of safety. Conditioning on the covariance raises success and lands the vehicle closer to the bay centre, yet the collision rate does not fall with it. Over the three conditions from which that behaviour was read it stands at 15.8% for the full method against 15.6% for the output-uncertainty arm. The additional parks are drawn instead from near-misses and stuck episodes, which decline from 38.1% and 23.2% to 21.7% and 8.1% of episodes. The behaviour therefore yields more precise parking and not safer driving.

4.9 Summary

A localisation pipeline that indicates its own error honestly, objective O1 (Section 1.3), is demonstrated by the calibration analysis reported in Section 4.2. Across the three conditions in which the GNSS tier changes mid-episode, the reported position standard deviation ranks the actual error at a Spearman correlation of 0.54.

Training to convergence, the first of the three counts on which objective O2 is assessed, was obtained on every seed once the variance-collapse pathology identified in Section 3.5 was mitigated, as confirmed by the training curves plotted in Fig. 3.8. The second, the behaviour of the uncertainty reported at runtime, is unsupported by the failure-prediction analysis presented in Section 4.3.1, where the head’s total uncertainty warns of failure only on the obstacle-free control. The third, separability, is refuted by the channel-coupling analysis set out in Section 4.3.2, where the evidence parameter ν holds constant across states, leaving the epistemic estimate a fixed multiple of the aleatoric one. The head therefore returns one signal under two names.

The 2×2 ablation reported in Section 4.4, aimed at the gap identified in Section 2.3, crosses a single factor pair, as objective O3 requires. Supplying the EKF covariance to the evidential actor raises parking success across all trainable conditions, with a gain of +26.8 pp on the deployment anchor and up to +36.2 pp on the empty layout. The advantage further persists through the seed draw, since all nine matched-seed comparisons on the per-seed record given in Section 4.7 favour the full method over the covariance-blind evidential arm, by +16.0 pp in the worst case. The benefit is lost without the evidential head, since the same covariance input on a standard Gaussian head shifts success by between -2.7 and $+1.3$ pp, a range that includes zero. The contribution is consequently a synergy, beneficial with neither component alone, and arises most plausibly through the state-conditioned action variance of the evidential actor, which no arm compares against another heteroscedastic head (Section 4.3.3).

The synergy extends to the driving itself, the subject of objective O4, and is demonstrated through the behavioural evidence presented in Section 4.6. Under the full method the vehicle slows, brakes harder, turns less sharply and changes its commands less abruptly as the reported uncertainty rises. The operating level is more conservative throughout as well, with an average brake command of 0.20, while all three baselines remain near 0.09. No part of the reward pays for any of it, so the behaviour was learned only because it leads to more successful parking.

Two of the three limitations examined, with their mechanisms, in Section 4.8 bound the scope of these findings without weakening their basis: the policies fail to transfer to the withheld irregular layout, and the total predictive uncertainty is too coarse to separate the episodes that end in a park from those that do not.

5 Conclusions and Future Work

The evidence in Chapter 4 affirmatively answers the research question posed in Section 1.1, within the two bounds recorded in the evaluation summary (Section 4.9).

5.1 Conclusions

5.1.1 The Objectives Revisited

The covariance computed by the pipeline (Section 3.4) is informative about that pipeline’s own error, satisfying objective O1. The calibration coefficient given in Section 4.2 licenses an ordering rather than a magnitude, in that a larger reported uncertainty marks a genuinely worse estimate, and the policy acts on that ordering alone. The conclusion is qualified by the coefficient being scored only where the fix state varies, so the ordering holds over the three conditions run on the trained lot and not over the unseen layout, which is held at RTK fixed throughout (Section 4.1.2).

A trustworthy input is only half of the architecture, and objective O2 is met on the output side for stability alone. The convergence recorded in Section 4.9 was obtained under the prior-anchoring regulariser introduced in Section 3.5, where the collapsed head is repaired against fixed references rather than against a label. An evidential action head can therefore be trained without a ground-truth action target, the first of the two questions raised in Section 2.3, since, once the likelihood is replaced by the PPO objective, the target presumed by the evidential loss is required by the regulariser alone. The second, over the usability of that head’s uncertainty output, is left unanswered by that stability. The epistemic estimate of a single-head NIG actor is the aleatoric estimate divided by the evidence parameter, and no supervisory signal for that divisor is available in model-free reinforcement learning, since a sampled action does not serve as its own label. The constant value identified in Section 4.3.2 is the consequence, and it bounds the use of the uncertainty outputs as a property of the parameterisation rather than of this particular training run.

The bound applies to those outputs alone and leaves the value of the two architectural layers open, which is settled by the two-by-two ablation designed in Section 3.8, as objective O3 requires. The design converts the gain reported in Section 4.9 into an attribution, since the same covariance input supplied to a standard Gaussian head moves success by an amount indistinguishable from zero. The benefit is thus carried by neither layer on its own but by the pairing, the interaction established in Section 4.4.2, and it should not be expected from the covariance input without an actor able to vary the spread of its action distribution from state to state.

The vehicle brakes in time with the live EKF standard deviation, which is the uncertainty-conditioned behaviour stipulated in objective O4 as the source of the gain. The timing is caused by the three covariance dimensions and by nothing else, since the covariance-blind arm shares every other input and grades the driving far more weakly. The reward formulation given in Section 3.6 contains no uncertainty term either, so a policy rewarded for parking alone acquires caution as an instrumental strategy where the observation supports it, the emergence argued in Section 4.6.3.

5.1.2 Contributions

Meeting the objectives revisited in Section 5.1.1 yields five contributions, ordered by dependence from the architecture to the evaluation on which its claims rest.

1. The first use of raw EKF covariance features as policy observations for ego-vehicle parking control, closing the input-side absence identified in Section 2.3. On the evidence of the O3 ablation the benefit is causal, although it holds only when the features are paired with an uncertainty-capable head.
2. The first actor-side NIG evidential head for a continuous-control policy-gradient agent, closing the output-side absence against the prior placements tabulated in Table 2.2. It is the uncertainty-capable output that the covariance benefit requires. Under this placement the output variance can collapse, a failure diagnosed against the DER limitations catalogued in Section 2.1.3 and mitigated by the three-part fix designed in Section 3.5.
3. Uncertainty-aware driving that emerges from training alone. The reward contains no uncertainty term and no slow-down is scripted, yet the vehicle brakes more, steers more smoothly and moves less abruptly as EKF uncertainty rises (Section 4.6).
4. A negative result on the output side and its cause. The epistemic and aleatoric channels of a single-head NIG actor conflate because the evidence parameter is unsupervised in model-free reinforcement learning (Section 4.3.2).
5. A reproducible evaluation asset, set out in Section 4.1, comprising a de-confounded four-condition sweep under controlled GNSS degradation in CARLA, declarative condition configurations, a fixed three-seed protocol and a deterministic analysis suite, with its reconstruction from a clean checkout documented in Appendix D.

5.2 Future Work

Every item below follows from a finding in Chapter 4 or a bound on its scope, ordered from what the present system can already support to what would require rebuilding it.

1. *Real-world validation.* The deployment pipeline is tested but never run on hardware, the bound recorded in Section 4.8.3, which leaves timing, actuation response and receiver behaviour unexamined. The staged validation documented in Appendix B is therefore the most direct extension available, on the parking area introduced in Section 1.2. Validation of the simulated noise levels against logged RTK traces from the physical receiver belongs to the same campaign. The hand-tuned fix-state transition matrix specified in Section 3.4 would then be replaced by one estimated from logged sequences, with the correction stream interrupted to induce the degraded tiers rather than left to chance.
2. *A separable epistemic channel.* The collapse identified in Section 4.3.2 arises because the evidence parameter is unconstrained in this head, and an unfamiliar state could still be recognised under a second head in which that parameter is tied to the training distribution. The most straightforward approach is to make the parameter a function of a density over the latent state, the posterior-network construction set out in Section 2.2.2 [63], whereas the same information is recoverable from the discrepancies among members of an ensemble of evidential heads [19], at the per-decision cost established in Section 2.1.2. Either would be trained under the current curriculum and the failure-gate analysis repeated. The safety layer would benefit accordingly, since a varying evidence parameter would let the rule defined in Section 3.9 tell an unfamiliar state from intrinsic noise, restoring the division drawn in Table 2.1 by reserving the handover for novel states and meeting a high aleatoric reading by slowing and turning less sharply.
3. *The heteroscedastic-Gaussian control arm.* In the fifth arm, proposed beyond the factorial in Sections 3.8 and 4.3.3, the single learned constant of the standard Gaussian head is replaced by a log standard deviation computed from the state, after the parameterisation of soft actor-critic [15]. Supplying the covariance input to that head would apportion the evidential head's advantage between the state-conditioned variance and the NIG evidence mechanism. No new apparatus is required for that comparison, as the arm is trained under the current curriculum and assessed on the same protocol, at one training run per seed.
4. *The cost axis as a trade curve.* Table 2.2 separates, in its passes column, the single-pass estimators from those requiring M ensemble members, and the saving recorded in that column motivates the head adopted in Section 3.5. A

count of passes nonetheless prices the estimators without ranking them, since the task performance purchased at each price is left undetermined. A sweep over ensemble size under the existing curriculum and evaluation protocol would recover that dimension, converting the column into a trade curve of success rate against per-decision latency, measured on the deployment hardware relative to the 200 ms budget fixed in Section 3.3.3. Each member carries a training run per seed and multiplies the compute of the ablation set out in Section 3.8 by M , making the sweep a campaign of its own rather than a further cell in the factorial.

5. *Out-of-distribution geometry and multiple spawn points.* Generalisation beyond the single geometry used for training would require a training distribution over layouts, given the complete failure on the unseen lot reported in Section 4.8.1. The layout generator that produced it already yields such a training set on the existing stack. Multiple spawn points would also replace the fixed pose described in Section 3.2.2, from which one rehearsed route can be learned in place of a transferable approach, a memorisation left open by the evaluation. Such training would target route-finding rather than parking, as the vehicle fails while traversing the unseen lot rather than during the final manoeuvre. The experiment would settle whether the covariance-conditioning benefit survives both changes.

5.3 Final Remarks

The problem set out in Chapter 1 was a vehicle required to park inside the markings of a single bay while its own position estimate varied in accuracy by two orders of magnitude. Supplying that estimate’s covariance to the policy, with an actor equipped to express uncertainty in its own actions, raised parking success on every trainable condition, through a response for which no reward term was specified.

Neither side of the architecture is specific to the setting that produced it. The input side requires only the covariance that any vehicle running an EKF already computes, and nothing about it depends on the degradation being a GNSS one, since a filter fusing any sensor set publishes a confidence signal in the same form. The output side is equally portable if the functional property is state-dependent action variance rather than the NIG parameterisation itself, which Section 4.3.3 argues but the factorial does not isolate. Any model conditioning its variance on the state would then satisfy it, a heteroscedastic Gaussian included.

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Appendix A Sensor Noise Models: Datasheet Derivations

This appendix records the calculations supporting the condensed noise parameters (Table 3.4), linking each simulated noise magnitude to its original manufacturer specification. The two rows marked † in that table have no corresponding datasheet figure, so the basis of the float-tier estimate and the rationale for the degraded tier are addressed separately in Appendix A.1. Every injection method in this appendix is governed by two master flags (`enable_gnss_noise`, `enable_imu_noise`). Noise on real hardware originates from the sensors themselves, so both flags are set to false for deployment and each mechanism reduces to a pass-through, as specified in Appendix B. The following derivations therefore describe only the simulated configuration, sensor by sensor.

A.1 RTK-GNSS: u-blox ZED-F9P

The GNSS datasheet [1] expresses horizontal position accuracy as CEP, so recovering a spread parameter requires a distributional assumption. Under a circularly symmetric zero-mean Gaussian with per-axis standard deviation σ_{axis} , the radial error follows a Rayleigh distribution characterised by two standard quantities. Its median is $\text{CEP} = \sqrt{2 \ln 2} \sigma_{\text{axis}} \approx 1.1774 \sigma_{\text{axis}}$, whereas its root-mean-square radial error, denoted by Distance Root Mean Square (DRMS), is $\sqrt{2} \sigma_{\text{axis}}$. Dividing the datasheet value by $\sqrt{\ln 2}$ therefore recovers the latter:

$$\text{DRMS} = \frac{\text{CEP}}{0.8326} = \sqrt{2} \sigma_{\text{axis}} \quad (\text{A.1})$$

where the divisor $\sqrt{\ln 2} \approx 0.8326$ is analytic rather than an engineering rule of thumb, so the conversion is exact wherever the circular assumption holds. The resulting DRMS is radial, yet it is adopted as the standard deviation of the zero-mean Gaussian injected on each axis independently. Substituting a radial figure for a per-axis one is deliberate and raises the per-axis standard deviation by a factor of $\sqrt{2}$ over the strict conversion. The factor reflects the datasheet figures having been measured in static open-sky conditions, free of the reflectors found in a typical car park (Section 3.4.1). The measurement was also more favourable than open sky alone suggests, since the quoted accuracy was obtained over 24 hours of static operation [1, Table 3] and holds only while the carrier-phase ambiguities remain resolved and the correction stream is uninterrupted. Both conditions are disrupted in a car park by the multipath and non-line-of-sight mechanisms described in Section 2.1.4, so a factor of $\sqrt{2}$ is a modest

allowance rather than a conservative one. The four tiers and their origins are given in Table A.1.

Table A.1 The four GNSS tiers and their noise derivations.

Tier	Derivation	Value	Provenance
rtk_fixed	$0.010/0.8326 = 0.0120$ m, raised	0.020 m	Datasheet [1, Table 3]
rtk_float	$0.300/0.8326 = 0.360$ m	0.360 m	Measured decimetre regime [34]
standalone	$1.5/0.8326 = 1.802$ m	1.802 m	Datasheet [1, Table 3]
degraded	Beyond specification	5.0 m	Stress condition

Taken in the order of the quality ladder in Section 2.1.4, from most accurate to least, the tiers of Table A.1 begin with the fixed tier, at the datasheet value of 0.010 m CEP plus one part per million of baseline length. The baseline-dependent term contributes only 0.0001 m over the 100 m typical of a parking lot, so it is disregarded and the CEP converts to a floor of 0.0120 m. That floor is optimistic, as the datasheet qualifies its figure by excluding antenna phase-centre offset errors [1, Table 3], which a real installation incurs alongside the multipath interference that surrounding vehicles and structures produce [32]. Quantifying either demands an antenna position measured on a physical platform and a surveyed site, neither of which this work instruments, as the mount poses listed later in Appendix B.2 remain site-specific placeholders. The floor is therefore raised conservatively to 0.020 m. One rung lower, no datasheet quotes a float figure, since float accuracy varies with atmospheric conditions and baseline length. The tier instead converts a 0.300 m CEP value from the decimetre regime that Reid et al. [34] measured for float solutions on highways (Section 3.4). The standalone tier is again taken from the datasheet, at 1.5 m CEP for an uncorrected fix, and marks the rung at which the error exceeds the geometry of the task. A 2.5 m bay spans only approximately $1.4 \sigma_{\text{axis}}$ of the resulting error, so a fix landing outside the target bay is a routine occurrence rather than a rare one. The degraded tier lies beyond the datasheet by construction, serving as the stress condition introduced in Section 3.4, and its 5.0 m figure models conditions outside the receiver's operational range under severe multipath and non-line-of-sight reception [32].

Injecting these tiers (Section 3.4.1) requires one further conversion, as their values are metric whereas the simulated receiver publishes latitude and longitude, and a metric displacement corresponds to a different angular offset at every location on the globe. The conversion is anchored to a local datum, derived from the geolocation of the spawn point and re-latched at every reset, and applies the flat-earth value of 111 320 m per degree on both axes across the parking area. Such a constant is exact only at the equator and for north-south displacement, so a small scale error enters the injected offset. The discrepancy is nonetheless immaterial to the filter, because the

EKF configured in Section 3.4.2 consumes the metric odometry output of the relay node together with a covariance stamped in metres, both of which remain correct regardless of the degree conversion.

The tier governs more than position, since velocity noise scales in the same way. At the fixed tier, the Doppler velocity standard deviation is set to the datasheet accuracy of 0.05 m/s [1, Table 1]. It is then multiplied by the same per-tier factors as the position noise (1, 18, 90 and 250 times the base), producing values of 0.05, 0.90, 4.5 and 12.5 m/s, respectively. The speed estimate therefore coarsens with the fix state.

The Doppler standard deviation also governs the third observable, the direction of travel, which is derived from position rather than from a moving-base receiver. It is obtained as COG from the direction of successive fixes, with the angle calculated as the two-argument arctangent of the per-callback displacement between the fixes before noise is injected, and Gaussian error is then added to it. Its variance, injected and reported alike, follows from that standard deviation and the current speed:

$$\text{var}(\psi_{\text{COG}}) = \left(\frac{\sigma_{\text{doppler}}}{\max(v, 0.1)} \right)^2 \quad (\text{A.2})$$

where σ_{doppler} denotes the tier's Doppler standard deviation and v is the current speed, floored at 0.1 m/s to keep the expression bounded near standstill. Scaling the variance by speed reproduces the precision of a Doppler-derived COG, which holds while the antenna is in motion and degrades as the vehicle slows. At low enough speeds the heading becomes unreliable altogether rather than merely imprecise, so three conditions gate publication. The displacement over the callback must exceed 0.05 m, no IMU ZUPT may be active, and the candidate variance must fall below $\pi^2/3$. An angle failing any one of them is discarded, and the filter runs without a heading until the next usable one arrives.

The three observables above take their magnitudes from the tier, whereas the error anisotropy and the dropping of fixes are randomised independently of it. Section 3.4 establishes what each reproduces, leaving the sampling that implements them to be set out alongside the tiers themselves. The anisotropy is governed by a ratio τ , drawn uniformly from [1.0, 1.5] at the start of an episode and divided between the axes as $\sqrt{\tau}$ in a randomly chosen direction and $1/\sqrt{\tau}$ in the other. Splitting it by the square root gives two factors with a product of one, so the tier's sigma is preserved as the geometric mean and the error changes shape without growing in magnitude. Dropped fixes are governed by a second and far shorter cycle, a Bernoulli draw with probability 0.02 resolved at every callback, which at 20 Hz costs a fix every 2.5 seconds on average. The rate leaves the fix available across the great majority of callbacks, yet interrupts a manoeuvre often enough for the condition to be encountered repeatedly during training.

A.2 IMU: VectorNav VN-100

The IMU datasheet [78] expresses noise as a spectral density, so a conversion is needed before either channel can be given a variance. For a white-noise process of density S sampled at a rate f , the variance of a single observation is:

$$\sigma_w^2 = S^2 f \quad (\text{A.3})$$

Evaluating Eq. (A.3) at the 20 Hz pipeline rate converts each datasheet density into the variance of one reading, starting from the gyroscope density of $0.0035^\circ/\text{s}/\sqrt{\text{Hz}}$:

$$\begin{aligned} S_{\text{gyro}} &= 0.0035^\circ/\text{s}/\sqrt{\text{Hz}} = 6.10865 \times 10^{-5} \text{ rad/s}/\sqrt{\text{Hz}} \\ \sigma_{\text{gyro}}^2 &= S_{\text{gyro}}^2 \cdot 20 = 7.46314 \times 10^{-8} \text{ rad}^2/\text{s}^2 \end{aligned}$$

The accelerometer density of $0.14 \text{ mg}/\sqrt{\text{Hz}}$ is treated the same way, once converted out of g units with $g = 9.81 \text{ m/s}^2$:

$$\begin{aligned} S_{\text{accel}} &= 0.14 \text{ mg}/\sqrt{\text{Hz}} = 1.37340 \times 10^{-3} \text{ m/s}^2/\sqrt{\text{Hz}} \\ \sigma_{\text{accel}}^2 &= S_{\text{accel}}^2 \cdot 20 = 3.77246 \times 10^{-5} \text{ m}^2/\text{s}^4 \end{aligned}$$

Both variances are recomputed at full floating-point precision rather than read back from the rounded figures above, and each serves two purposes. The first is to parameterise the additive Gaussian noise that corrupts the reading. The second is to populate the covariance field of the outgoing message, which the simulator would otherwise report as zero. The downstream filter therefore receives a corrupted reading together with an honest account of how far it may have been displaced. Expressed as standard deviations, which are easier to compare with the measured quantities, the two values are $2.73 \times 10^{-4} \text{ rad/s}$ for the gyroscope and $6.14 \times 10^{-3} \text{ m/s}^2$ for the accelerometer.

Both figures describe scatter redrawn independently at every sample, one of the three error terms carried by a real inertial sensor. The other two persist across many samples and are therefore modelled separately from the white noise. The first is in-run bias stability, treated as a fixed offset that holds for an entire episode and is redrawn at each boundary. Its value on every axis is sampled from a uniform distribution symmetric about zero and bounded by the stability figure specified in the sensor datasheet, which gives $5^\circ/\text{h}$ for the gyroscope, the lower end of the typical 5 to 7 range, and 0.04 mg for the accelerometer. Neither is quoted in the units required by the model, so converting the first from degrees per hour and the second from g gives the

bounds actually sampled:

$$\begin{aligned} b_{\text{gyro}}^{\text{max}} &= 5 \cdot \pi / 180 / 3600 = 2.42407 \times 10^{-5} \text{ rad/s} \\ b_{\text{accel}}^{\text{max}} &= 0.04 \times 10^{-3} \cdot 9.81 = 3.924 \times 10^{-4} \text{ m/s}^2 \end{aligned}$$

Alongside that offset, the second persistent term is a residual scale-factor error of $\pm 0.5\%$ per axis, taken from the factory calibration specification. It is applied multiplicatively rather than additively and is resampled together with the bias at every episode boundary. Combining both terms with the white noise above, each measured channel follows:

$$\tilde{\chi} = \zeta \chi + b + \eta \quad (\text{A.4})$$

where χ is the true value, $\tilde{\chi}$ the measured value, ζ the scale factor, b the per-episode bias and η the white noise.

The channel model given by Eq. (A.4) corrupts whatever motion the simulator reports. A stationary vehicle is not reported as exactly stationary, as the physics solver leaves small residual rates on a body at rest, and the channel model would corrupt those residuals as though they were genuine motion. A ZUPT guard limits the drift that would otherwise accumulate. The yaw rate $\dot{\psi}$ is set to zero when $|\dot{\psi}| < 0.015 \text{ rad/s}$, and the planar accelerations are set to zero when the gyroscope ZUPT is active and both magnitudes fall below 0.2 m/s^2 . Noise injection is skipped for every channel clamped in this manner, ensuring that the guard and injection remain consistent.

A.3 2D LiDAR: SICK TiM571

The LiDAR datasheet [73] separates the range error of the scanner into an offset that persists and a scatter that does not, quoting $\pm 60 \text{ mm}$ for the systematic component and below 20 mm at 90% remission for the statistical one. The systematic term varies between units and between mountings while holding constant during a session, so it is represented by a range bias sampled uniformly from $\pm 0.060 \text{ m}$ once per episode. The statistical term instead describes per-point Gaussian range noise with a standard deviation of 0.020 m , taken from the higher 90% remission figure rather than the 0.010 m figure quoted at 10% remission. Car-park surfaces span the full remission range, so the larger value is the more cautious choice for the clearance features.

Both terms act on range and are applied in the polar form in which the scan is published, so a true range r_{true} is reported as:

$$r_{\text{rep}} = r_{\text{true}} + b_r + \mathcal{N}(0, \sigma_r^2) \quad (\text{A.5})$$

where r_{rep} is the reported range, b_r is the per-episode bias and σ_r is the per-point standard deviation. Returns below the sensor's 0.05 m minimum operational range are then discarded. The simulator provides a per-sensor noise attribute of its own, which is set to zero at spawn so that Eq. (A.5) remains the only corruption applied to the scan.

A.4 Extended Kalman Filter Process Noise

The process-noise covariance Q determines the rate at which reported uncertainty rises during a lost fix. The entries for position, yaw and yaw rate govern that rise, and this section records them alongside the remaining estimated states, justifying each value relative to the others.

Those relative values are interpretable only against the structure of the matrix, which the filter holds as a continuous-time spectral density over a fifteen-element state. It contains the pose (three positions and three orientations), the six rates of change of those quantities, which together form the twist, and three linear accelerations. Each prediction step adds the matrix to the propagated covariance in proportion to the timestep, so a larger diagonal entry inflates the uncertainty of its associated state more rapidly between corrections. Only part of that state is live, as the filter's planar mode estimates the pose $[x, y, \psi]$ together with the two planar velocities and the yaw rate $\dot{\psi}$, leaving the remaining rows inert. The shipped diagonal is listed in Table A.2.

Table A.2 Diagonal of the EKF process-noise covariance Q .

State	Role	Value	Rationale
x, y	Active	1.0×10^{-2}	Residual constant-velocity model error on position
ψ (yaw)	Active	1.0×10^{-4}	Orientation drift consistent with the VN-100 noise floor
$\dot{\psi}$ (yaw rate)	Active	1.0×10^{-3}	Gyro noise density; dominant source of between-fix growth
Planar velocities	Active	1.0×10^{-4}	Longitudinal component constrained by the Doppler update each tick
z , roll, pitch	Suppressed	1.0×10^{-6}	Positive-definite floor; not estimated in planar mode

The yaw-rate entry $\dot{\psi}$, the largest of the rate terms, sets the reference point for the whole diagonal. Yaw rate is the only fused IMU channel, so the uncertainty allowed by this entry propagates to the reported yaw and, through kinematic coupling, to the position estimate. Its value is therefore tuned well above the VN-100 gyroscope noise density derived in Appendix A.2, so that the filter admits the noise of the sensor and also the far larger error of propagating a single rate channel through the motion model for the duration of a lost fix. The remaining active groups are sized against the yaw-rate

entry according to the correction each one receives. The yaw entry ψ sits an order of magnitude lower, since the yaw rate already supplies most of the orientation inflation and only a small additional drift is required. The planar velocities are held at that level as well, the longitudinal one being corrected by the Doppler speed delivered at every update. The position entries, by contrast, are the largest of the estimated states, as the fix that constrains them is the correction lost during a degraded tier, and their value therefore covers the drift of the constant-velocity motion model across an unbounded gap.

Together these entries set how rapidly the posterior expands between measurements, and with it the slope of the rising edges in the sawtooth covariance profile plotted in Fig. 3.5. Only the yaw-rate and yaw terms would be retuned on the vehicle described in Appendix B, against the measured noise floor of its fitted IMU in place of the VN-100 figures. The position and velocity entries instead quantify how far the motion model departs from the true trajectory between fixes, an error added to the covariance through the Q_k term of the prediction step in Eq. (2.10). A change of hardware leaves this mismatch unaffected, so their values would carry over unchanged.

Appendix B Real-World Deployment

This appendix describes the sim-to-real pathway outlined in Chapter 3, which forms the basis for the first item of future work in Section 5.2. The pathway is an implemented and tested software pipeline that has not yet been run on hardware (Section 4.8.3). The work remaining before an initial trial therefore concerns measuring the site-specific physical quantities expected by the pipeline, including the survey of the parking area and calibration of the vehicle's actuators. Those quantities are supplied through a single machine-verified interface, a set of configuration files gathered under one deployment directory in the repository layout shown later in Fig. D.1.

B.1 Code-Path Parity

On the localisation side, deployment is restricted to one modification, the replacement of the publishers of two topics. The position of that substitution within the data flow is shown in Fig. B.1.

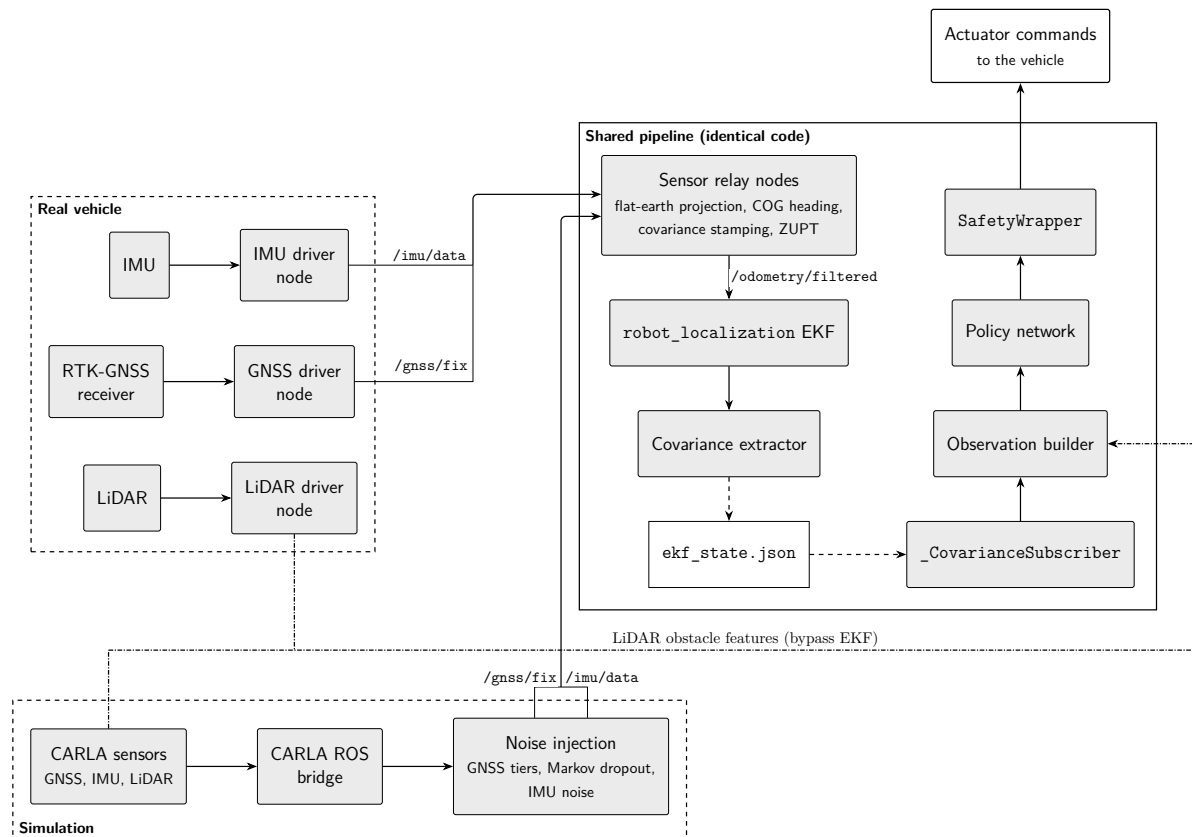


Figure B.1 Sensor data flow in simulation and on the real vehicle.

The CARLA bridge publishes the GNSS fix and IMU stream in simulation, whereas the vehicle's hardware driver nodes publish them on the real platform.

Everything downstream of those two topics belongs to the localisation and decision pipeline described in Section 3.1, which carries over unchanged and differs between the two settings only through the master-switch values listed later in Appendix B.2. The remaining difference is at the far end of the flow, where the resulting command is applied either to the simulated vehicle in CARLA or to the physical actuators through the calibration set out later in Appendix B.4.

Two properties follow from that parity. First, the observation supplied to the policy is produced by the same builder and scaled according to the same predetermined physical ranges used during training, so the learned weights are compatible with that interface by construction, although performance without retraining remains untested (Section 4.8.3). Second, simulator ground truth never enters the observation path, ensuring that every input consumed by the policy in simulation is also available on the real vehicle.

B.2 Robot Operating System Interface Parity

The interface between the hardware and the pipeline consists of two standard message types. The relay nodes expect `sensor_msgs/NavSatFix` and `sensor_msgs/Imu`, formats already published by off-the-shelf ROS 2 drivers such as `ublox_ros` and `nmea_navsat_driver`. The two sides therefore connect directly without custom driver code to translate between them.

The interface is bound to the running system by a real-vehicle launch description that forms part of the repository. It starts the same relay executable as the simulation stack and constructs the static sensor transforms under conventional frame names (`base_link`, `imu_link`, `lidar_link`, `gnss_link`). The mounting poses encoded in those transforms are read from `sensor_config.yaml`, which also positions the sensors on the simulated vehicle. A mount measured on the physical platform is therefore recorded once and governs both the simulator and the real vehicle.

The relays perform two functions, only one of which survives deployment. Noise injection becomes redundant on a real receiver, since its measurements already carry genuine error, whereas signal conditioning is required on the vehicle as much as in simulation. The division is settled by the launch description, which sets the master switches (Appendix B.1) to their deployment values. The noise, Markov-transition and anisotropy flags are set to false and the dropout probability to zero, disabling every noise mechanism derived in Appendix A. The signal conditioning established in Section 3.4 remains, comprising the flat-earth projection, COG derivation, covariance stamping and ZUPT clamping, none of which is an artefact of simulation. Real receiver values therefore pass through unchanged in every other respect.

B.3 Extended Kalman Filter Frame Calibration

The EKF reports poses in an odometry frame anchored at the node's most recent initialisation. The bays and boundaries targeted by the policy are instead specified in the world coordinates of the layout files. Both frames describe the same physical ground, so distances and angles are preserved between them and only their origins and headings differ. Connecting them is therefore a matter of calibration, using one known correspondence to solve for the two-dimensional rigid transform (t_x, t_y, ψ_0) satisfying:

$$\begin{aligned} x_{\text{world}} &= \cos \psi_0 \cdot x_{\text{odom}} - \sin \psi_0 \cdot y_{\text{odom}} + t_x \\ y_{\text{world}} &= \sin \psi_0 \cdot x_{\text{odom}} + \cos \psi_0 \cdot y_{\text{odom}} + t_y \\ \psi_{\text{world}} &= \psi_{\text{odom}} + \psi_0 \end{aligned} \tag{B.1}$$

where $(x_{\text{odom}}, y_{\text{odom}}, \psi_{\text{odom}})$ is a pose in the EKF odometry frame and $(x_{\text{world}}, y_{\text{world}}, \psi_{\text{world}})$ is the equivalent pose in world coordinates. The translation (t_x, t_y) between the origins and yaw offset ψ_0 between the headings account for the three unknowns. Observing the vehicle once at a location where both its odometry and world poses are known therefore fixes all three simultaneously, and obtaining those readings is the task of the calibration procedure.

The odometry reading is obtained once the EKF pose has converged, a condition met when consecutive measurements remain within defined position and yaw stability thresholds. Its y follows the ROS coordinate convention REP-103, which takes y as positive to the left, so both the inference loop and simulation environment negate it to align the pose with the lot frame. The corrected pose is then matched with a known reference pose, and together they supply the correspondence from which Eq. (B.1) is solved for the offset (t_x, t_y, ψ_0) . In simulation that reference pose is the spawn position specified in the layout file, whereas on the vehicle it is a physical marker at the lot datum, surveyed on site and recorded in `real_world_datum.yaml`. The record locates the marker in the lot frame, where its position and heading form the reference pose against which the converged odometry pose is matched. The same point is also recorded in global coordinates as latitude and longitude on the WGS84 ellipsoid, which the sensor relay uses as its projection origin. That same position is retained once more as an easting and northing in the Universal Transverse Mercator projection, which no node reads and which serves instead as a cross-check that exposes errors in the surveyed figures before deployment.

B.4 Actuation Calibration

Physical actuators respond non-linearly to normalised policy outputs. That mismatch takes a different form on the bipolar steering axis than on the two non-negative pedal axes (Section 3.3.3), so the actuation-calibration layer holds a separate parameter set per channel. Let u_{policy} denote the raw policy output on a channel and u_c the input at which its deadband is centred. An actuator that responds asymmetrically about its neutral position takes a u_c displaced from zero. The shifted output $\Delta u = u_{\text{policy}} - u_c$ is the signed distance of the request from that centre, from which the physical command is obtained as:

$$u_{\text{physical}} = \text{clip}(u_{\text{raw}}, u_{\text{min}}, u_{\text{max}}), \quad u_{\text{raw}} = \begin{cases} b_{\text{act}} & \text{if } |\Delta u| < d_{\text{db}} \\ g_{\text{act}} \cdot \Delta u + b_{\text{act}} & \text{otherwise} \end{cases} \quad (\text{B.2})$$

where u_{raw} is the unclipped command, g_{act} is the gain, b_{act} is the bias, d_{db} is the deadband half-width about u_c , and $[u_{\text{min}}, u_{\text{max}}]$ is the physical output range imposed on each channel. The deadband branch holds the command at the bias regardless of the input, suppressing jitter around the centre, while the clip bounds the affine branch to the range physically admitted by the channel.

Each of those four parameters describes a physical response that can be established only by measuring the actuator, which leaves all four unmeasured until the vehicle is available. Determining them per channel requires commanding a range of policy outputs and recording the physical response to each, which traces the response curve of that channel. All four are obtained from a single run of that procedure, each being read off a distinct feature of the same curve. The gain is derived from its slope and the deadband from the smallest input that initiates motion. Any asymmetry between the two directions then determines the deadband centre, leaving the bias to be read from any residual output at zero input.

No such run has yet been possible, so every parameter in `actuation_calibration.yaml` retains the identity value for its role. The gain is unity, the bias and deadband are zero, and the output bounds remain at the full physical range. This combination reduces Eq. (B.2) to its input, as does the complete absence of the file. An uncalibrated vehicle therefore transmits each command as issued by the policy rather than applying a correction unsupported by measurement.

Delivering the calibrated command presumes a drive-by-wire connection to the vehicle, marking the end of the symmetry with the sensor interface described in Appendix B.2. An incoming fix arrives in a standard message type already published by off-the-shelf drivers, whereas an outgoing command is written to a Controller Area Network (CAN) bus through a suitable interface adapter. The frame format is defined

by the manufacturer for that platform alone. Code producing those frames can therefore be written only for a selected vehicle, and the repository contains none. The actuation calibration is thus the last stage of the pipeline that carries over to any platform.

B.5 Mission and Pre-Deployment Sequence

The parity of the code paths and the two calibrations establish the conditions under which the vehicle may be driven, while the objective of a particular deployment run is stated separately. The objective is recorded in `mission.yaml`, which is read only on the vehicle and names the target bay identifier alongside the site's layout file. The deployment entry point reads both fields, loads the layout through the same cached loader used by the simulator and resolves the bay within it. Either step raises an immediate error if its field is unset, as does a bay identifier absent from the layout. A misconfigured mission is therefore rejected before the vehicle moves. These checks confirm that the mission is well formed but do not establish correct behaviour on hardware. Five staged tests are intended to provide that evidence before any autonomous deployment, with a driver at the wheel for the first three so that each component can be observed while the policy has no control over the vehicle:

1. Verify the EKF pipeline in isolation against its covariance output.
2. Verify the target-bay pose pipeline against the relative target observations.
3. Verify the actuation calibration against known inputs.
4. Run the first closed-loop test in a single bay under RTK fixed conditions.
5. Test progressively across fix-state tiers and conditions.

Appendix C Hyperparameter Reference

Every arm and seed was trained under the same set of hyperparameter values, summarised in Table 3.10. Those values were fixed in advance rather than searched over, since a separate search per arm would make the configuration a third variable and leave any measured difference partly attributable to four searches converging unevenly, as Section 3.8.1 sets out. This appendix therefore records the committed values under the parameter names of the configuration files, so that every figure reported in Chapter 3 can be traced to the entry governing it.

C.1 Committed Configuration

Every value governing a training run is declared in a single configuration file, `train_config.yaml`, and its committed contents are reproduced in Table C.1. Those entries restate Table 3.10, listing separately the batch size and epoch count paired there and adding the network architecture and the value-function clipping setting that the summary omits.

Table C.1 Committed values of `train_config.yaml`.

Parameter	Value	Note
<code>n_steps</code>	2048	Steps per update
<code>batch_size</code>	256	Minibatch size for the epoch loop
<code>n_epochs</code>	5	Optimisation passes per rollout
<code>gamma</code>	0.997	Discount
<code>gae_lambda</code>	0.95	GAE bias/variance trade-off
<code>clip_range</code>	0.2	Policy trust region; <code>clip_range_vf</code> null (no value-function clipping)
<code>learning_rate</code>	$0.0003 \rightarrow 0.00001$	Stage-owned linear decay; see Table C.2
<code>ent_coef</code>	$0.02 \rightarrow 0.006$	Stage-owned linear decay
<code>vf_coef</code>	0.5	Value-loss weight
<code>max_grad_norm</code>	0.5	Global gradient-norm clip
<code>target_kl</code>	0.04	Early-stops epochs above this KL; backup to the clip
<code>net_arch</code>	[256, 256], ReLU	Actor and critic hidden layers
<code>log_std_init</code>	-0.6	Initial action standard deviation ≈ 0.55 ; standard head only
<code>lambda_reg</code>	0.02	Prior-anchor coefficient; linear warm-up over 50 000 steps
<code>aleatoric_floor</code>	0.04	Sampling standard deviation floored at $\sqrt{0.04} = 0.2$

Two tabulated entries, the learning rate and entropy coefficient, carry only a base value in that file. Every curriculum stage defined in Section 3.7.2 supplies its own schedule in place of that base. Each stage therefore contains a `standard_overrides` block for the vanilla and input-uncertainty arms alongside an `evidential_overrides` block for the output-uncertainty and full-method arms. The blocks differ only in the entropy coefficient and are restricted to schedule keys by the governing allowlist. Their values across the curriculum are consequently presented separately in Table C.2, which reproduces the budget and learning-rate columns of Table 3.8.

Table C.2 Per-stage schedules. Each value decays linearly from start to final over the stage budget.

Stage	Budget (10^6 decisions)	Learning rate (start $\rightarrow$ final)	Entropy, standard (start $\rightarrow$ final)	Entropy, evidential (start $\rightarrow$ final)
1	1.5	0.0003 $\rightarrow$ 0.00005	0.003 $\rightarrow$ 0.0005	0.003 $\rightarrow$ 0.0015
2	1.5	0.00015 $\rightarrow$ 0.00005	0.0015 $\rightarrow$ 0.0005	0.0015 $\rightarrow$ 0.0015
3	2.0	0.00015 $\rightarrow$ 0.00004	0.0015 $\rightarrow$ 0.0005	0.0015 $\rightarrow$ 0.0015
4	1.5	0.00012 $\rightarrow$ 0.00004	0.001 $\rightarrow$ 0.0005	0.002 $\rightarrow$ 0.0015
5	2.0	0.00012 $\rightarrow$ 0.00003	0.001 $\rightarrow$ 0.0005	0.002 $\rightarrow$ 0.0015
6	1.5	0.0001 $\rightarrow$ 0.00002	0.001 $\rightarrow$ 0.0005	0.002 $\rightarrow$ 0.0015

The pattern of those schedules follows from two design choices made in Section 3.7.2. The learning rate restarts each stage at approximately three times the final value of the preceding one, a level low enough that a nearly converged policy is not destabilised at a difficulty boundary. The entropy floor then varies by head type, with the standard blocks descending to 0.0005 in every stage while the evidential blocks remain at 0.0015. The evidential schedules also begin at a higher value from stage 4 onwards. Exploration on the evidential head is tied to the aleatoric estimate, so both of those departures are required to keep it from collapsing.

No entry of Table C.1 is conditioned on an arm, so the one declaration governs all four and no contrast reported in Chapter 4 can be traced to a configuration difference between them. The entropy floor listed in Table C.2 is the single exception, and it is set from the head type alone, being the guard against variance collapse on the evidential head rather than a value derived from any arm’s measured performance.

Appendix D Reproduction Guide

This appendix documents the procedure for reproducing the result tables reported in Chapter 4 from a freshly cloned copy of the source, covering the container stack, the cross-container file bridge, the supporting tooling, the seed conventions, the make targets that constitute the pipeline and the repository structure they operate on. Those targets span every workflow in the project and are described in full, with their variables, in the repository’s `COMMANDS.md`, from which the subset required for reproduction is restated in Appendix D.5. Both that file and the full source it accompanies are available [on GitHub](#). The main README of the repository also contains animations of the vehicle driving autonomously in CARLA, together with a two-dimensional visualiser that traces its behaviour under each of the four receiver tiers defined in Section 3.4.1.

D.1 Container Stack

The experiment runs in the three Docker services listed in Table D.1, and Table D.2 gives the full version set. The services are split by the Ubuntu release each component needs, 24.04 for the Jazzy bridge and 22.04, the base of the NVIDIA GPU Cloud (NGC) PyTorch image, for the training stack. The state estimate therefore crosses between the two through the file bridge described in Appendix D.2. The sensors are spawned and the simulation clock is advanced only from the training container, so simulation time has a single writer and an episode replays identically from the same seed. The bridge container also restarts automatically after a headless CARLA failure and reconnects without human intervention.

Table D.1 The three container services orchestrated by `docker-compose`.

Service	Base image	Contents
<code>carla-server</code>	CARLA 0.9.16	Headless simulator with GPU passthrough
<code>ros2-bridge</code>	ROS 2 Jazzy	CARLA bridge, <code>robot_localization</code> EKF, sensor relay nodes
<code>training</code>	NGC PyTorch 24.10	Stable-Baselines3 2.8.0, Gymnasium 1.2.3, the evidential networks, <code>rclpy</code>

Simulation, training and evaluation run within Docker, so the host requires only Git, Docker Engine, Make and the NVIDIA Container Toolkit to produce the raw records, with no local installation of CARLA [4] or ROS. The analysis behind the reported tables and figures runs outside the containers, in a virtual environment created by `make install` that needs neither a GPU nor the simulator. Everything the containers depend on is pinned inside the images, where `pyproject.toml` fixes the

Python dependencies to the exact versions captured from the working training image. Code directories are the exception, being bind-mounted into each container from the host so that an edit takes effect without a rebuild.

Parallel data collection is also available across these three services, scaled by replicating the `carla-server` and `ros2-bridge` pair rather than the training service. The `parallel_workers` key in `train_config.yaml` fixes the replica count, and the worker scripts launch the corresponding number of simulator and bridge instances. Each pair is isolated by a distinct port block, a separate Data Distribution Service (DDS) domain and its own EKF state file, so that no traffic passes between workers.

Table D.2 Software configuration of the experiment stack.

Component	Version	Container
CARLA simulator	0.9.16	carla-server
ROS 2 (bridge, EKF, relays)	Jazzy	ros2-bridge
ROS 2 client library	Humble	training
PyTorch base image	NGC 24.10	training
Stable-Baselines3	2.8.0	training
Gymnasium	1.2.3	training
Python	3.10 / 3.12	training / ros2-bridge

D.2 The Jazzy-Humble File Bridge

The state crosses from the Jazzy bridge to the Humble training container as three JSON files on a shared volume listed in Table D.3. The two releases use different DDS versions and build their message types separately, so a subscription across them proved unreliable.

Each file carries data in one direction only, and the three follow two cadences. The continuous cadence belongs to `ekf_state.json`, which the covariance extractor writes at every EKF update and the environment reads at every decision step. The episodic cadence instead covers the two training-to-bridge files, both rewritten at each reset. The covariance extractor polls `initial_pose.json` for the spawn pose, then issues the `/set_pose` call that anchors the EKF state to the new episode. The noise relay in turn reads `episode_config.json` before the vehicle moves, and the same file carries the optional drift override defined with the rest of the fix-state model in Section 3.4.1. Each run is thus configured from its own draw and does not inherit the relay's state from the previous run.

Table D.3 The three shared files that cross the container boundary.

File	Writer	Reader	Purpose
<code>ekf_state.json</code>	Covariance extractor (bridge)	Environment (training)	Latest EKF pose, twist and 3×3 covariance
<code>initial_pose.json</code>	Environment (training)	Covariance extractor (bridge)	Spawn pose; triggers an EKF state reset via <code>/set_pose</code>
<code>episode_config.json</code>	Environment (training)	GNSS noise relay (bridge)	Starting fix tier, spawn datum and drift override (Section 3.4.1)

The files are accessed without a shared lock, so two mechanisms safeguard the exchange against race conditions. Each write is atomic, as the record is serialised to a temporary file and subsequently renamed over the target using `os.replace`, the atomic POSIX rename, ensuring that a partially written file is never visible to a reader. Atomicity by itself leaves a complete but stale record indistinguishable from a fresh one, so each record is also marked with a monotonically increasing sequence number alongside a modification timestamp, allowing an unchanged record to be skipped without reparsing. The reader records the latest sequence at each episode reset and subsequently accepts only strictly higher values, so the first observation of an episode is withheld until the EKF has been re-initialised.

Each file holds a single sub-kilobyte record on a page-cache-backed volume, so the exchange completes in memory and takes almost none of the 200 ms decision interval fixed in Section 3.3.3. The records are also plain text and can therefore be logged, inspected and replayed when a fault spans the two containers. The most recent values persist on the shared volume as well, so they survive a bridge relaunch after a headless CARLA failure.

D.3 Supporting Tooling and Verification

Three supporting tools were built around the stack to speed up development and make each design decision verifiable. A layout generator converts the geometry of each lot from a local-frame floor-plan module into a world-frame YAML description and draws a matching bird's-eye plot, so every layout can be defined and checked by eye before it enters the simulator. Once the layout is loaded, a detachable two-dimensional episode visualiser attaches to a training or evaluation run without altering it and shows the EKF estimate, the ground truth and the live covariance ellipse together as the episode is stepped. A drift between the estimate and the truth can thus be read against the

uncertainty claimed for it, which confirms that the filter responds to each fault modelled in Section 3.4.1 as intended.

Beyond these two views, each configuration is examined in a unified lot inspector across the four fidelity levels set out in Table D.4, from static geometry up to a fully stepped environment. A discrepancy absent at one level and present at the next can only come from what that level added, so each error is traced to the stage that introduced it. Faults in the geometry, the sensor placement and the observation pipeline are therefore separated before costly training begins.

Table D.4 The four fidelity levels of the lot inspector.

Level	Adds	Verifies
1	The bare lot geometry	Bay placement and lot dimensions
2	The sensor mounts and their fields of view	Mounting poses and coverage
3	A live LiDAR, visualised against the simulated environment	Returns consistent with the modelled scene
4	A stepped training environment under a predetermined or manually selected action, with no policy loaded	The exact observation the policy would receive

All four levels are assessed visually, as judging whether a rendered scene is faithful demands an observer. The underlying implementation is instead checked by an automated suite as the code changes. The repository, laid out in Fig. D.1, holds more than 470 unit and integration tests, run in the training container by the `docker-test-unit` target, and a continuous integration pipeline enforces linting, formatting, static typing and a clean package import on each push. Beyond code quality, the properties on which the methodology depends are encoded as tests, including the channel-liveness principle established in Section 3.7.1, which is asserted by checking that every observation channel varies from the first stage. The arm definitions given in Section 3.8 are verified in the same way, by merging each configuration file onto the base and confirming that the result is the intended arm. An edit that would compromise the experimental design therefore fails the corresponding test, allowing the fault to be traced and corrected rather than passing undetected. Reproducibility is further supported at the level of the runs themselves, as every reported run was launched through one of the `make` targets listed in Table D.5, with explicit parameters, and every tuneable value is declared in a YAML file rather than left as a hidden code default.

D.4 Seeds and Determinism

Every run takes its random state from the seed field of `configs/deployment/agent_config.yaml`, which both the training process and the noise container read. Editing that one entry therefore updates both processes, so the network initialisation and the sensor-noise realisations derive from the same value.

Each of the three legs set out in Section 3.8.2 uses a different setting of that field, and its results are kept apart within `outputs/raw/evaluation_results/`, where each `seed_<N>` directory holds one folder per ablation arm (Section 3.8). The target location is taken from the name of the trained run, so records from different legs cannot overwrite one another.

Reproducing the full study is nonetheless costly, as each of the twelve training chains takes 23 to 33 hours of wall-clock time at the observed 85 to 120 decisions per second on the stack listed in Table D.2. The complete set therefore requires around two weeks of continuous GPU time.

D.5 Pipeline Reference

The make targets that reproduce the reported results are listed in pipeline order in Table D.5. Two variables recur across them, `BASELINE` designating one of the four arms and `CHECKPOINT` identifying the directory of a single trained run, termed its leaf. Both are bare names and not paths, the full location of each within the output hierarchy being reconstructed by the recipes.

The same pair locates a trained run and its evaluation in two separate trees. Checkpoints are saved under `checkpoints/<baseline>/<leaf>/`, whereas results are written to `outputs/raw/evaluation_results/seed_<N>/<baseline>/<leaf>/`, in which the leading `seed_<N>` layer provides the per-seed separation described in Appendix D.4. Each evaluation leaf is then split by the mode of the safety wrapper into `with_wrapper` and `without_wrapper` directories. The protocol set out in Section 4.1.3 is run with the wrapper disabled, so every result in Chapter 4 is read from the `without-wrapper` directory.

D.6 Repository Structure

Every path named in this appendix belongs to the directory tree given in Fig. D.1. Its top level keeps three things apart, the Python package, the configuration that parameterises it, and the artefacts a run produces, written to directories absent from a fresh clone. Each subpackage and script directory has its own README.

Table D.5 Make targets for the full pipeline.

Target	Purpose
docker-build	The three service images and the inspection image
docker-up / docker-down	Start or stop the worker pairs and training stack
docker-train STAGE=N BASELINE=<arm>	One curriculum stage; stages 2 to 6 add CHECKPOINT=<leaf> to resume
docker-eval BASELINE=<arm> CHECKPOINT=<leaf>	The 200-episode sweep over eval_config.yaml, four conditions analysed (Section 4.1.3); NO_SAFETY=1 disables the wrapper
run-seed-leg	Every arm and seed through all six stages, with evaluation and analysis; idempotent, so a re-run resumes a crash
analyse-ablation / analyse-gate	Per-seed arm contrasts and gate ROC tables (STAGE=6 SEED=<s>)
analyse-calibration / handover-timing	Per-checkpoint EKF calibration and handover timing (ARM=<arm> CHECKPOINT=<leaf>)
analyse-cross-seed	Pools every seed's stage-6 evaluation into the headline tables
figures / analysis-bundle	Chapter 4 figures and tables, in outputs/main_analysis/
docker-test-unit / verify	Tests in the training container; the lint, type and import checks on the host

Figure D.1 Repository layout.

uncertainty_rl/.....	Python package
├─ envs/	Environment: sim/, real/, shared core
├─ networks/	NIG head, policies, EvidentialPPO
├─ training/	train_ppo.py
├─ evaluation/	evaluate.py, conditions, metrics
├─ ros2/	Relays, EKF launch, covariance extractor
├─ utils/	Constants, covariance, geometry
├─ configs/	
│ └─ train_config.yaml	PPO and evidential settings
│ └─ eval_config.yaml	Evaluation conditions
│ └─ baselines/	The four arm definitions
│ └─ deployment/	Agent, sensor, sim/ and real/ configs
│ └─ layouts/	Generated lot geometry
├─ scripts/	Tooling, analysis, figures, workers
├─ tests/	Unit and integration tests
├─ docs/	Dissertation, design notes, media
├─ outputs/	raw/, raw_derived/, main_analysis/
├─ checkpoints/, logs/	Models and training logs
├─ Makefile, COMMANDS.md	Pipeline targets
└─ docker-compose*.yaml	Service definitions

Appendix E Per-Seed Evaluation Results

The per-seed success rates and mean final position errors underlying the pooled task figures reported in Chapter 4 are recorded in Table E.1, so that each step of aggregation can be checked rather than taken on trust. Each row corresponds to one combination of arm, condition and training seed, summarising the 200 deterministic evaluation episodes run under it. Both are derived from these rows, as the pooled figures reported in Table 4.2 are the mean over the three seeds of a given arm and condition, and the per-seed ranges quoted in Section 4.7 are the extrema.

Table E.1: Per-seed success rate and mean final position error ($n = 200$ episodes per row).

Arm	Condition	Seed	Success (%)	Error (m)
vanilla_ppo	anchor_deployment	7	10.0	11.59
vanilla_ppo	anchor_deployment	42	32.5	8.72
vanilla_ppo	anchor_deployment	123	29.0	11.43
vanilla_ppo	anchor_empty	7	18.5	4.82
vanilla_ppo	anchor_empty	42	41.5	2.36
vanilla_ppo	anchor_empty	123	40.0	1.87
vanilla_ppo	gnss_degrade_one_way	7	12.5	9.36
vanilla_ppo	gnss_degrade_one_way	42	28.0	11.33
vanilla_ppo	gnss_degrade_one_way	123	41.5	7.65
vanilla_ppo	ood_irregular_rtk_fixed	7	0.0	36.01
vanilla_ppo	ood_irregular_rtk_fixed	42	0.0	19.22
vanilla_ppo	ood_irregular_rtk_fixed	123	0.0	23.44
input_uncertainty	anchor_deployment	7	36.5	11.25
input_uncertainty	anchor_deployment	42	23.0	10.45
input_uncertainty	anchor_deployment	123	16.0	10.24
input_uncertainty	anchor_empty	7	21.0	1.75
input_uncertainty	anchor_empty	42	43.5	1.03

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Table E.1: Per-seed success rate and mean final position error
($n = 200$ episodes per row). (Continued)

Arm	Condition	Seed	Success (%)	Error (m)
input_uncertainty	anchor_empty	123	27.5	1.98
input_uncertainty	gnss_degrade_one_way	7	41.5	9.08
input_uncertainty	gnss_degrade_one_way	42	24.5	8.71
input_uncertainty	gnss_degrade_one_way	123	19.5	7.19
input_uncertainty	ood_irregular_rtk_fixed	7	0.0	25.11
input_uncertainty	ood_irregular_rtk_fixed	42	0.0	21.32
input_uncertainty	ood_irregular_rtk_fixed	123	0.0	25.08
output_uncertainty	anchor_deployment	7	18.0	4.80
output_uncertainty	anchor_deployment	42	22.0	5.42
output_uncertainty	anchor_deployment	123	17.0	6.13
output_uncertainty	anchor_empty	7	32.0	1.80
output_uncertainty	anchor_empty	42	19.5	1.54
output_uncertainty	anchor_empty	123	34.0	1.44
output_uncertainty	gnss_degrade_one_way	7	23.0	4.76
output_uncertainty	gnss_degrade_one_way	42	20.5	5.58
output_uncertainty	gnss_degrade_one_way	123	16.5	6.49
output_uncertainty	ood_irregular_rtk_fixed	7	0.0	18.65
output_uncertainty	ood_irregular_rtk_fixed	42	0.0	23.31
output_uncertainty	ood_irregular_rtk_fixed	123	0.0	23.11
full_method	anchor_deployment	7	39.5	7.54
full_method	anchor_deployment	42	47.0	4.08
full_method	anchor_deployment	123	51.0	12.45
full_method	anchor_empty	7	48.0	0.37
full_method	anchor_empty	42	78.5	0.71

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Table E.1: Per-seed success rate and mean final position error
($n = 200$ episodes per row). (Continued)

Arm	Condition	Seed	Success (%)	Error (m)
full_method	anchor_empty	123	67.5	0.52
full_method	gnss_degrade_one_way	7	40.0	8.30
full_method	gnss_degrade_one_way	42	55.0	3.13
full_method	gnss_degrade_one_way	123	62.5	9.55
full_method	ood_irregular_rtk_fixed	7	0.0	19.64
full_method	ood_irregular_rtk_fixed	42	0.0	19.37
full_method	ood_irregular_rtk_fixed	123	0.0	20.46